MAXIMO ENTERPRISE ASSET MANAGEMENT AT OTORITA IBU KOTA NEGARA (OIKN)

Training Material – Work Management (part 1)

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1 Person, Labor, User, & Craft

1.1 Overview

Person represent an individual that related to EMS system. Person record is created for any individual whose name appears anywhere on any data. Person record is required when creating a user or a labor. A person data contain basic information about an individual's name, address, contact information, and other generic information.

Labor represent employees and contractors who perform work on work orders. Labor records contain information about an individual's skills and qualifications. These records are used to plan and schedule work, and to track labor costs for work orders. A labor record must have a person record associated with it.

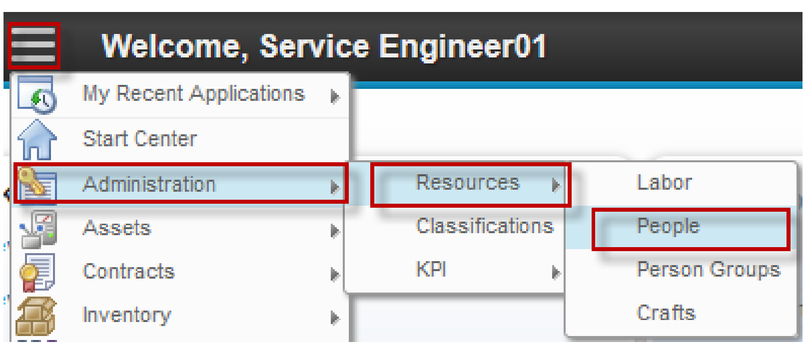
User represent authentication to access the system. User records contain user names, passwords, and security profiles that determine which applications, options, and data a user can access. A user must have a person record. A labor record and user record can be associated to the same person record.

Crafts represents an occupation or trade, and typically the craft name reflects the type of work done by members of the trade. If pay rates differ based on expertise, multiple skill levels with different pay rates for each skill level can be assigned to the craft. For each craft user can specify standard rates, and add premium pay codes for premium pay rates.

1.2 People Application

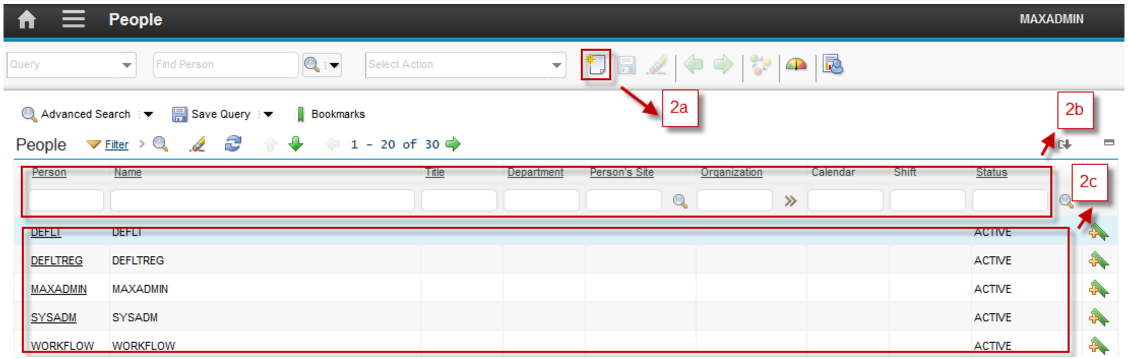
People application is used to create, modify, view, and delete records person data. The **People** application stores personal information, employee information, and delegation configurations.

Followings are details about **People** applications accessibility:

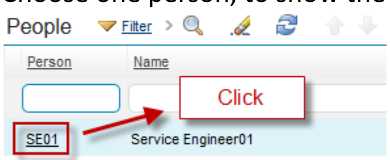
<table>
<tr>
<td>1</td>
<td>


From **Go To** menu, choose **Administration – Resources – People**
</td>
</tr>
</table>

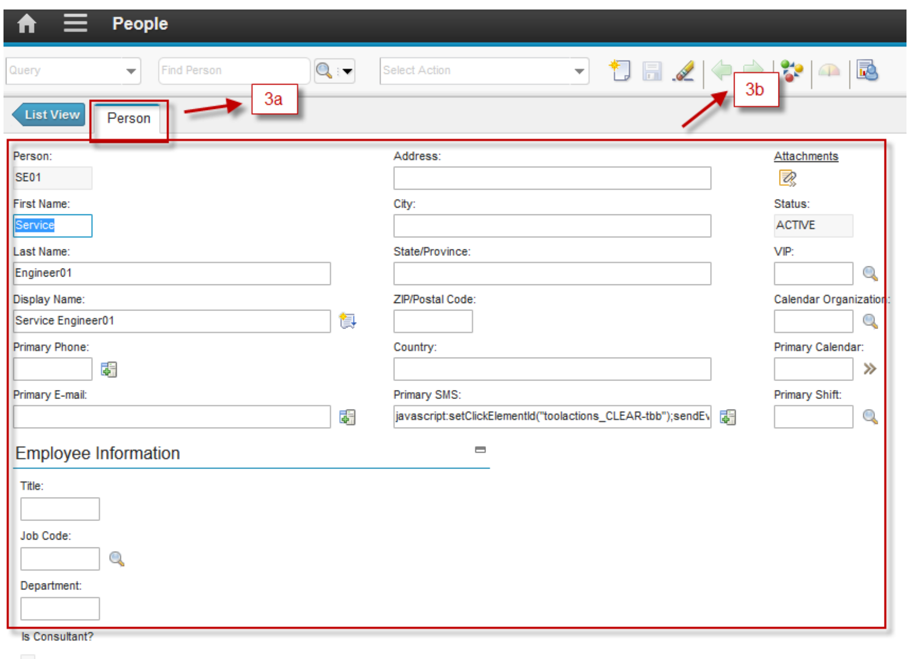
<table>
<tr>
<td>2</td>
<td>



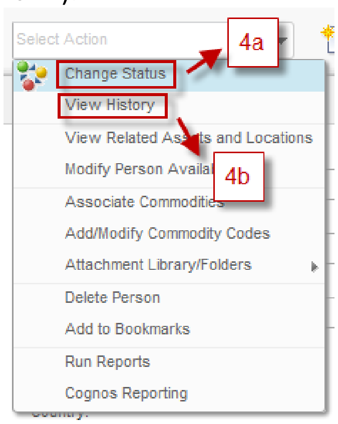
Header Bar will give information that user is in the **People** application.
2.a **New Person** button, to add new person
2.b **Filter** person by **Person ID**, **Name**, **Title**, **Status,** etc.
2.c List of all persons based of filtering result.
</td>
</tr>
<tr>
<td>3</td>
<td>
Choose one person, to show the details of the person



Will show display like below picture.



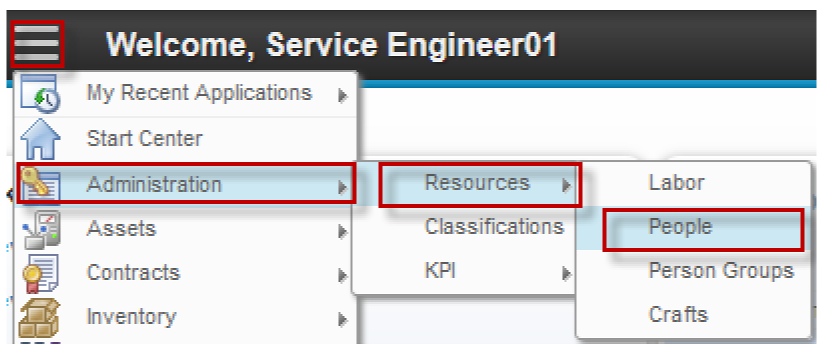
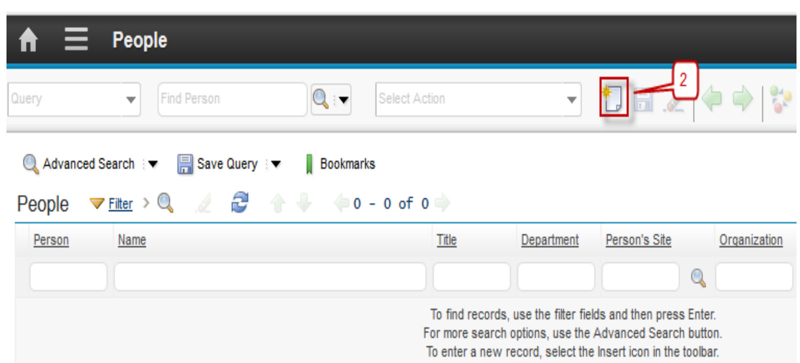
3.a **Person** tab, to view and update details of person
3.b Detail information of the person, such as first name, last name, display name, primary phone, primary email, etc.
</td>
</tr>
</table>

<table>
<tr><td>4</td><td>User can perform some action related to the person from Select Action menu (on top of the form).


4.a Change status action to change status this person
4.b View History, to view the change status history of the person.</td></tr>
</table>

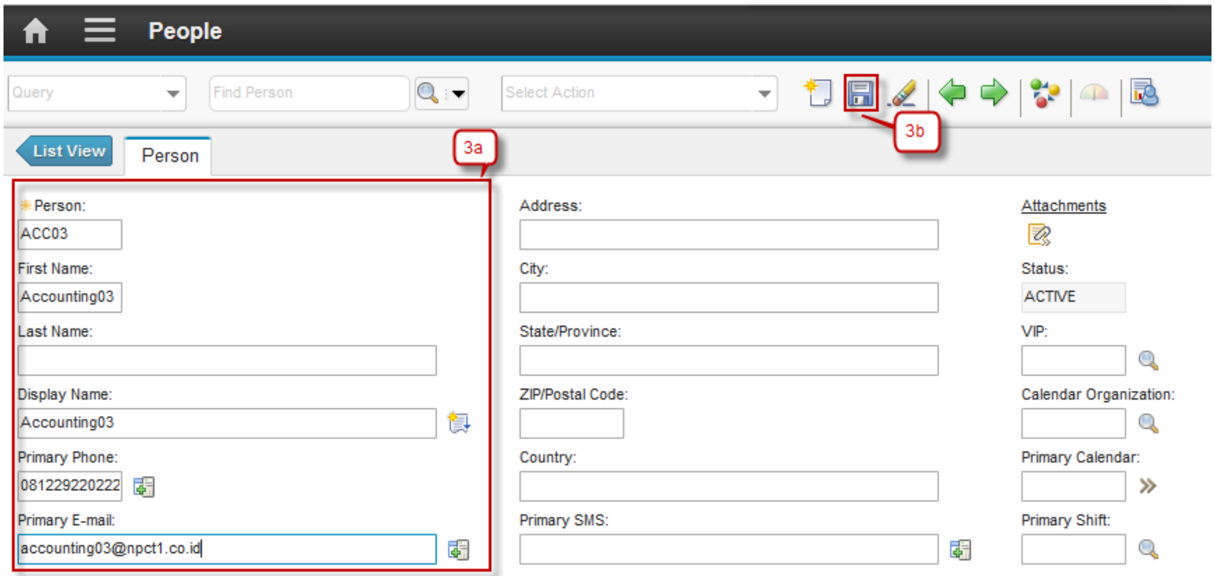
1.2.1 Create Persons

Person creation data can be created by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create new person:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Go to People application by click ☰ (GoTo) – Administration – Resources – People.</td><td>
</td></tr>
<tr><td>2. Click New Person.</td><td>
</td></tr>
</table>

3a. Input the Person Code and any additional information about Person.

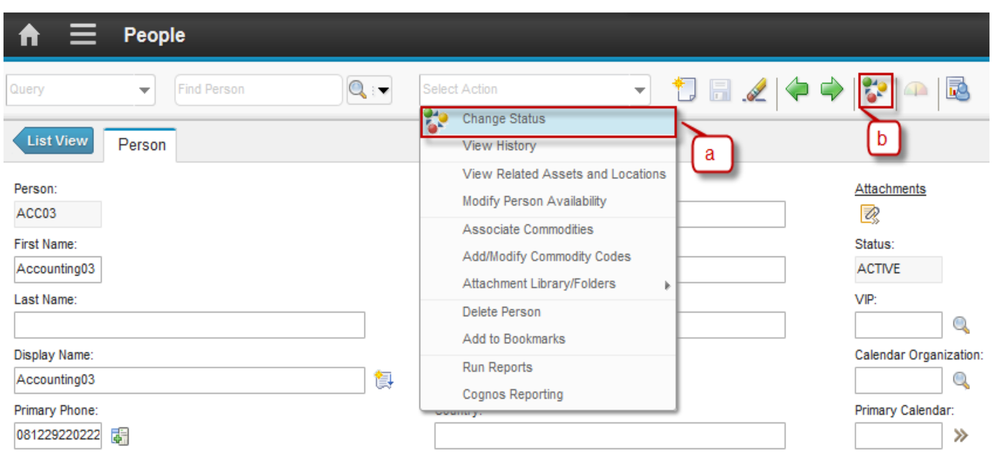
3b. Click **Save.**



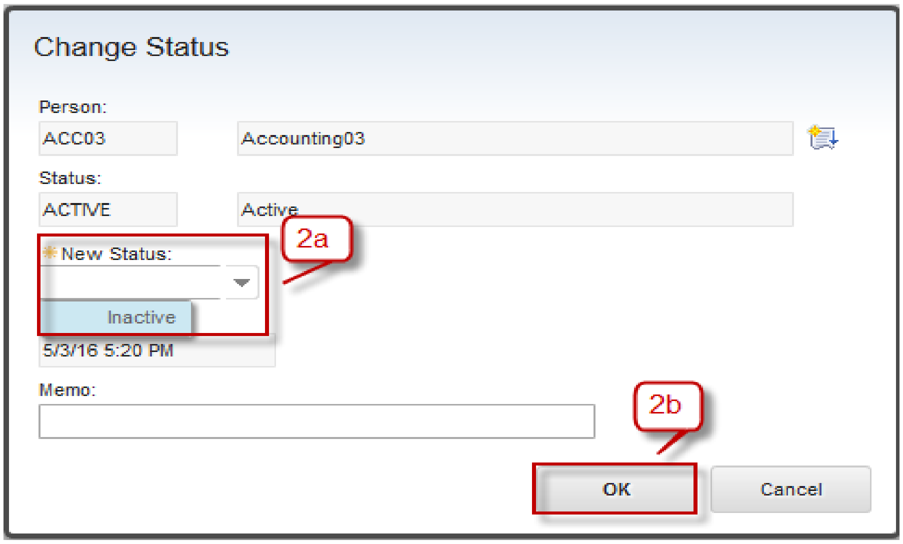
1.2.2 Change Status of Person

Person status change can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Change Status of a person data:

Step	Action
1. Change Status of Person have two ways: a. From **Select Action → Change Status** b. Click button 	

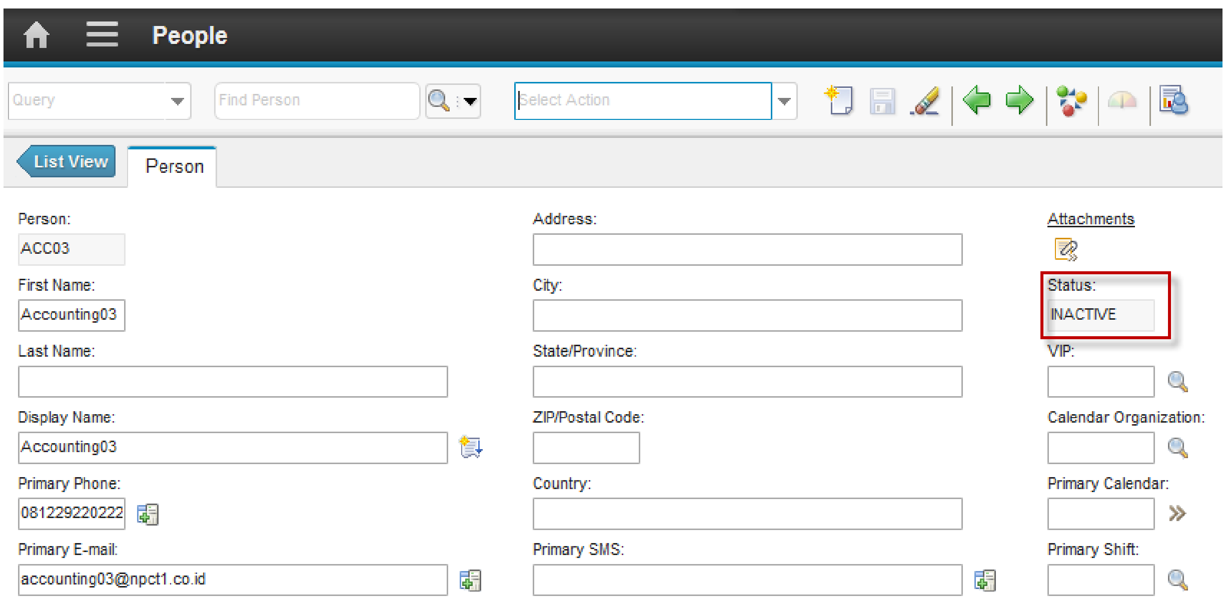


2a. Dialog box **Change Status** will be appeared, see the figure below :



2b. Click button **OK**

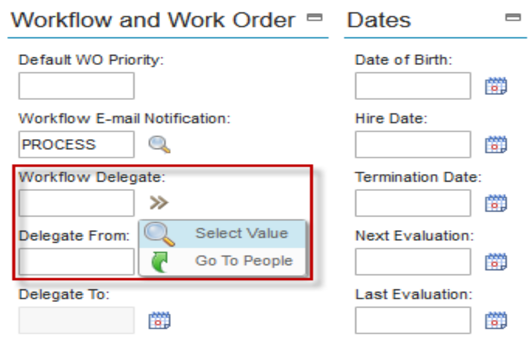
3. Status will be changed to **INACTIVE**



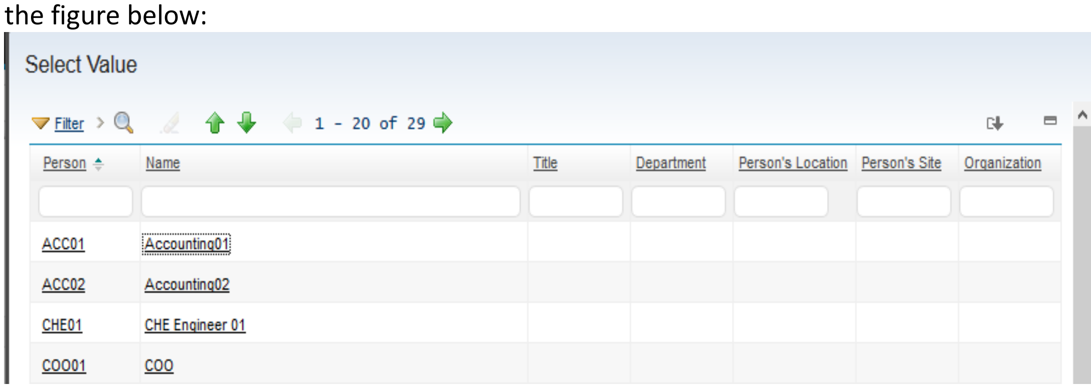
1.2.3 Set Person's Delegation

When a person is off, delegation can set toward those person, so any upcoming assignment can be redirected to the delegation. Person's delegation can be configured by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to set person's delegation:

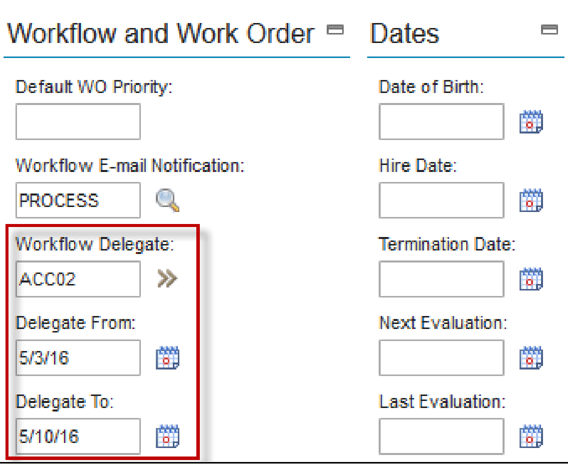
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Go to section Workflow and Work Order → click button » → Select Value or Go To People to select the person who will be delegated</td><td>



</td></tr>
<tr><td colspan="2">2. When click button Select Value will be appeared list of all person and select one of record, see the figure below:



</td></tr>
<tr><td colspan="2">3. Input field Delegate From and field Delegate To using button

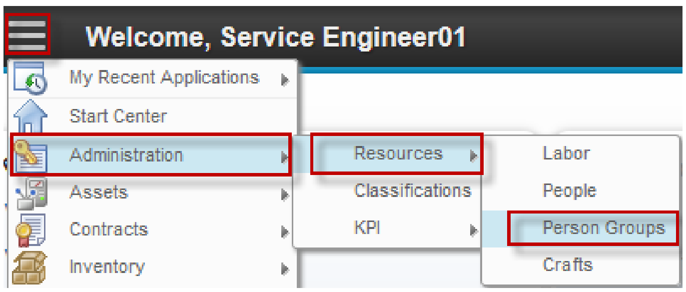


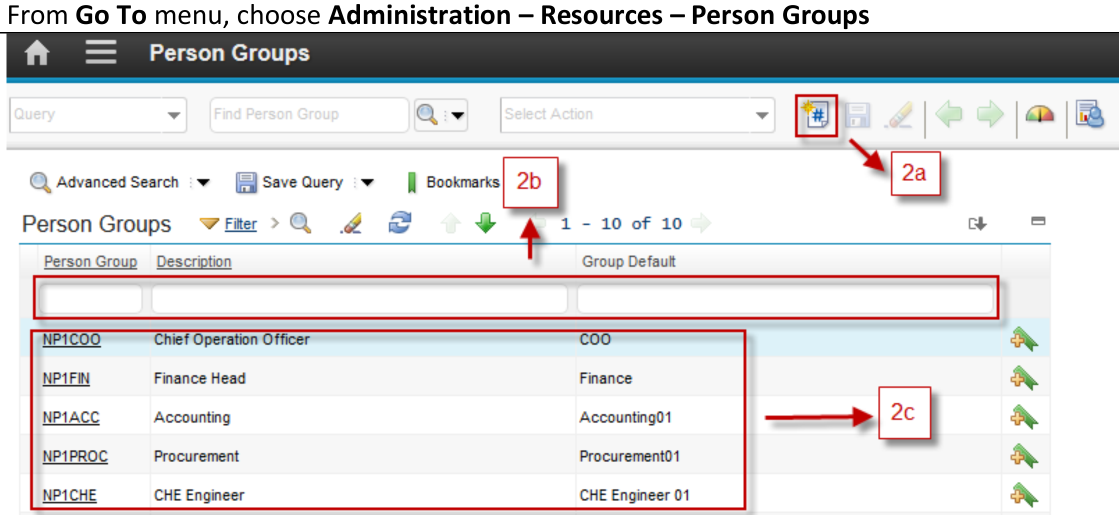
</td></tr>
</table>

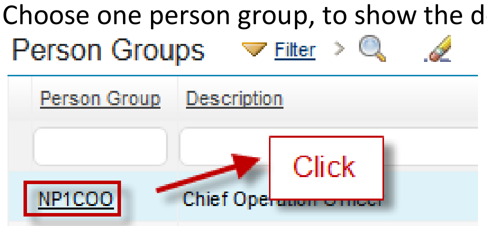
1.3 Person Groups Application

Person Groups application is used to manage the groups of person within organization. A person group can be the recipient of a document that is routed by a workflow process. If a document is routed to a person group, everyone in the group receives the document, unless it is configured to send the document only to the person whose calendar indicates availability.

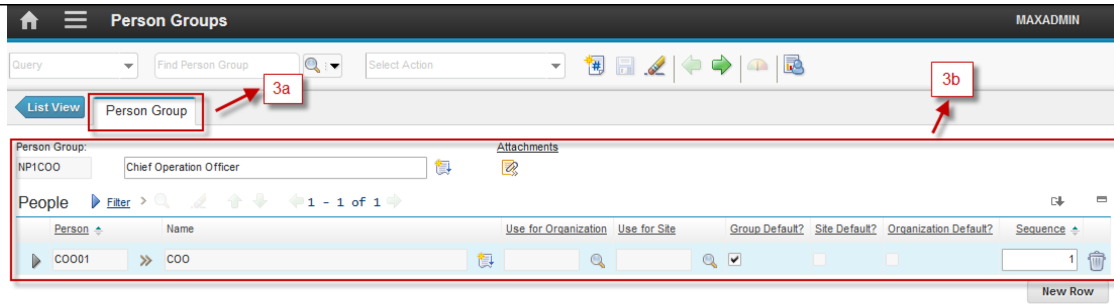
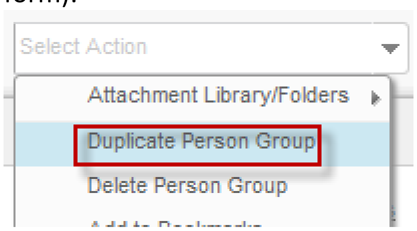
Followings are details about **Person Groups** applications accessibility:

<table>
<tr><td>1</td><td>


From Go To menu, choose Administration – Resources – Person Groups
</td></tr>
<tr><td>2</td><td>


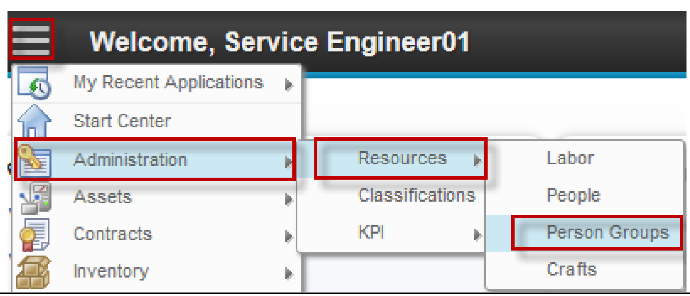
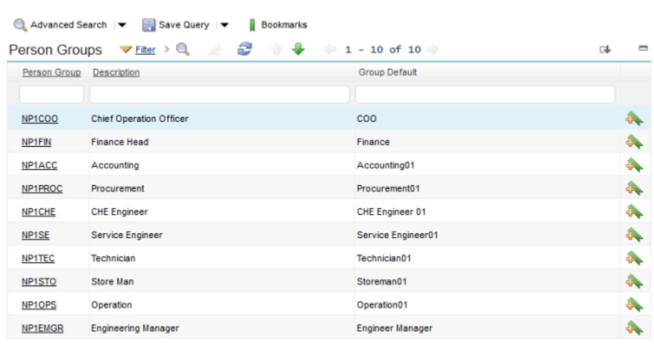
Header Bar will give information that user is in the Person Group application.
2.a New Person Group button, to add new person
2.b Filter person group by Person Group Code, Description, etc.
2.c List of all person group based of filtering result.
</td></tr>
<tr><td>3</td><td>
Choose one person group, to show the details of the person group


Will show display like below picture.
</td></tr>
</table>

	 3.a **Person Group** tab, to view and update details of person group 3.b Detail information of the person group and all person listed in the person group.
4	User can perform some action related to the person from Select Action menu (on top of the form).  **Duplicate Person group** action to create new person group based on opened person group.

1.3.1 Add/Remove Person in a Person Group

Person group's member modification can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to add/remove person groups:

Step	Action
1. Go to Person Groups application by click (GoTo) – **Administration – Resources – Person Groups.**	
2. Select one of Person Group to be added or removed	

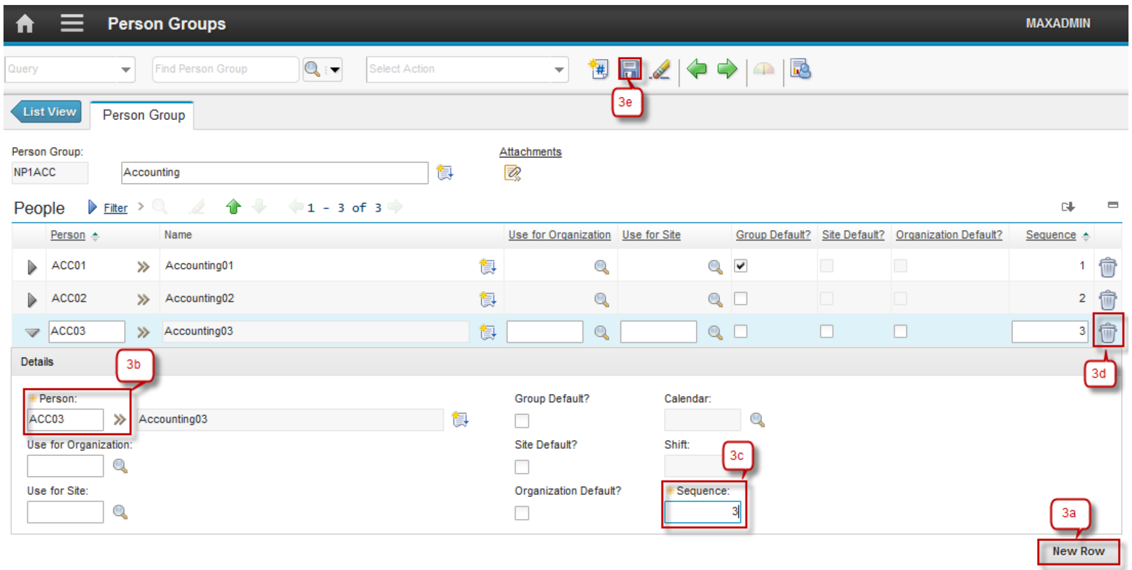
3a. Click **New Row** to add person as member of this person group.

3b. Select the person as member of this person group.

3c. Input the sequence.

3d. Click button to remove a member of this person group

3e. Click **Save.**



1.4 Crafts Application

Crafts application is used to define craft records for a work plan, and to define skill levels, standard rates. The craft code reflects the type of work that employees and contractors perform.

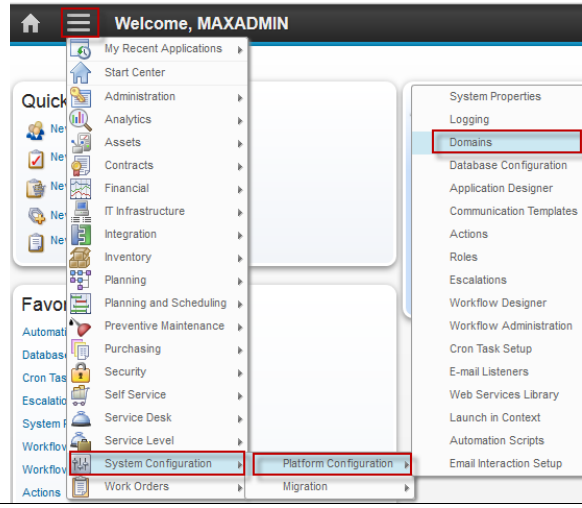
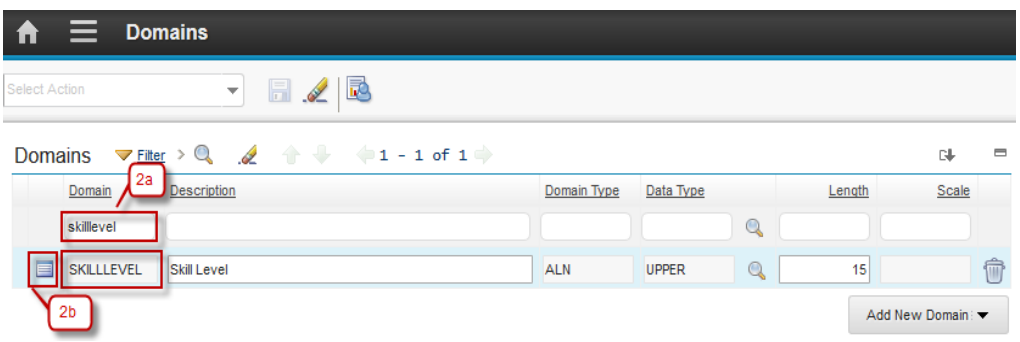
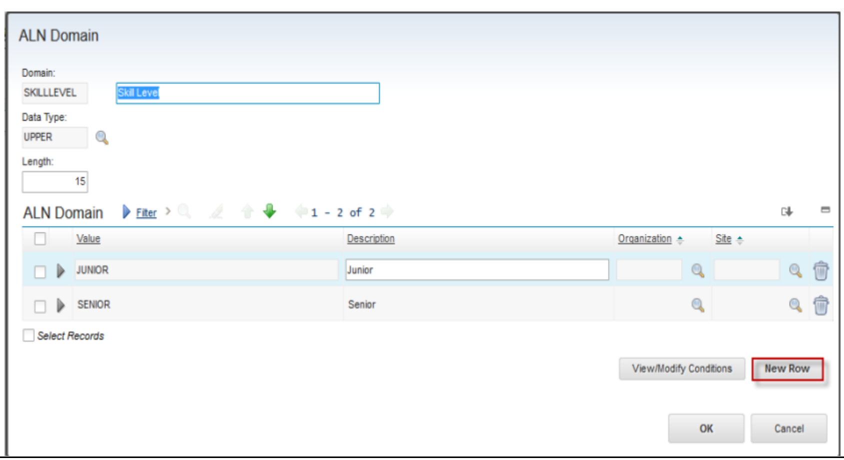
Besides, in **Crafts** application user also can:

- Associate labor records with crafts and skill levels
- Specify multiple skill levels for each craft

To distinguish between a junior level mechanic and a senior level mechanic, it is not required to create two separate craft records. Only one mechanic craft record is needed and set up junior and senior skill levels within the mechanic craft (skill level 1 is highest, and 5 is lowest). Each skill level within the mechanic craft can have different standard rates.

1.4.1 Define Skill Level

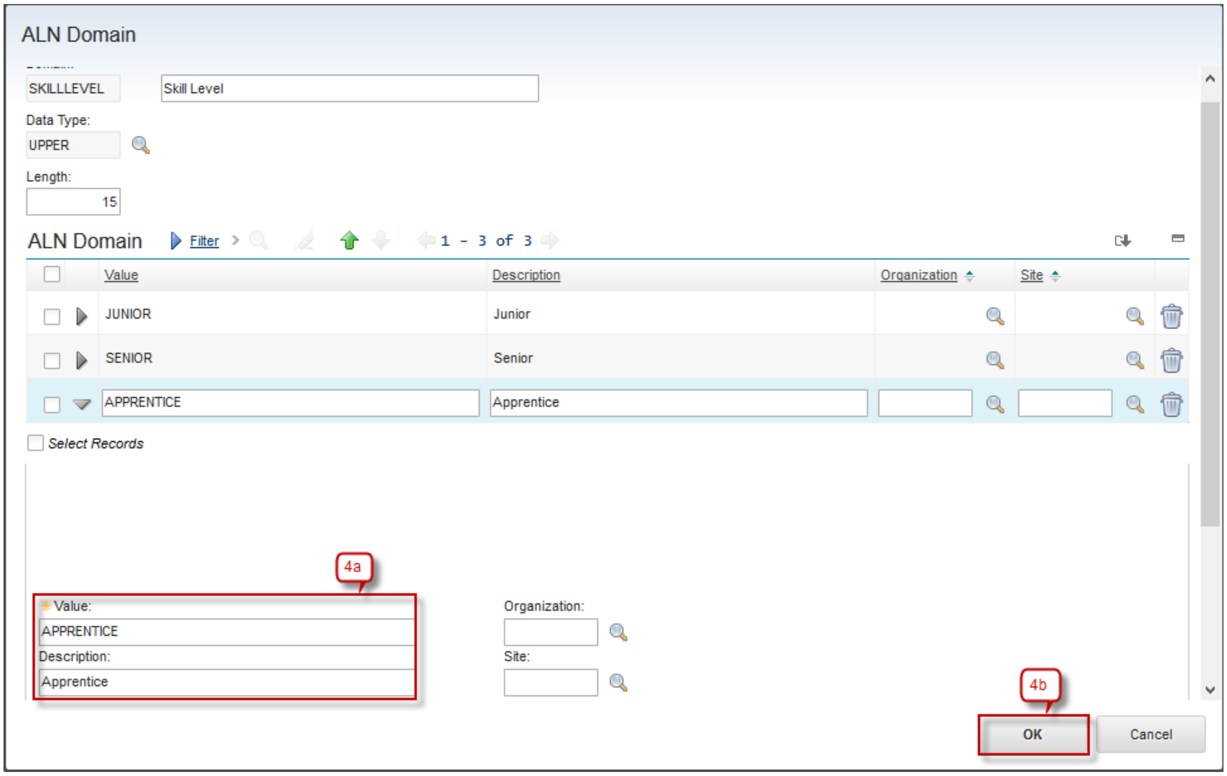
Skill level definition can only be done by *Administrator*. Below are step by step to define Skill Level:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Go to Domains application by click (GoTo) – System Configuration – Platform Configuration – Domains.</td><td>
</td></tr>
<tr><td colspan="2">2a. Search Domains name in field Domains = ‘skilllevel’
2b. Click button to add skill level.

</td></tr>
<tr><td>3. Click button New Row</td><td>
</td></tr>
</table>

4a. Input the value of Skill Level.

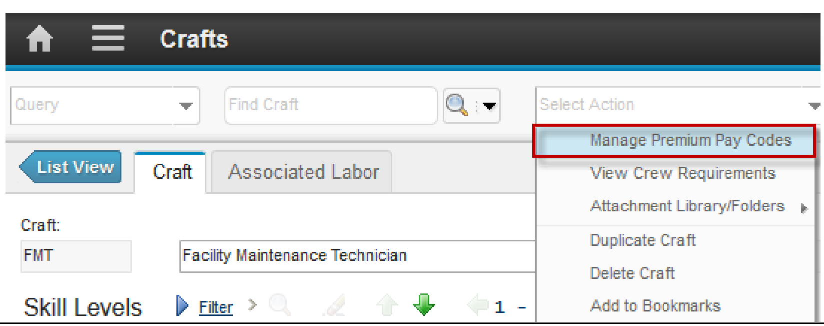
4b. Input the value description.

4c. Click button **OK.**



1.4.2 Define Premium Pay Codes

Managing premium pay codes can be done by *Service Engineer, CHE Engineer,* and *Engineering Manager*. Below are step by step to define Premium Pay Codes:

Step	Action
1. Click Craft Tab → **Select Action → Manage Premium Pay Codes**	

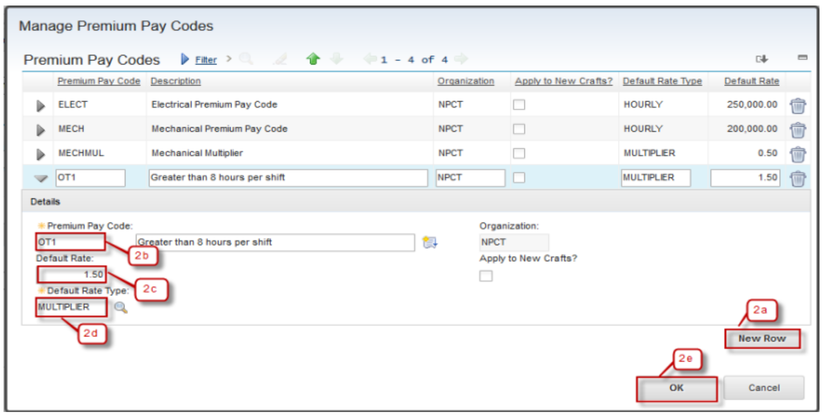
2a. Click **New Row.**

2b. Input the Premium Pay Code.

2c. Input Default Rate.

2d. Input Default Rate Type.

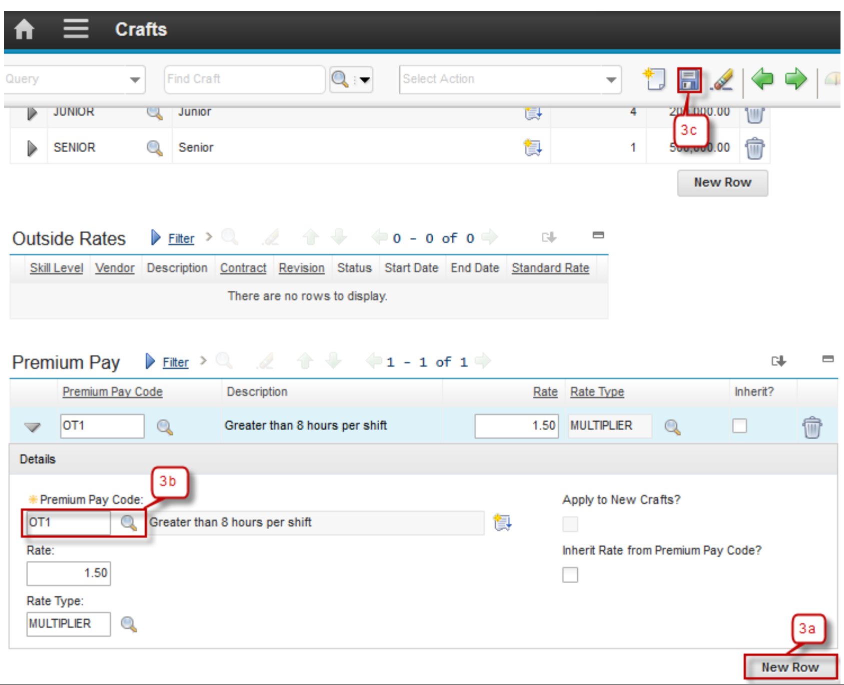
2e. Click **OK**.



3a. Click **New Row** to add Premium Pay in this Craft.

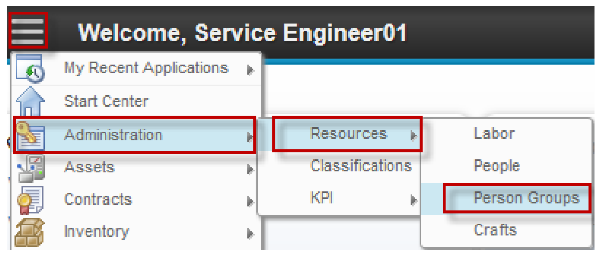
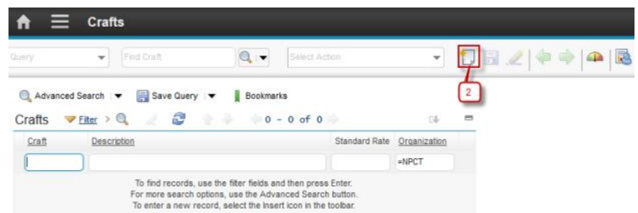
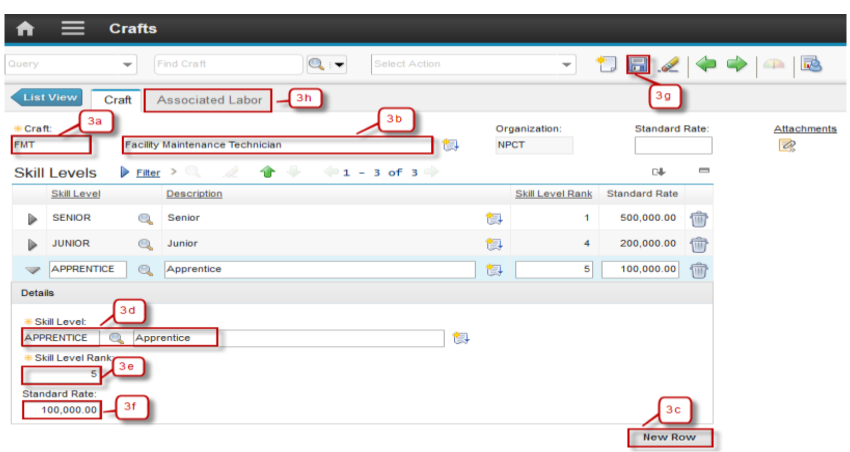
3b. Select Premium Pay Code using button 

3c. Click **Save**



1.4.3 Create Crafts

Craft creation can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Crafts:

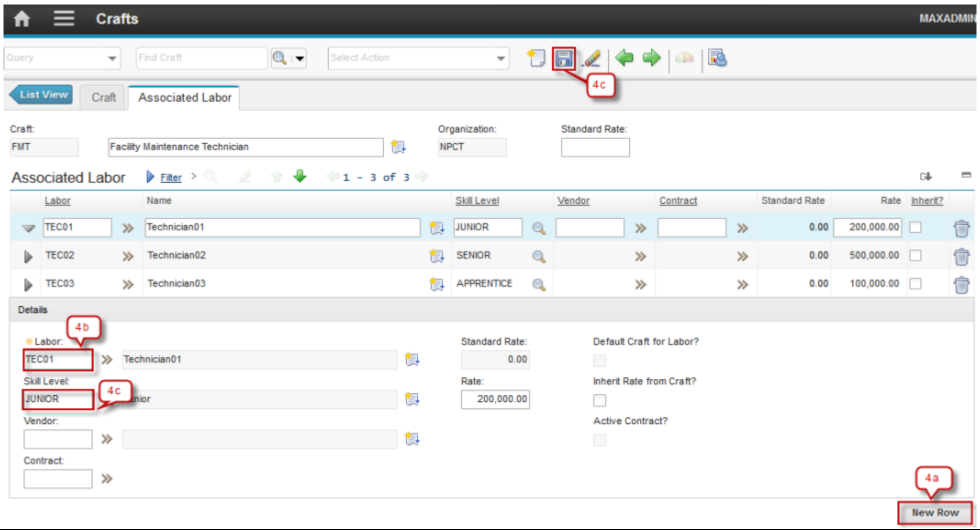
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Go to Crafts application by click (GoTo) – Administration – Resources – Crafts.</td><td>
</td></tr>
<tr><td>2. Click New Craft .</td><td>
</td></tr>
<tr><td colspan="2">3a. Input the Craft name.
3b. Input the Craft description.
3c. Click New Row to add Skill Level of this craft.
3d. Input the Skill Level name and the description.
3e. Input the Skill Level Rank (1 is highest, 5 is lowest).
3f. Input Standard Rate for that skill level of that craft.
3g. Click Save.
3h. Click Associated Labor tab to input labor that has this craft skill.

</td></tr>
</table>

4a. Click **New Row** to associate labor with this craft skill.

4b. Input the labor name that has this craft skill.

4c. Input the skill level of this craft that owned by the labor.

4d. Click **Save.**

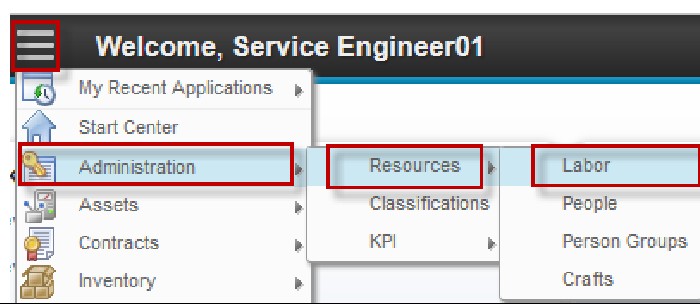


1.5 Labor Application

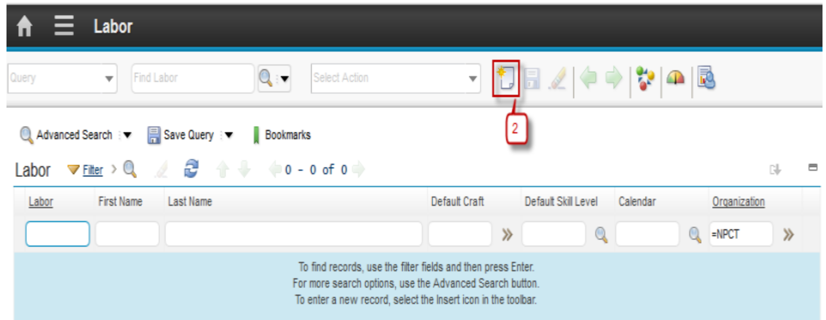
Labor application is used to create, change, view, and delete labor records for employees and contractors. Labor records can contain personal and work-related information, as well as the type and location of work. In addition, crafts, skill levels, qualifications, and certifications can be specified that are associated with labor records. Before labor record can be created, a person record corresponding to the labor must be created first.

1.5.1 Create Labors

Labor data creation can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Labor:

Step	Action
1. Go to Labor application by click ☰ **(GoTo) – Administration – Resources – Labor.**	

2. Click **New Labor.**



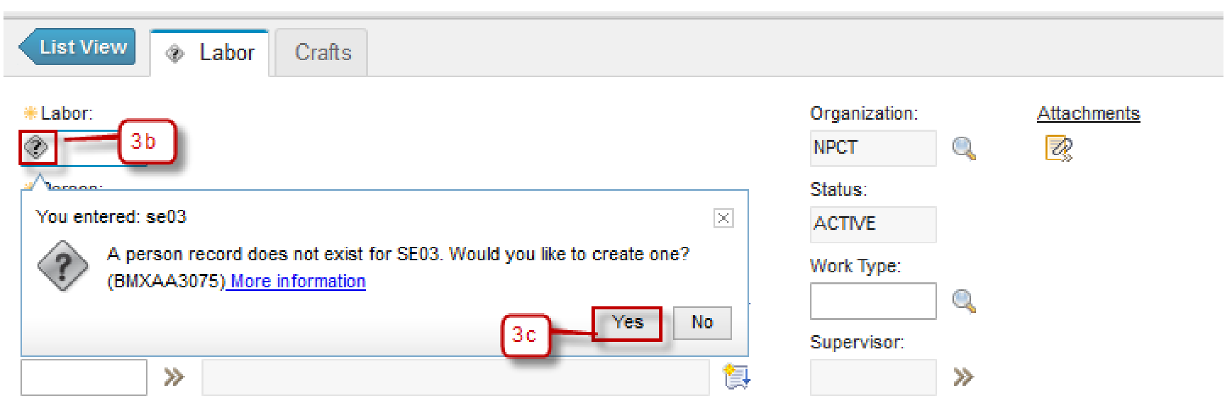
3a. Input the Labor Code.

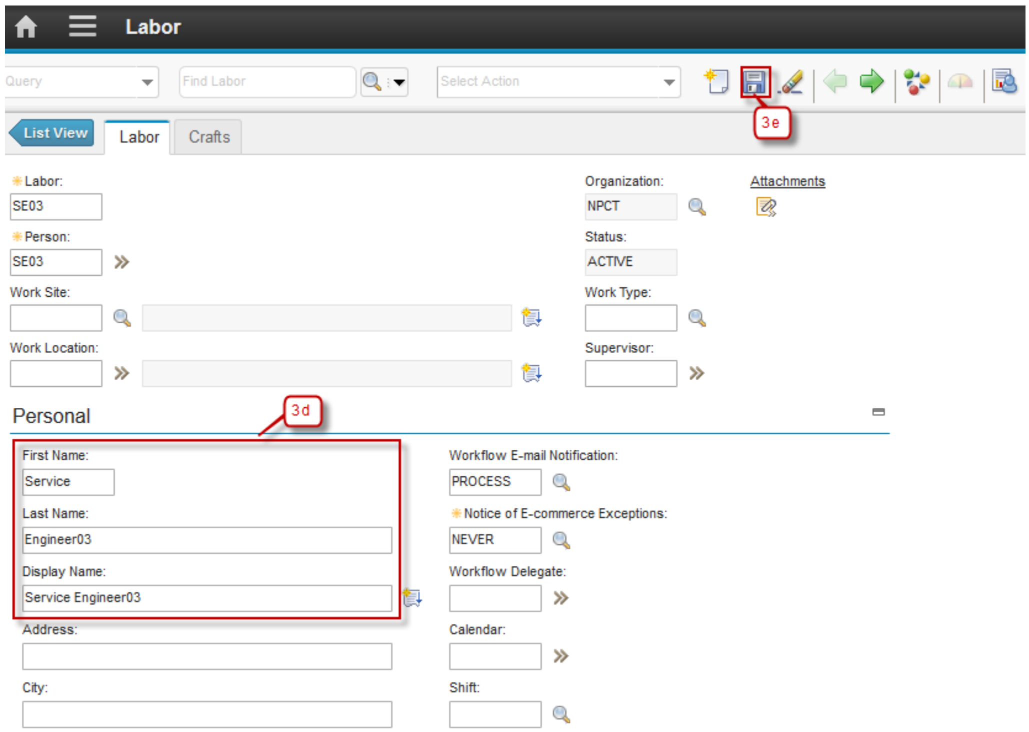
3b. If the labor code that inputted above doesn't have a person data yet, the question mark icon will appear, Click the question mark.

3c. Click **Yes** to create the person data based on the labor code that inputted above

3d. Input any additional information about Labor

3e. Click **Save.**



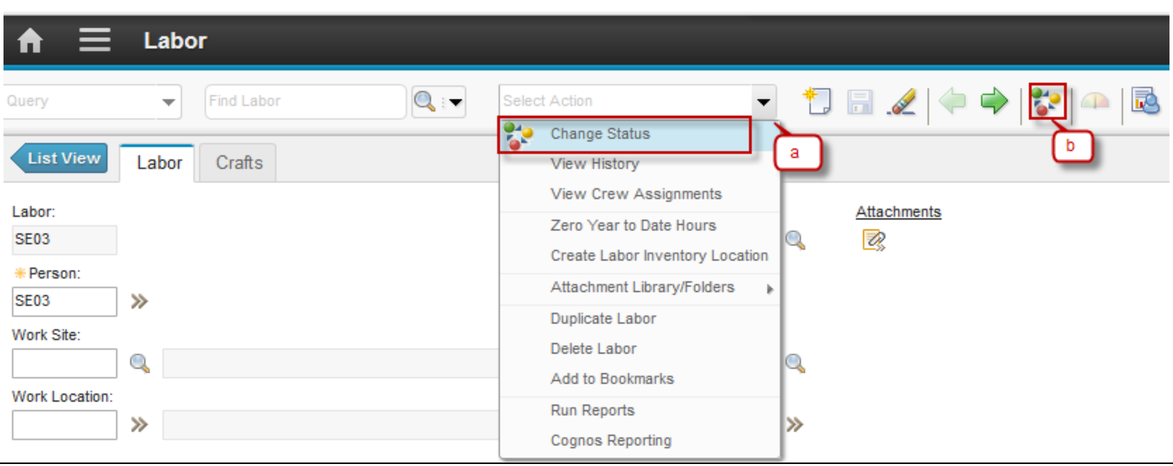


1.5.2 Change Status of Labor

Labor status change can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Change Status of Person

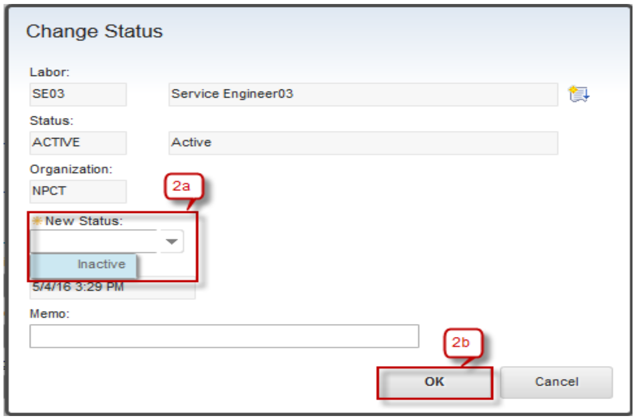
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td colspan="2">

1. Change Status of Person have two ways:
   a. From **Select Action → Change Status**
   b. Click button 



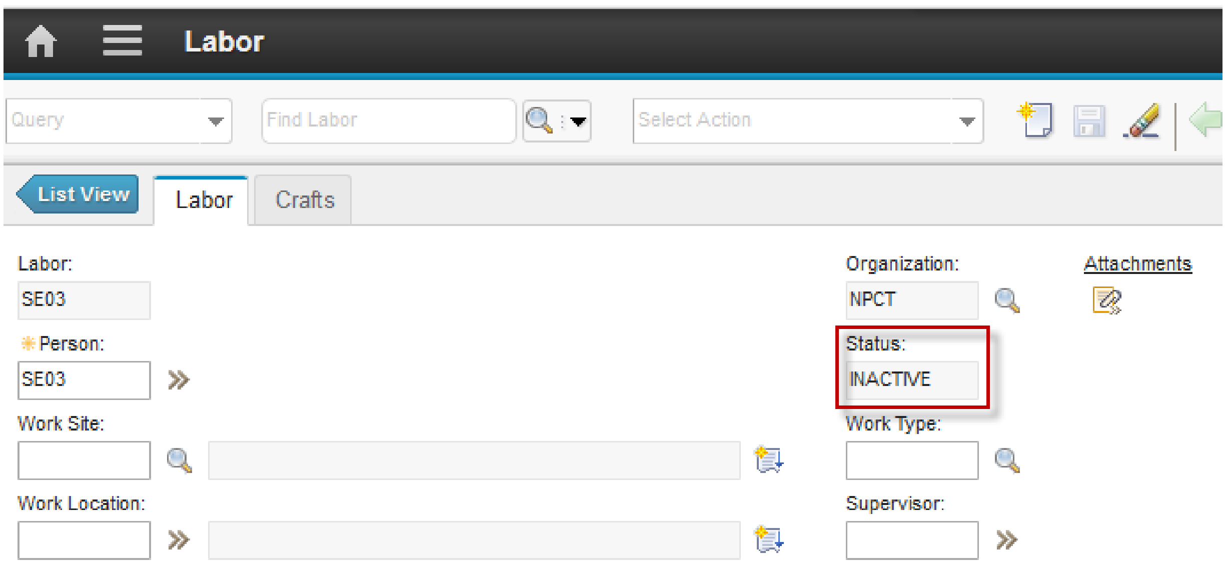
</td></tr>
</table>

2a. Dialog box **Change Status** will be appeared, see the figure below :



2b. Click button **OK**

4. Status will be changed to **INACTIVE**



1.6 Exercise Phase

2 Asset & Location

2.1 Overview

Asset is an item of economic value that a company own, benefit from, or has use of, in generating income. In EMS, an item is treated as asset when user wants to track its moving history, and maintenance work can occurs toward it.

Asset can be either rotating or non-rotating. Rotating assets are assets that are interchangeable, such as motors, pumps, or fire extinguishers. In EMS, remarkable difference between rotating asset and non-rotating asset is that rotating assets have both an asset number and an inventory item number, while non-rotating just have an asset number, without an inventory item number. The item number lets user to track assets as a group as they are moved in and out of inventory and other types of locations, while the asset number is useful to track individual instances of asset as it is moved from one location to another and from one site to another.

Here is example to help understanding inventory item number and asset. A company might have four identical (same make, same model) centrifugal pumps, so all four pumps would have the same item number. To track the use, repair, and locations of each individual pump, each pump would have its own, unique asset number.

Location is generally defined as a place where assets are operated, stored, and repaired. Generally locations are defined as a means of tracking assets. Another consideration, a maintenance work can be done toward **Locations**.

2.2 Classification Application

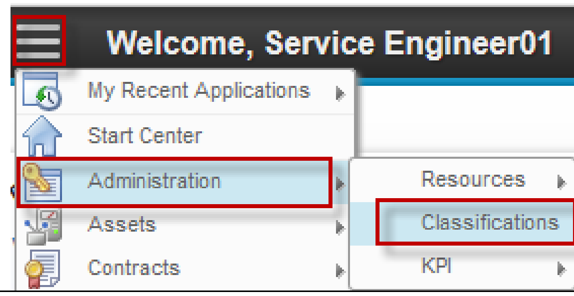
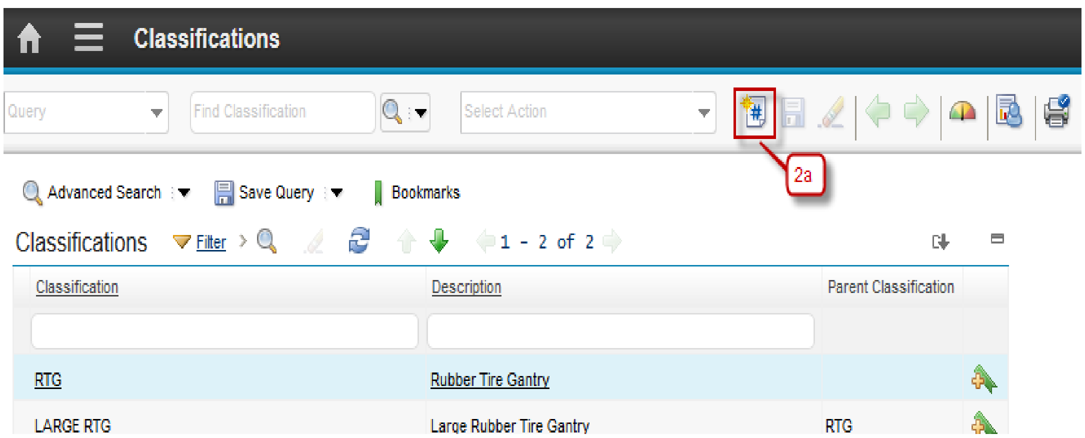
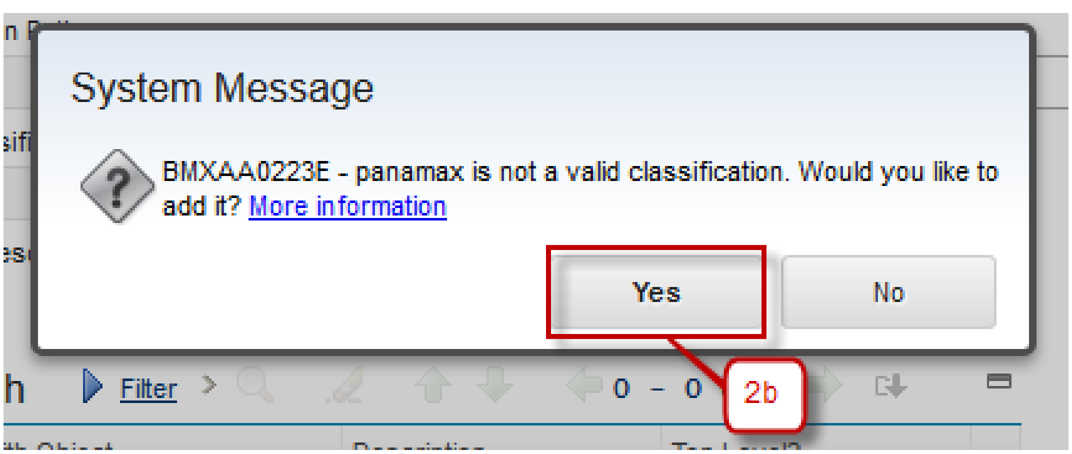
The Specifications tab on asset and location records can be used to add Classification information to asset and location records. Specification is available on item data too. Using Classifications lets user to structure assets and locations in organized hierarchies so that:

- Records can be easily located via Classifications.
- Records are not duplicated unintentionally.
- Asset and item lists are consistent with vendor's lists.

Classifications use attributes to define the classification category. When you list a rotating item that has a classification on an asset or location record, Maximo copies the Classification and Attributes to **Specifications** tab of the asset or location record.

2.2.1 Create Classifications

Classification creation can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Classification:

Step	Action
1. Go to Classification application by click ☰ **(GoTo) – Administration – Classifications.**	
2a. Click **New Classification.**   2b. If the classification that inputted above doesn't valid, the message box to added the classification will appear. Click button **Yes** and then classification will be added.	

3a. Input the Classification name.

3b. Input the Classification description.

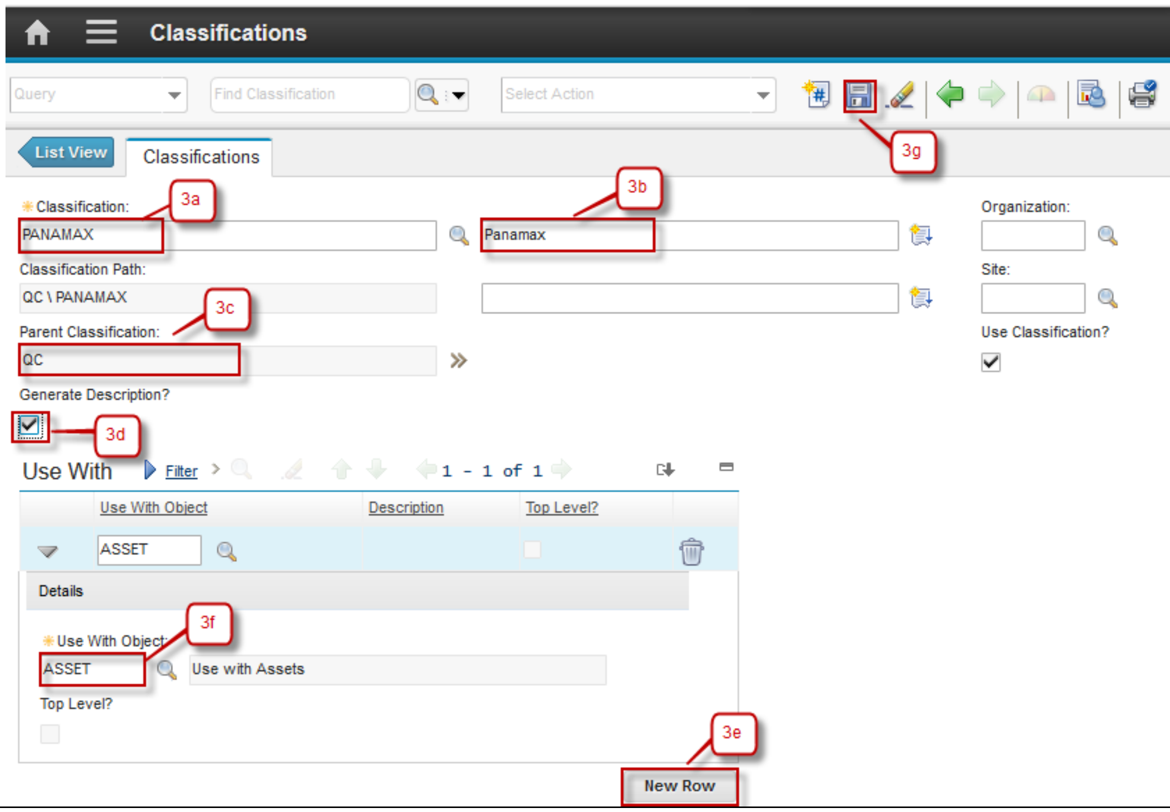
3c. Input the Parent Classificication if this classification has a parent.

3d. Default from **Generate Description** is checked, if description classification don't want generated when the classification in use make sure the check box is unchecked.

3e. Click **New Row** to add the object that will use this classification.

3f. Input the object that will use this classification.

3g. Click **Save.**



2.2.2 Define Attributes to a Classification

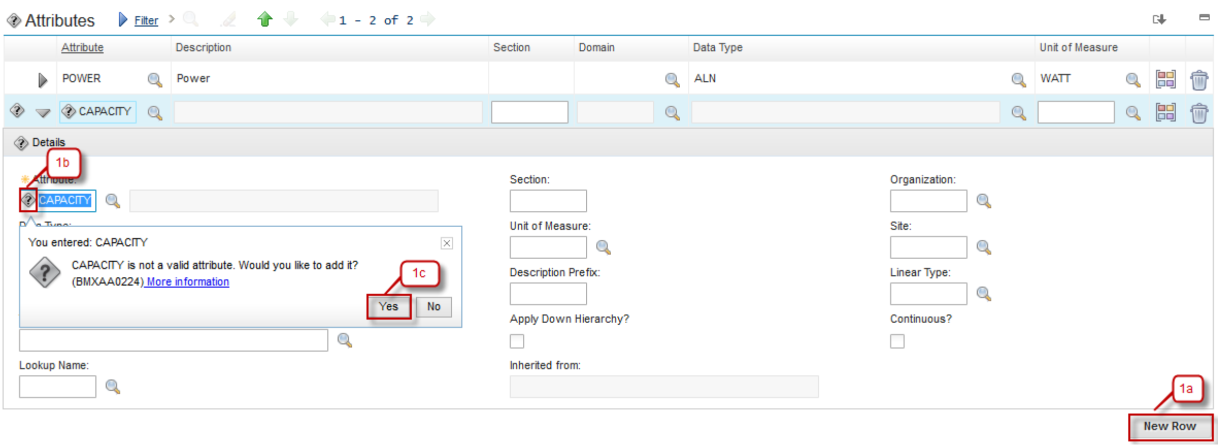
Defining attributes to a classification can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Define Attribute to a Classification:

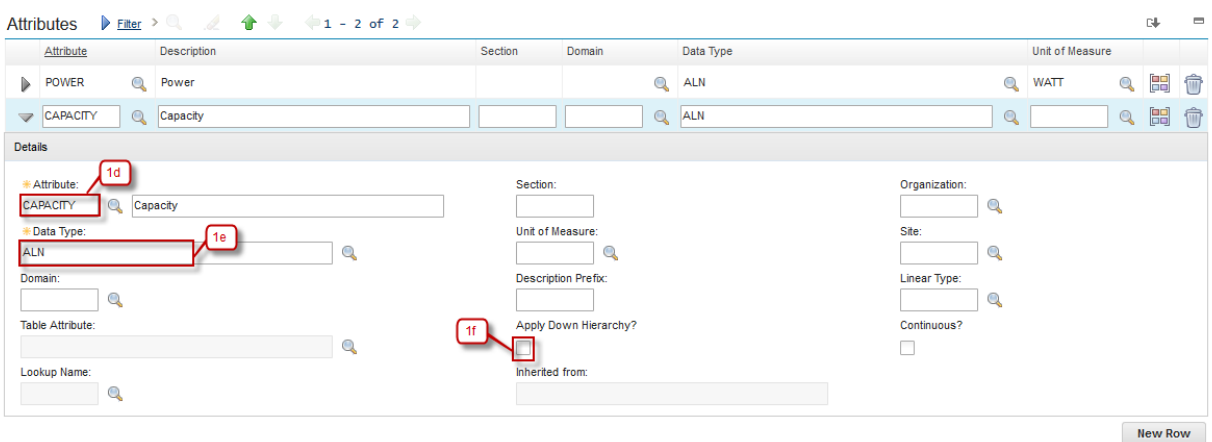
Step	Action
1a. Click **New Row** to add attribute on this classification.	
1b. If the attribute you added is a new attribute, question mark icon will appear.	
1c. Click **Yes** to add the attribute.	

1d. Input the Description of that attribute.

1e. Input the Data Type of that attribute.

1f. Check the **Apply Down Hierarchy** checkbox if you want this attribute will appear in any child classification of this classification and click **Save**.





2.3 Locations Application

Locations application is used to add, view, modify, and delete location records for assets, and organize these location into systems. Systems are used to maintain location's hierarchical. When location is organized into systems, user can quickly find a location using the drilldown and then identify the assets at a specific location.

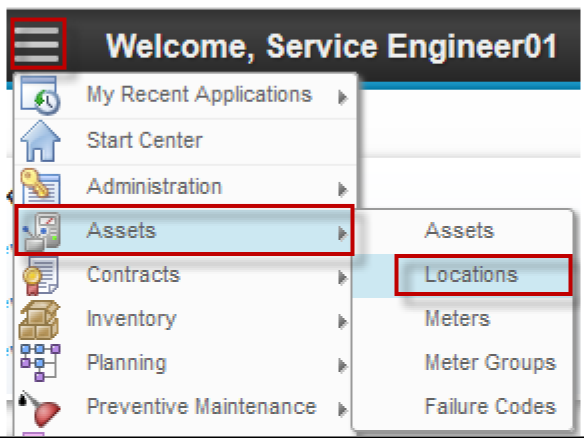
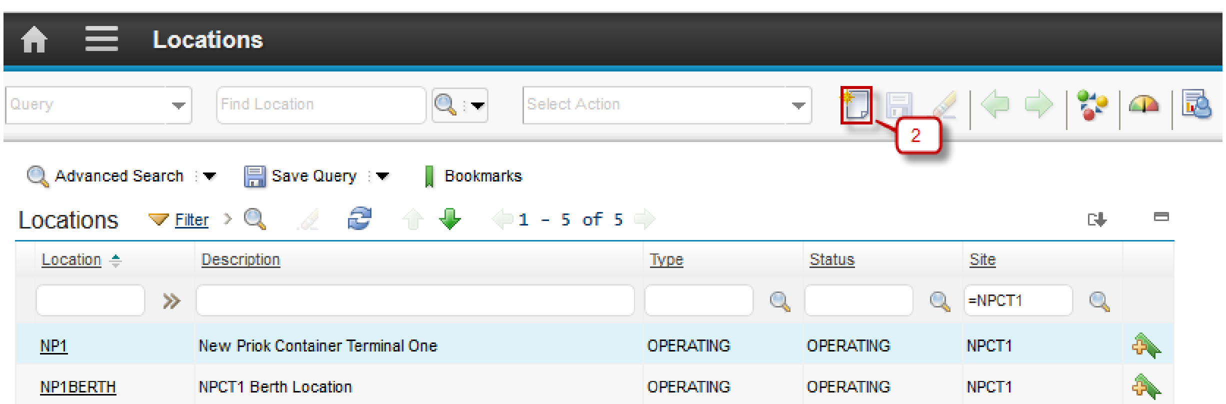
Locations are categorized into several types:

- Courier - This is an inventory location, used to indicate an individual who is responsible for inventory items while in transit from one inventory location to another.

- Holding - This is an inventory location, used to hold items received with a status of WINSP (waiting for inspection). One site can only have one holding location.
- Operating - This is an asset location, used to indicate a location where assets are located or equipment is operated.
- Repair - This is an asset location, used to indicate a location where assets are repaired or refurbished.
- Salvage - This is an asset location, used to indicate a location used to store retired or decommissioned assets.
- Vendor - This is an inventory location, used to indicate the source of items, material, and rotating assets, and may be used to indicate the location of an asset returned to the vendor for repair or refurbishment.

2.3.1 Create Locations

Location creation can be performed by *Service Engineer*. Below are step by step to Create Locations:

Step	Action
1. Go to Locations application by click (GoTo) – **Assets** – **Locations.**	
2. Click **New Locations .** 	

3a. Input the Locations name.

3b. Input the Locations description.

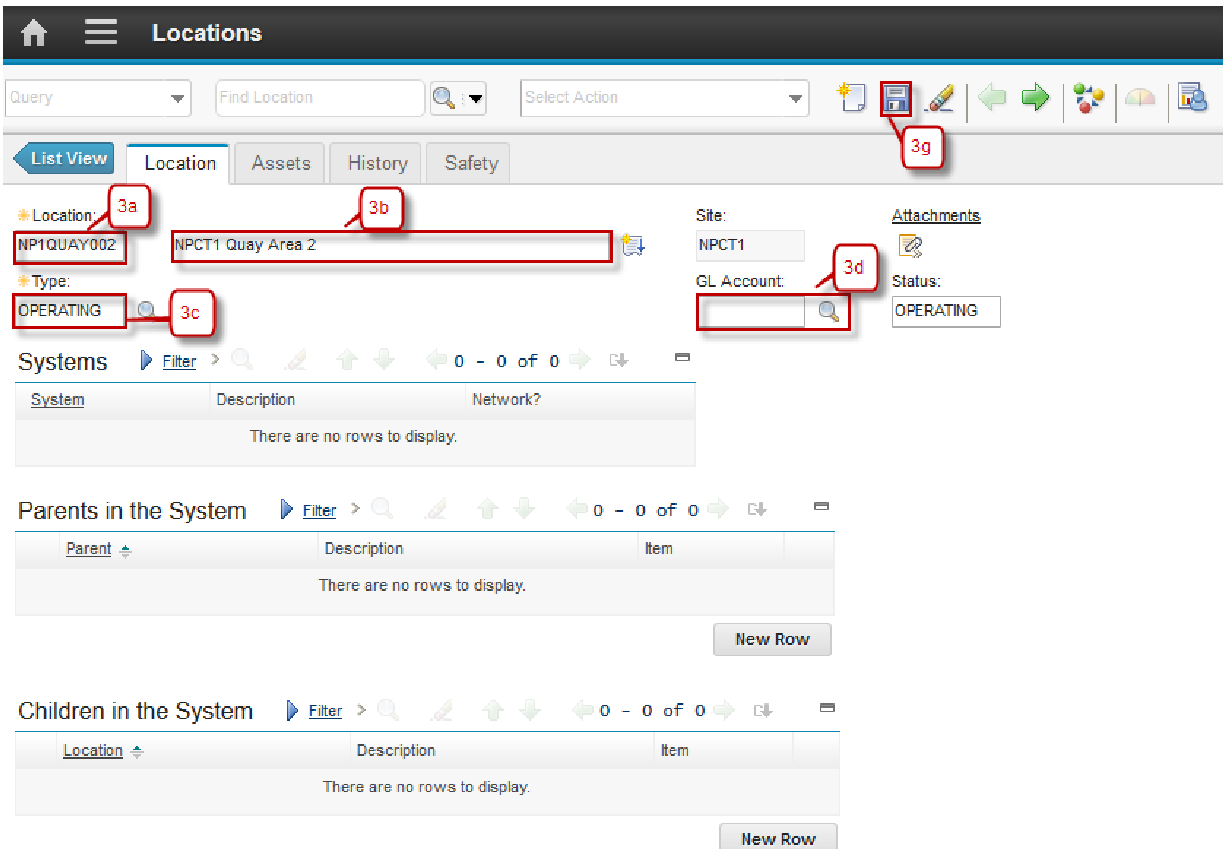
3c. Select the type of Locations, default type is ‘OPERATING’.

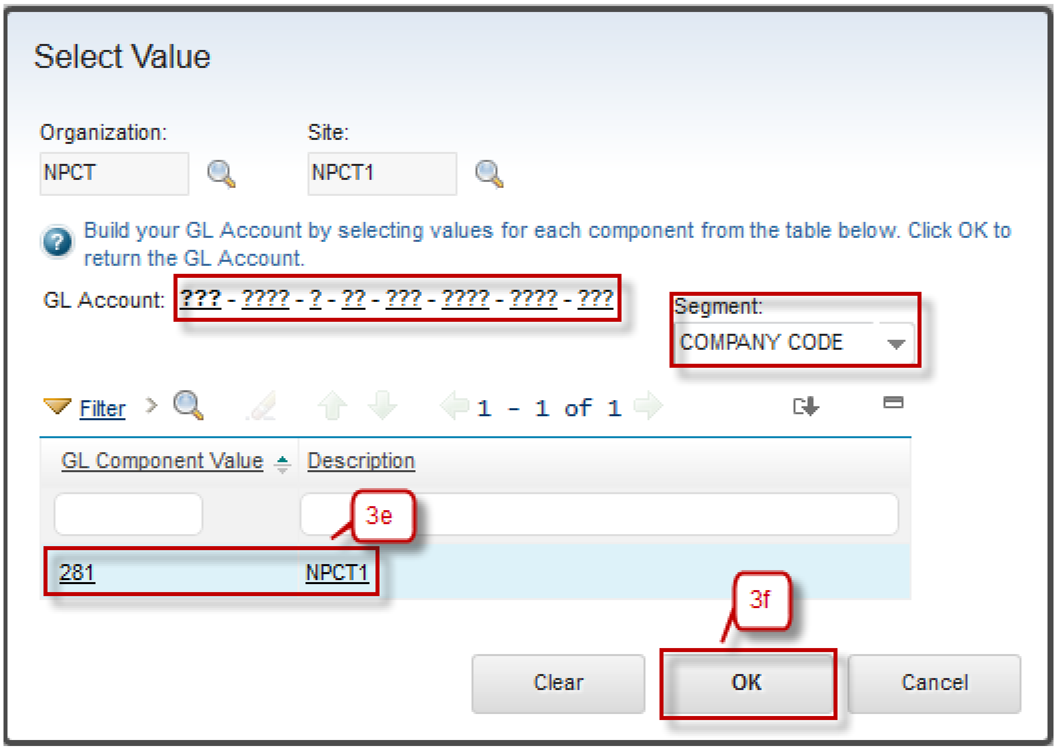
3d. Define GL Account of Locations, click button

3e. Select account value for every segment of GL Account.

3f. Click **OK**.

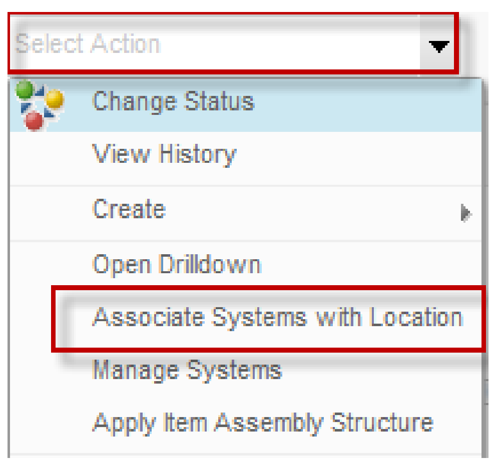
3g. Click **Save.**





2.3.2 Associate Systems with Locations

Below are step by step to Associate Systems with Locations by *Service Engineer*:

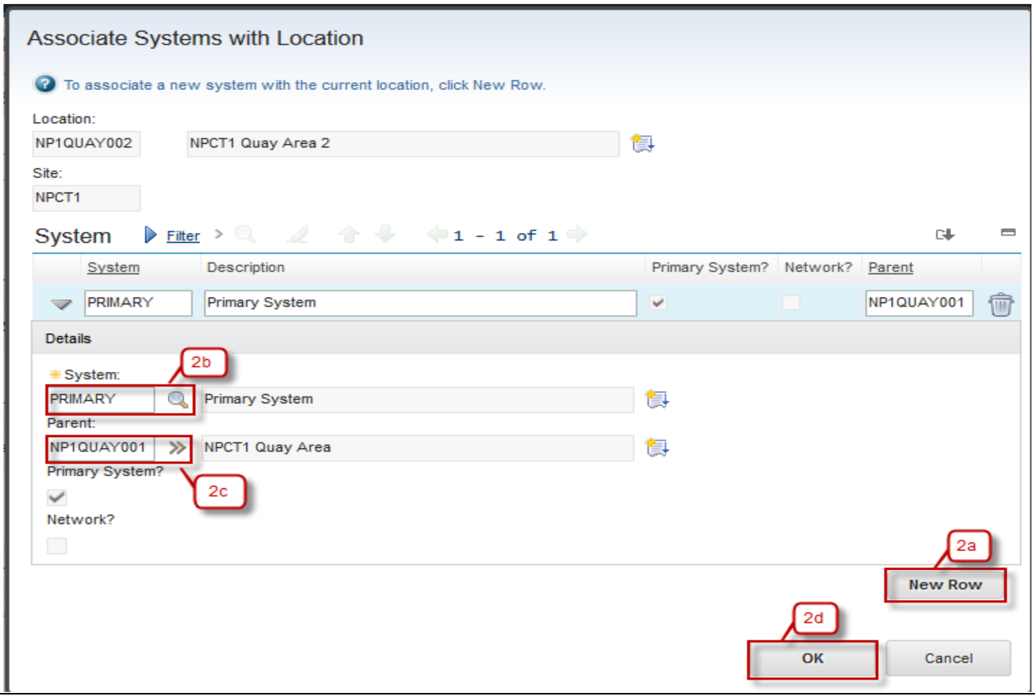
Step	Action
1. Go to **Select Action – Associate Systems with Location.**	

2a. Click **New Row.**

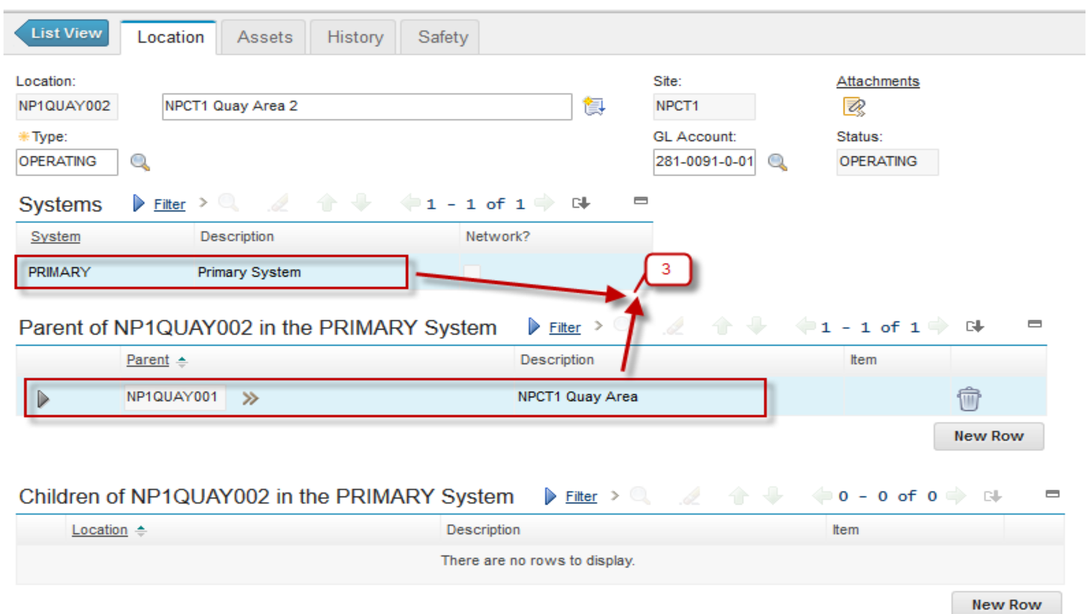
2b. Select System of Locations.

2c. Select Parent of Locations.

2d. Click button **OK.**



3. Field System and Parent will defined

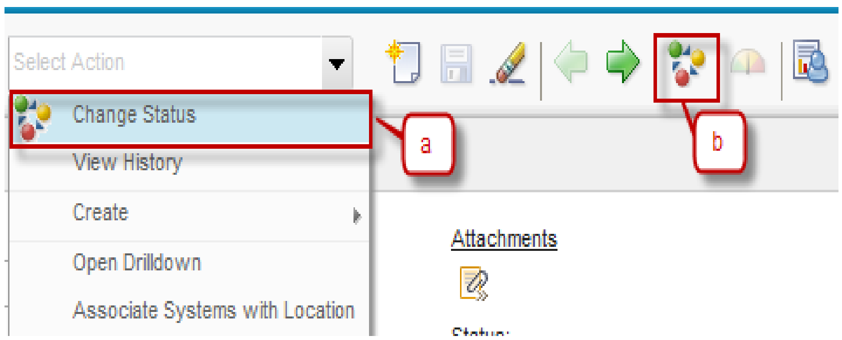


2.3.3 Change Status of a Location

Below are step by step to Change Status of a Location by *Service Engineer*:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td colspan="2">

1. Change Status of Person have two ways:
    a. From **Select Action → Change Status**
    b. Click button 

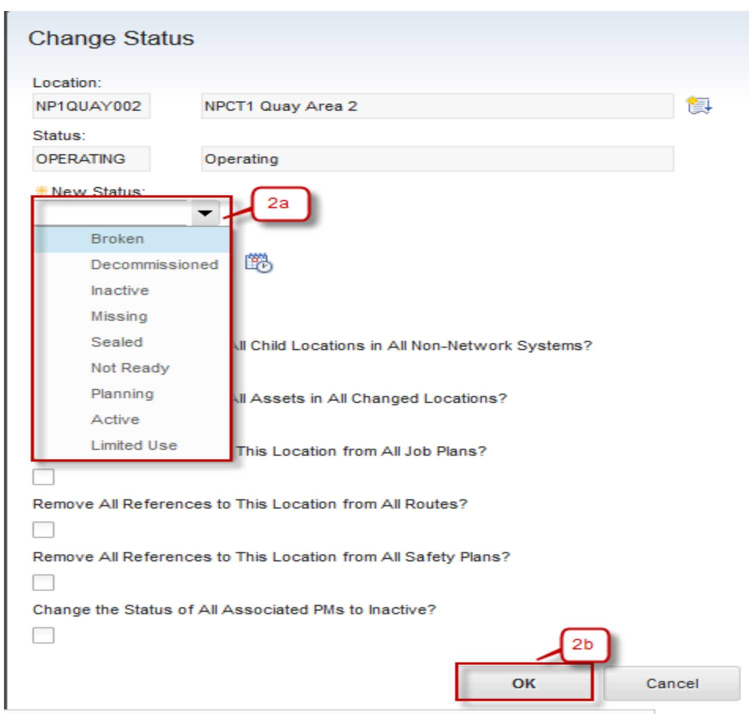


</td></tr>
<tr><td colspan="2">

2. Dialog box **Change Status** will be appeared, see the figure below :

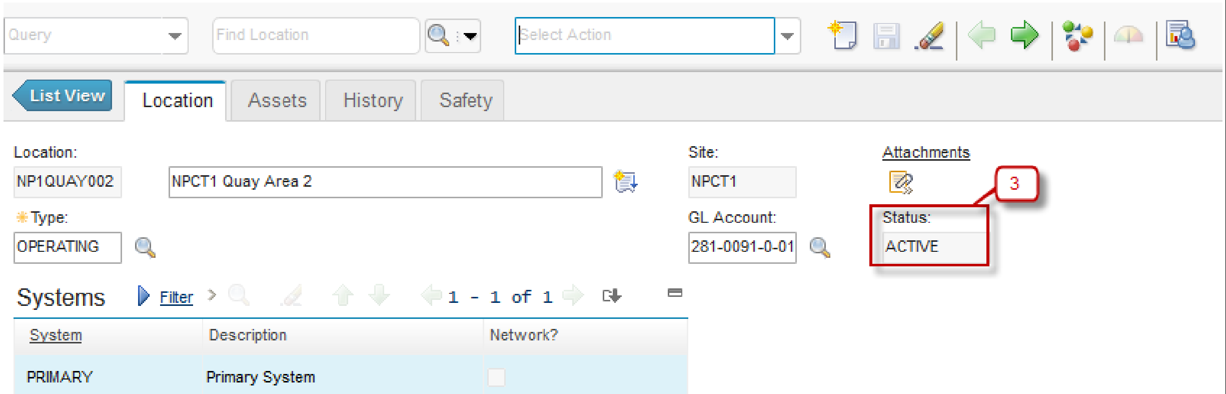
2a. Select New Status of Locations.

2b. Click **OK.**



</td></tr>
</table>

3. Status will be change to **ACTIVE**



2.4 Meters & Meter Groups Application

A meter record in EMS is virtual identifier used to record measurements. meter records is associated with assets and locations to record measurements, for example run hours, temperature, or pressure, that are used to track asset and location performance. **Meters** application is used to add, view, modify, and delete meter records.

Meters can have one of three types:

- Continuous - Meters where the readings increase continuously, for example odometers. These meters might be used to track miles, hours, engine starts, pieces produced, fuel consumed, and other continuous readings.
- Gauge - Meters where the readings may fluctuate, for example thermometers or pressure gauges. These meters might be used to track temperature, pressure, noise levels, oil levels, and other readings that fluctuate,
- Characteristic - Meters where an observed state is being tracked, for example a color change. These meters might be used to track Brick/Refractory Color (yellow, orange, white), or oil color (clear, turbid, dark).

Meter readings may be recorded for a variety of reasons, including:

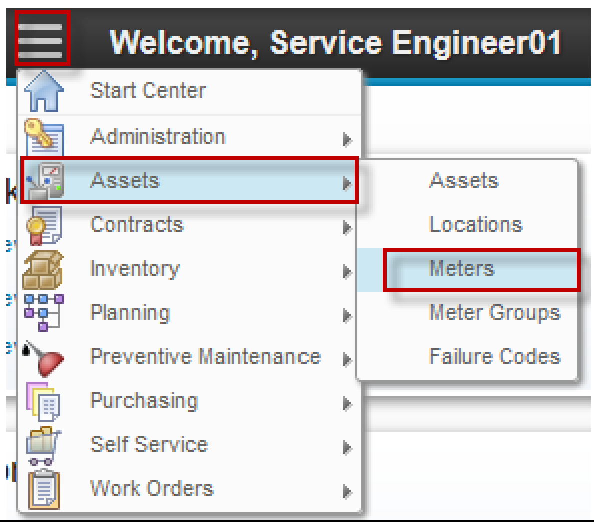
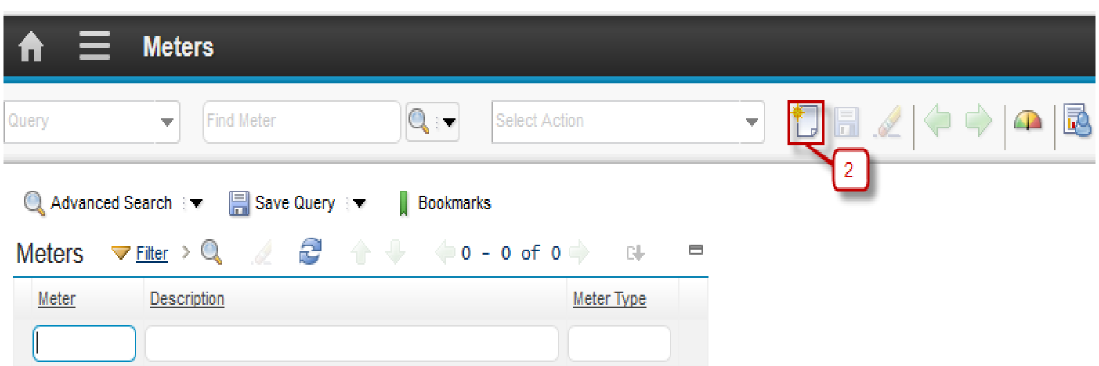
- determining the next time an asset is due for maintenance
- performing efficiency comparisons between similar asset performing similar work
- tracking production costs, for example tons per hour in the mining industry
- tracking service costs, for example odometer readings for a trucking fleet
- tracking efficiency indicators, for example miles per gallon
- conforming to regulatory requirements, for example maintaining cradle to grave details of time controlled components in the aviation industry

In **Meters** application, when user create a meter record, user can define the type of meter, the unit of measure used by the meter, and the type of reading (delta or actual) that the meter should record. Meter records can be associated with assets, inventory items, and locations, and can be used to drive condition monitoring and preventive maintenance. Meters can be applied individually, or as part of a meter group.

A meter group is two or more meters that are commonly used on the same types of assets or locations. **Meter Groups** application is used to create groups for meters that are commonly used together. For example all of the vehicles in a fleet use meters to track mileage, usage, fuel, and oil consumption. Rather than attach these four meters to a vehicle one at a time, user can define a meter group and use it to associate all four meters to the vehicle at the same time.

2.4.1 Create Meters

Below are step by step to Create Meters by *Service Engineer*:

Step	Action
1. Go to Meters application by click ☰ **(GoTo) – Assets – Meters.**	
2. Click **New Row.** 	

3a. Input the Meter name.

3b. Input the Meter description.

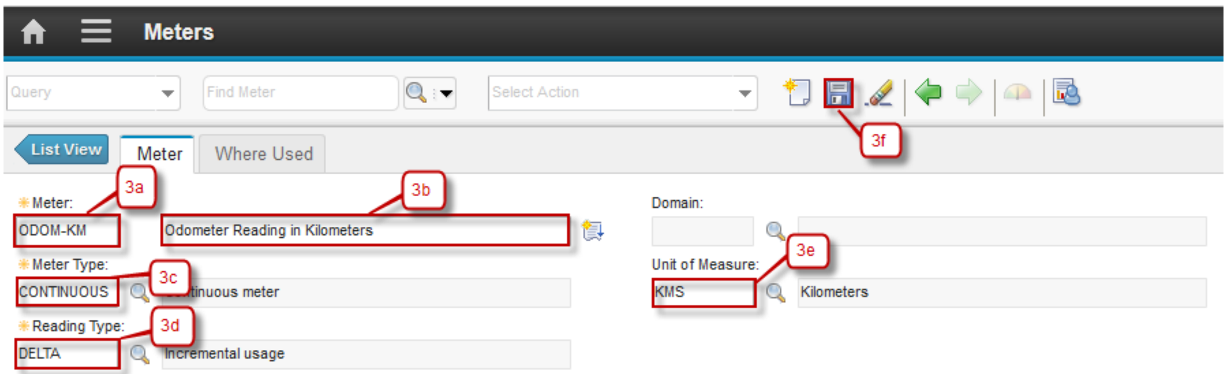
3c. Select the type of Meter.

3d. Input the Reading Type.

- If the type of Meter is ‘CONTINUOUS’, field Reading Type is mandatory.
- If the type of Meter is ‘CHARACTERISTIC’, field Domain is mandatory.

3e. Input the Unit of Measure

3f. Click **Save.**

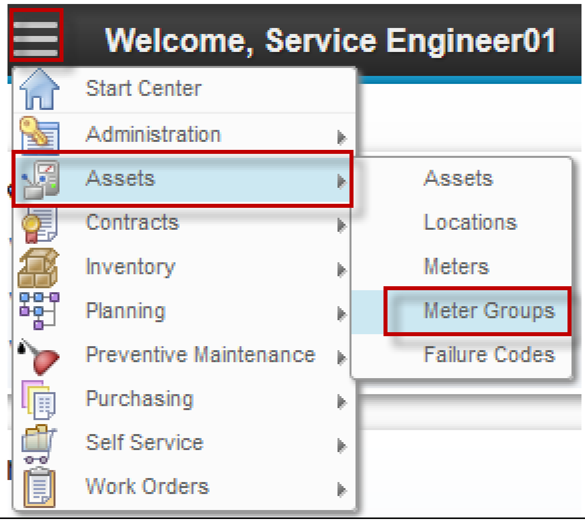


2.4.2 Create Meter Groups

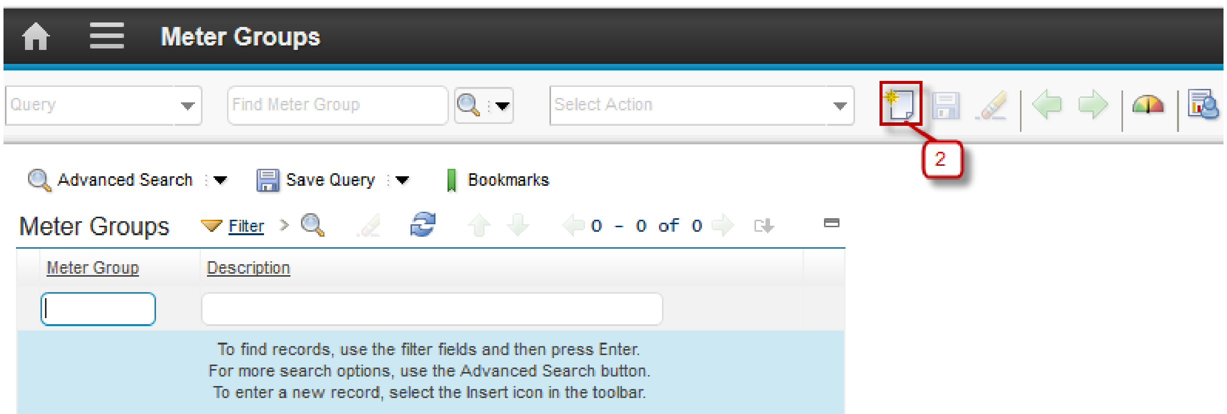
Below are step by step to Create Meter Groups by *Service Engineer*:

Step	Action

1. Go to Meters application by click (GoTo) – **Assets – Meter Groups.**



2. Click **New Row.**



3a. Input the Meter Group name.

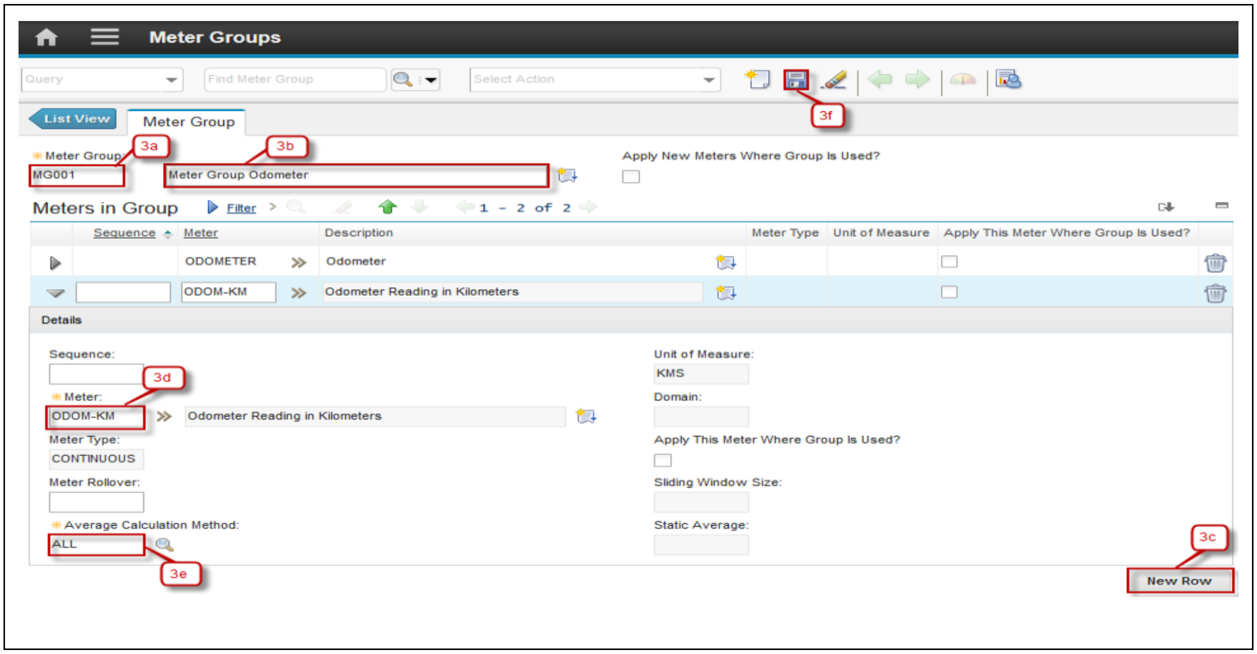
3b. Input the Meter Group description.

3c. Click **New Row**.

3d. Select Meter using button

3e. Select Average Calculation Method using button

3f. Click **Save.**



2.5 Assets Application

Asset application is used to create and store asset numbers and corresponding information, such as parent, location, vendor, up/down status and maintenance costs.

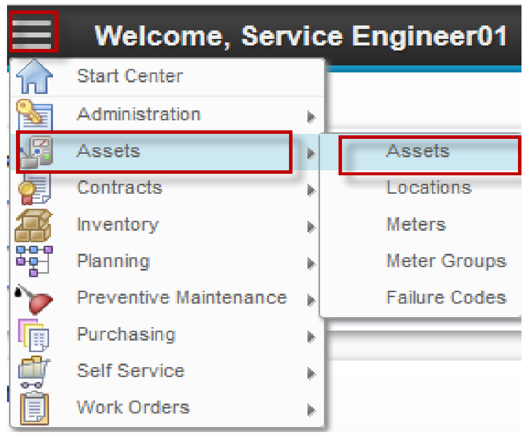
In **Asset** application user can build asset hierarchy, an arrangement of buildings, departments, assets and subassemblies. The asset hierarchy provides a convenient way to roll up maintenance costs so that user can check accumulated costs at any level, at any time. It also makes it easy to find a particular asset number.

2.5.1 Create Assets

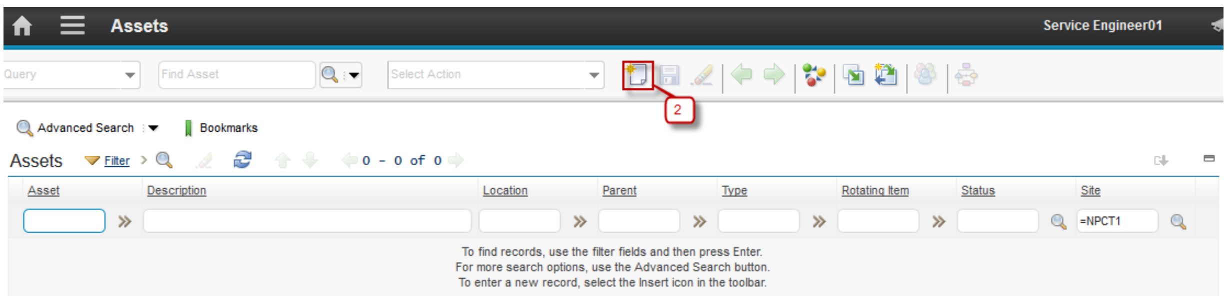
Asset creation can be done by *Service Engineer*. Below are step by step to Create Assets:

Step	Action

1. Go to Assets Application by click (GoTo) – **Assets – Assets.**



2. Click **New Row.**



3a. Input the Asset name and make sure default status is 'PLANNING'.

3b. Input the Asset description.

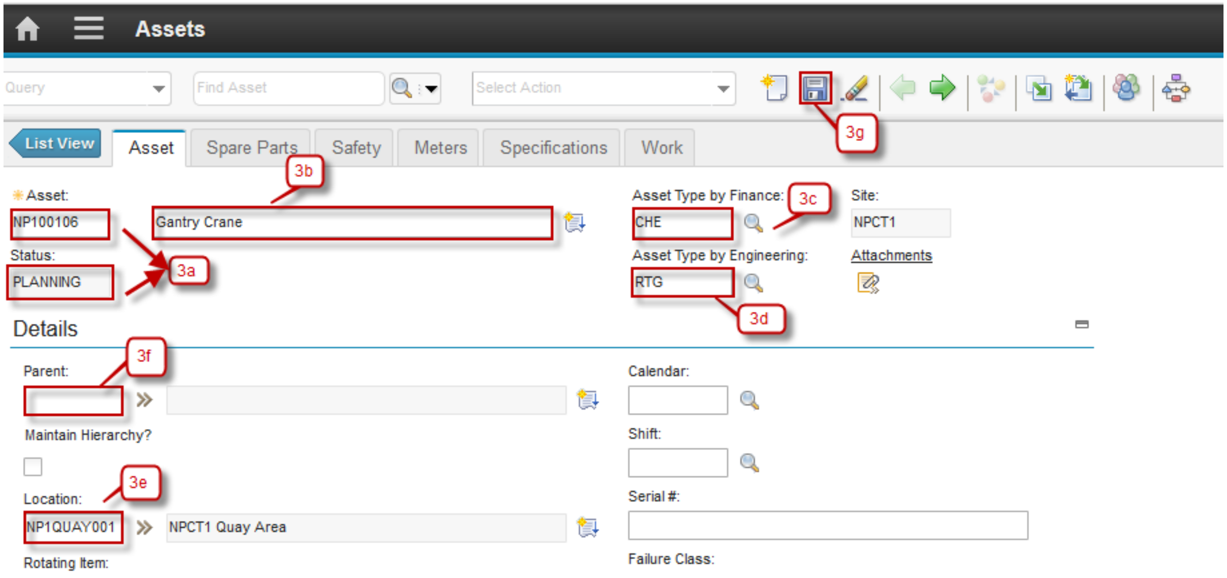
3c. Select Asset Type by Finance using button .

3d. Select Asset Type by Engineering using button .

3e. Input Parent of Asset if the Asset have a parent.

3f. Input Location of Asset**.**

3g. Click **Save** and then after save the Asset button **Route Workflow** will be started automatically

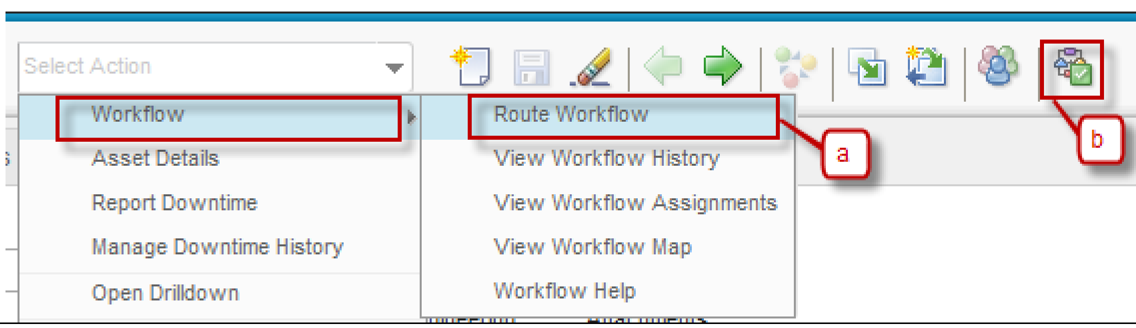


2.5.2 Asset Creation Approval by CHE Engineer

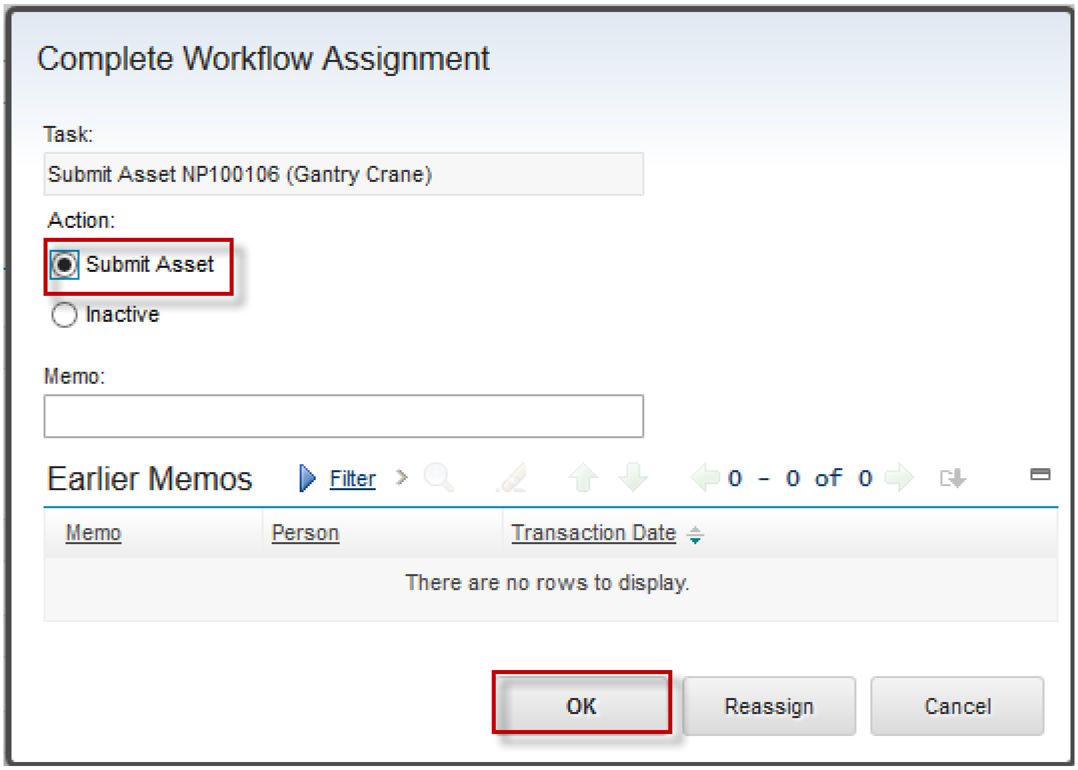
Created Asset must be approved by *CHE Engineer*. Below are step by step to Asset Creation Approval:

Step	Action

1. Asset Creation Approval have two ways:
   a. From **Select Action → Workflow → Route Workflow**
   b. Click button 

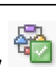


2. Click **Route Workflow** and dialog box will be appeared, see the figure below:



3a. Login as CHE Engineer 01.

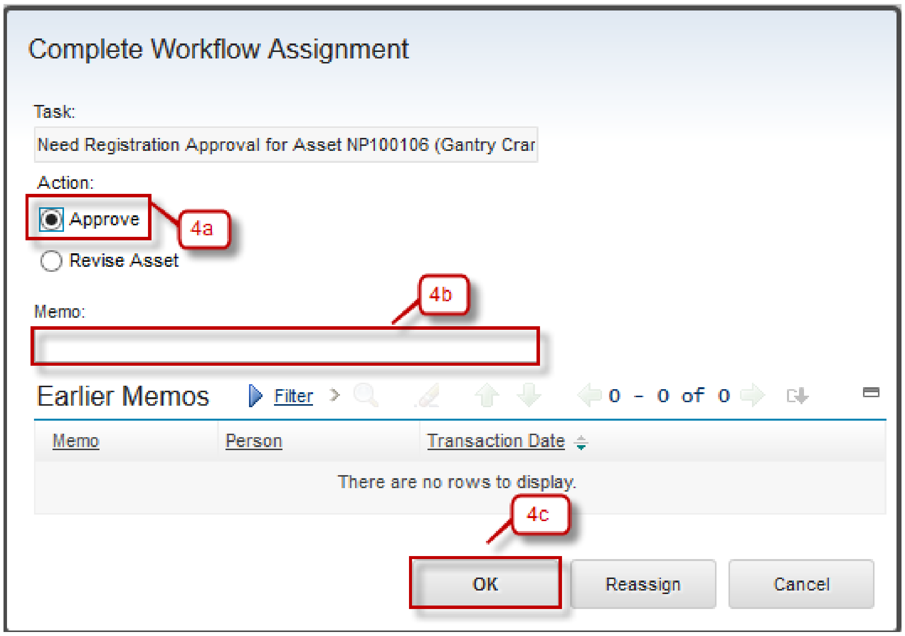
3b. Assigment will be appeared in column **Inbox/Assignments** for approval Asset.

4. Click button **Route Workflow**  and dialog box will be appeared see the figure below:

4a. Select Action Approve

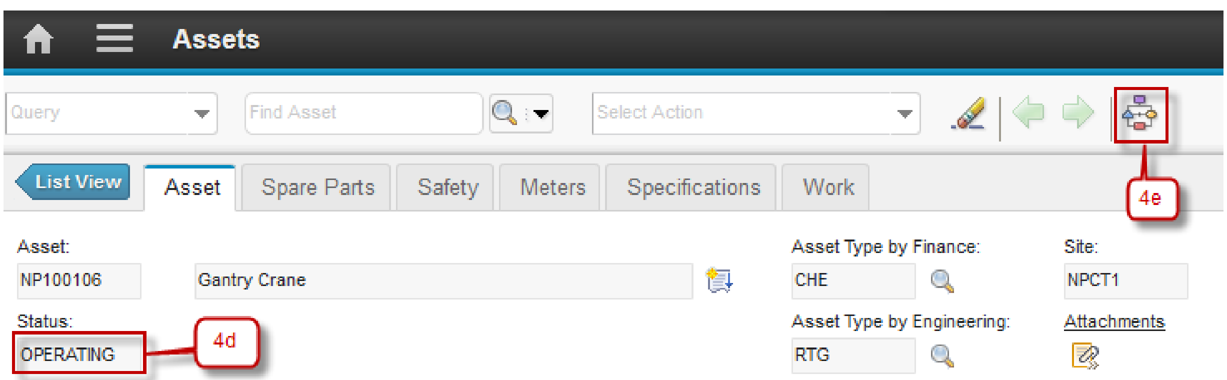
4b. Input Memo (if needed)

4c. Click **OK**



4d. Status changed to 'OPERATING'.

4e. Workflow stopped.

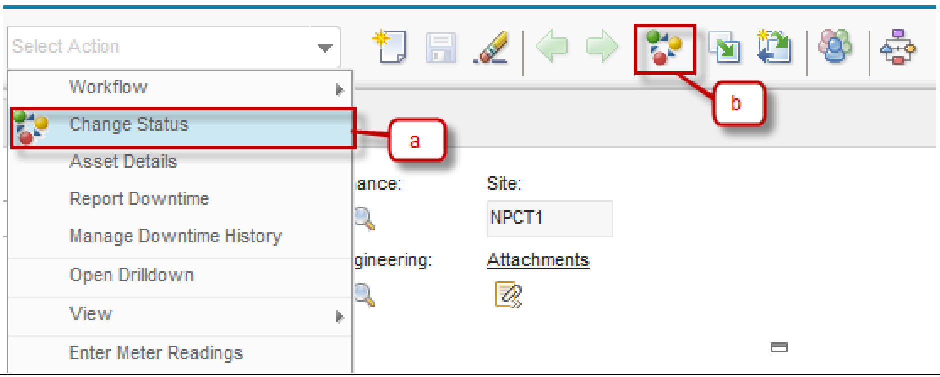


2.5.3 Change Status of an Asset

Sometimes, there is a needs to change the status of an asset. For example, when an asset is broken or decommissioned, user should change the status, to indicate that the asset is now in broken or decommissioned status. Below are step by step to Change Status of an Asset by *Service Engineer*:

Step	Action

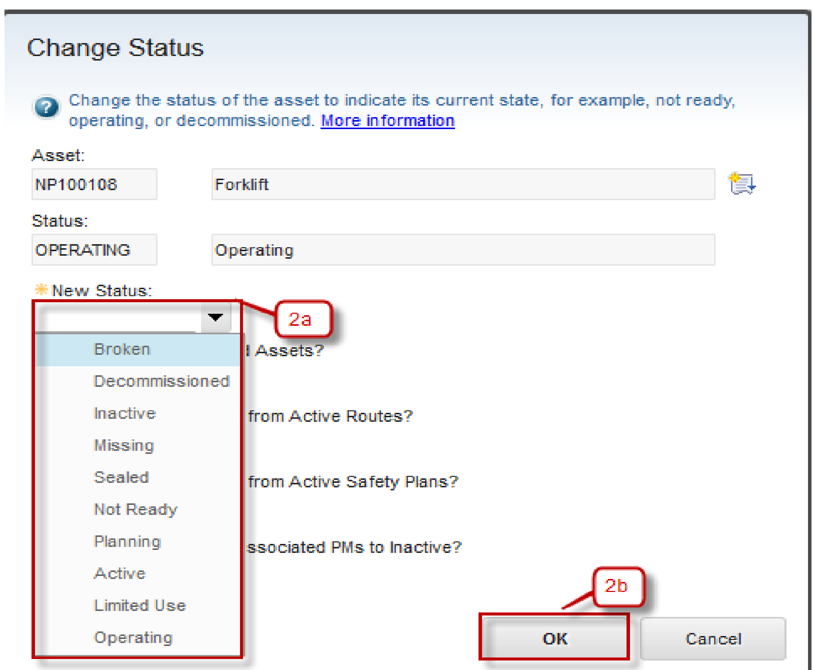
1. Change Status of Person have two ways:
   a. From **Select Action → Change Status**
   b. Click button 



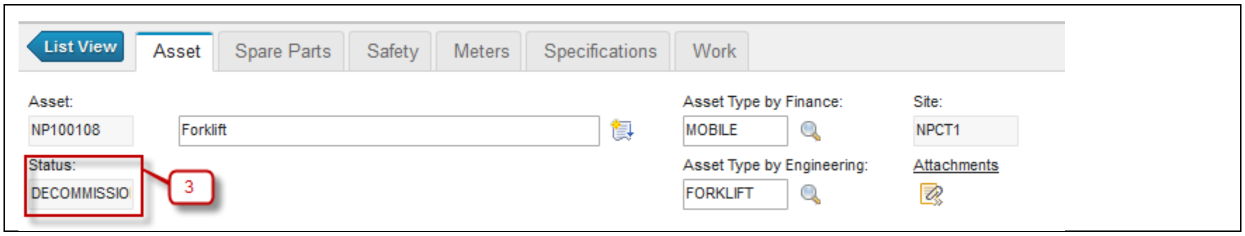
2. Dialog box **Change Status** will be appeared, see the figure below :

   2a. Select New Status of Asset.

   2b. Click **OK.**



3. Status changed to DECOMMISSIONED

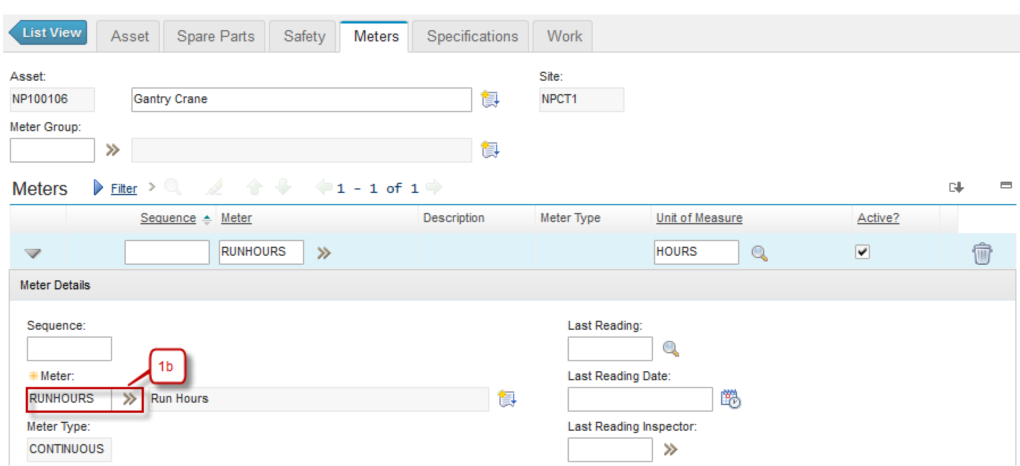


2.5.4 Assets Meters Registration & Meter Readings

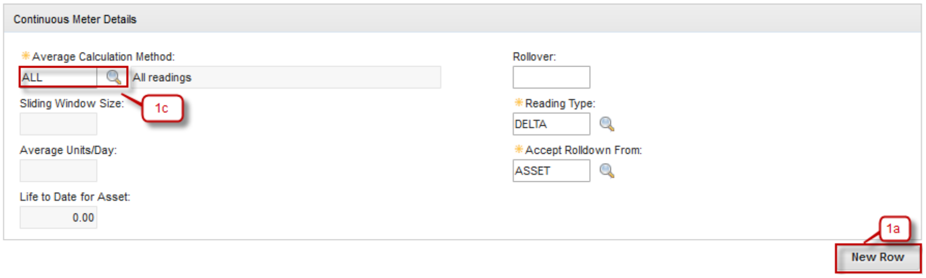
User can register all meters that apply to an asset in **Assets** application. Meters registration in an asset can be performed by registering a meter group that will automatically register all meters in meter group to the asset, or manually registering each meter to the asset.

In **Asset** application, user also can perform **Enter Meter Reading** action to report meter readings for the selected asset. This meter reading will be used as a measurement about the assets performance based the meters. This meter reading can be used to determine when a maintenance toward those asset should be performed. As additional notes, Enter Meter Reading action can be performed in **Work Order Tracking** application too (This application will be discussed later).

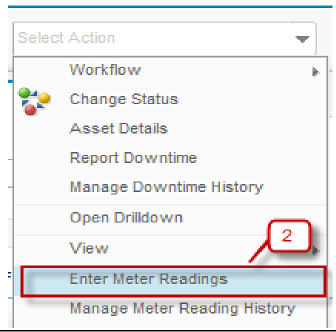
Below are step by step to register a meter to an asset, and performs **Enter Meter Reading** action, by *Service Engineer*:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td colspan="2">
1. Select one of Asset, go to Tab Meters .
1a. Click New Row.
1b. Select the Meter to use in Asset using button », field Meter Type and field Unit of Measure automatically fill.
1c. If the type of Asset is ‘CONTINUOUS’ , field Average Calculation Method is mandatory.


</td></tr>
</table>



2. To perform enter meter reading action, choose **Select Action → Enter Meter Readings.**

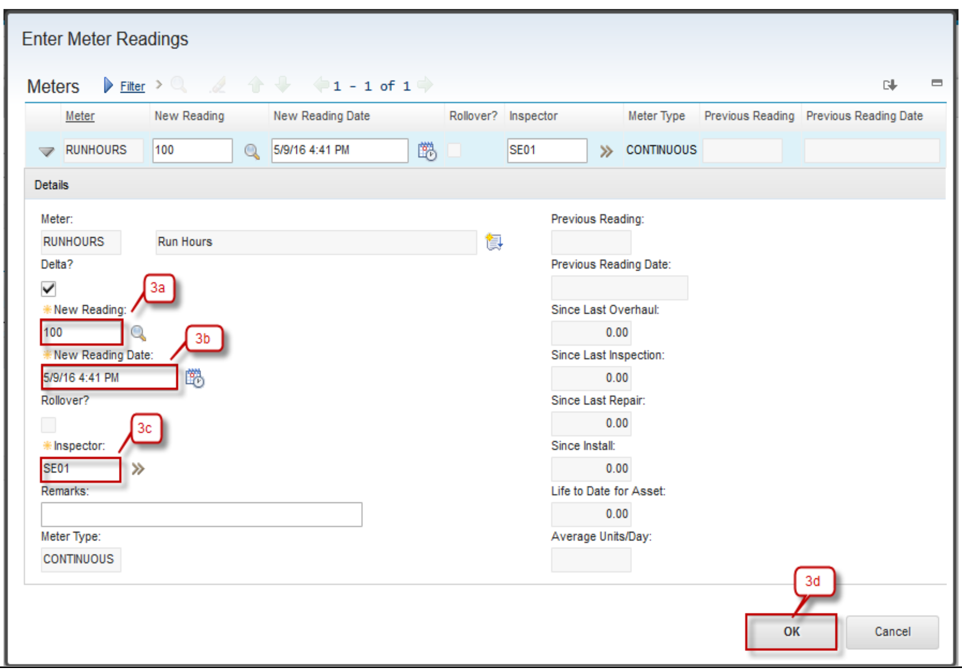


3a. Input value New Reading.

3b. New Reading Date automatically fill when input New Reading.

3c. Inspector automatically fill with user login.

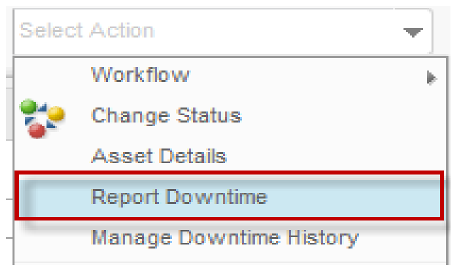
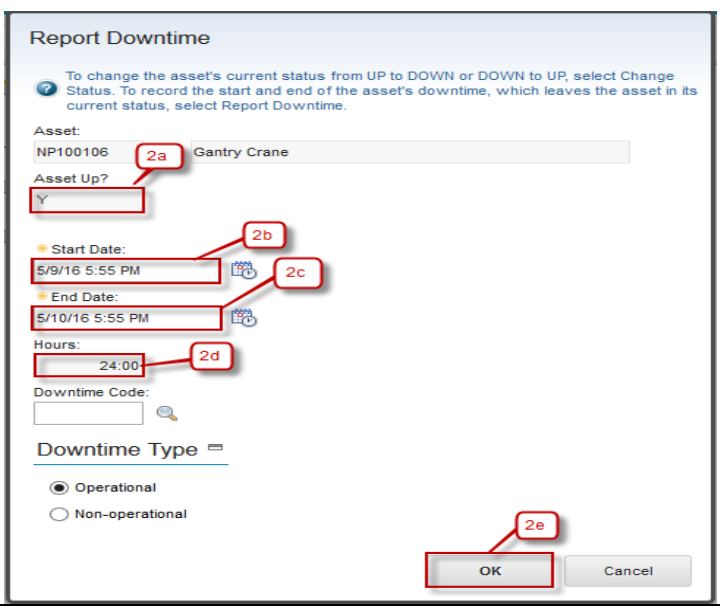
3d. Click **OK.**



2.5.5 Downtime

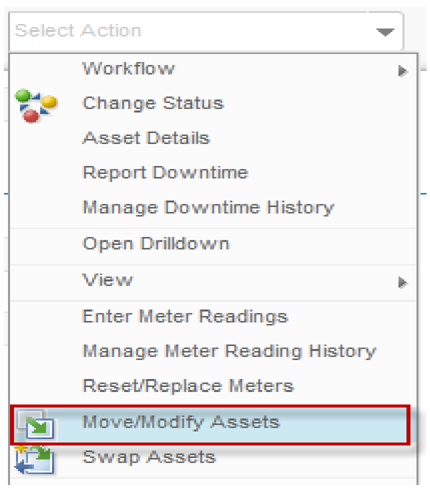
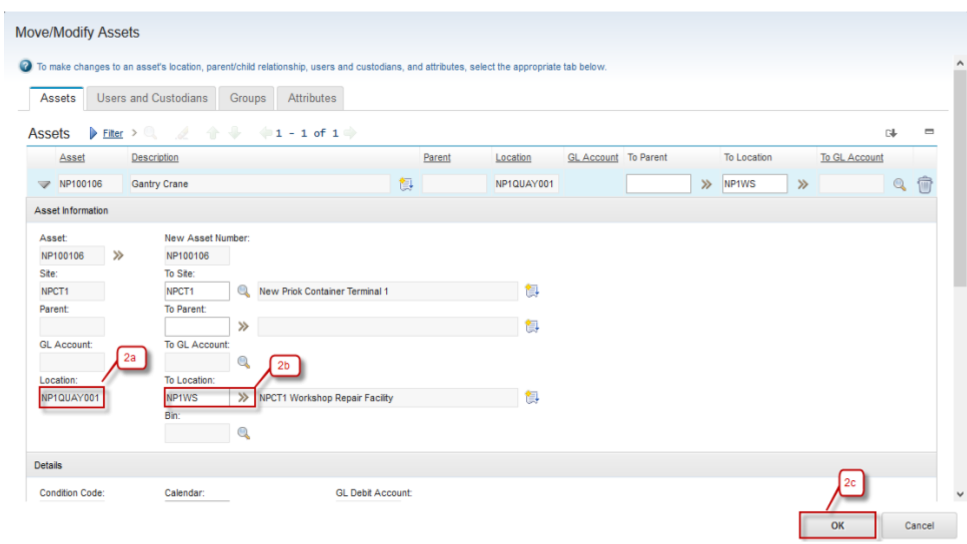
Downtime is another feature that available in an asset. This feature can be used to determine and track how long an asset has been out of service. With this feature, user can be informed about how effective an asset is. A field of total accumulated downtime that ever happens to the asset is available in **Asset** application too.

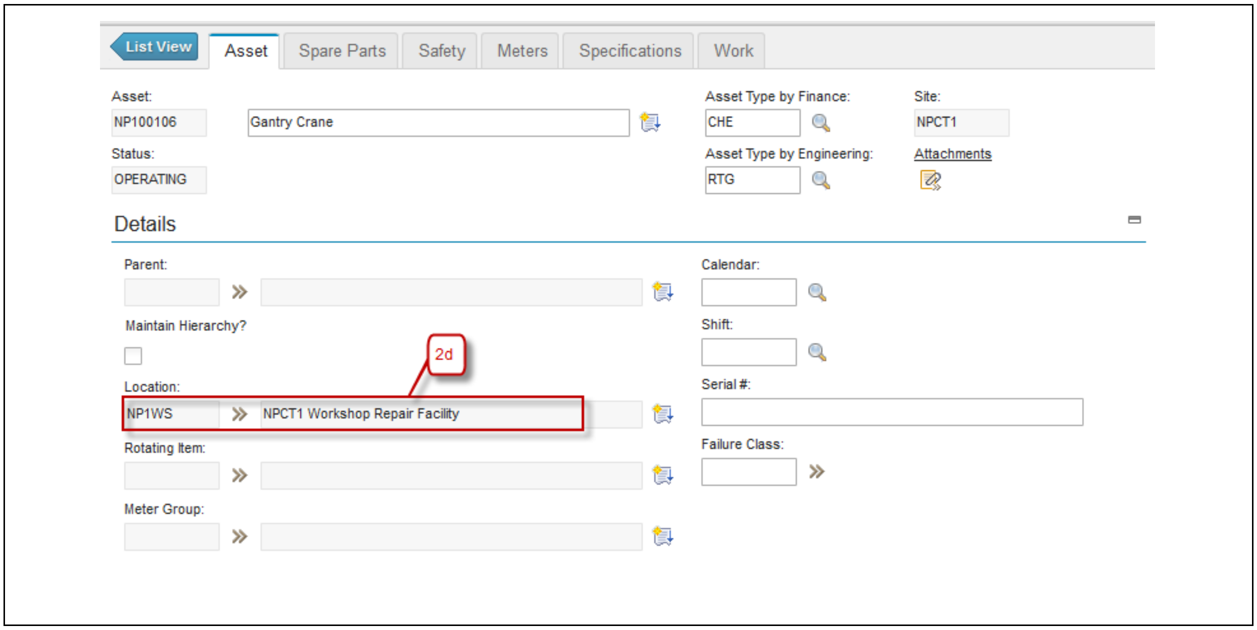
Downtime is reported in **Assets** application using **Report Downtime** action. In this action, user should define how long a downtime occurs to the asset, the start time, and the end time. As a note, this report downtime feature also available in **Work Order Tracking** application. Below are step by step to report downtime of an asset:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Select one of Asset. Select Action → Report Downtime</td><td>
</td></tr>
<tr><td colspan="2">2a. Make sure default Asset Up? is Y.
2b. Start Date usually automatically fill with current date, but it’s editable.
2c. Input End Date until when the asset will be up again.
2d. Value of Hours is interval between Start Date and End Date.
2e. Click OK.

</td></tr>
</table>

2.5.6 Move Assets

When an asset need to be moved from a location to another location, for example, from operating location to repair location, user can do that from **Asset** application via **Move/Modify Assets** action. Besides locations, parent asset and condition code (for rotating asset) can be changed through this action too. Below are step by step to move location of an asset:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Select one of Asset. Select Action → Move/Modify Assets</td><td>
</td></tr>
<tr><td colspan="2">2a. Current Location before being moved.
2b. Field To Location where the Asset move.
2c. Click OK.
2d. Asset already move to New Location.

</td></tr>
</table>



2.6 Exercise Phase

3 Work Order

3.1 Work Order Overview

Work Orders are used to identify, plan, schedule, execute and track the work that is performed at Locations and/or on Assets throughout organization. Initially, a work order is a request for work to be performed. Once completed, a work order is a record of activities performed that make up the history of the Locations or Assets.

Work order record contains all information about a work, from work plan until work result. With a work order, a work can be tracked about its current status, all resource required, cost prediction, all resource used, and all task performed in the work.

3.2 Work Order Tracking Application

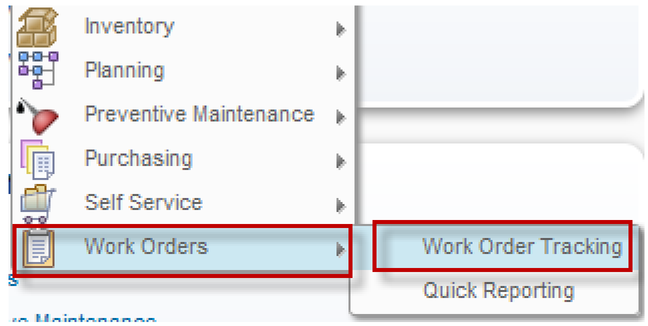
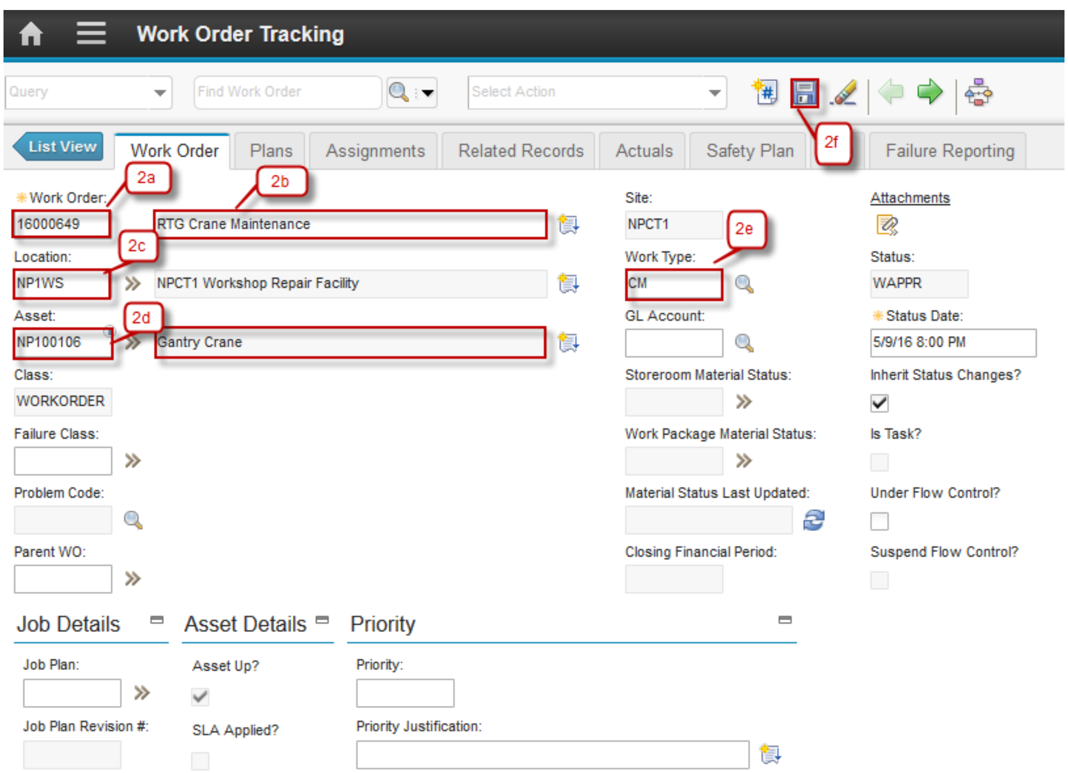
You use the **Work Order Tracking** application to plan, review and approve work orders for assets and locations. When you create a work order in Maximo, you initiate the maintenance process and create a historical record of work being performed. You can create work orders in several Maximo applications.

The **Work Order Tracking** application contains the following tabs:

- **List**: To search work orders.
- **Work Order**: To create, view and modify work order; view PM and scheduling information; see which job plan and safety plan are applied; view the origination work order for a follow-up work order; identify the failure hierarchy for the asset or location.
- **Plans:** To enter, view and modify job tasks and labor, material, services and tool requirements for the work plan.
- **Related Records**: to view, add and delete related work orders, and tickets; to view follow-up records for the current record.
- **Actuals**: to enter, view and modify actual work order start and finish times labor hours and costs, material quantities, locations, costs and tool quantities, hours and cost.
- **Safety Plan**: To enter, view and modify safety information on work order.
- **Failure Reporting**: to report asset and location failures to help identify breakdown patterns or trends.
- **Log**: to view and create work log and communication entries about the current record.

3.2.1 Materials Planning

Below are step by step to create Materials Planning:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Go to Work Orders Tracking Application by click (GoTo) – Work Orders – Work Order Tracking.</td><td>
</td></tr>
<tr><td colspan="2">2a. Work Order number is autonumber.
2b. Input Work Order description.
2c. Location automatically fill when selecting asset.
2d. Select Asset.
2e. Select Work Type.
2f. Click Save.

</td></tr>
</table>

3. Move to Tab **Plan**. Go to Tab **Material**.

3a. Click **New Row**

3b. Select the Item and description automatically fill.

3c. Default Line Type is Item.

3d. Input Quantity, default value of Quantity is 1.00

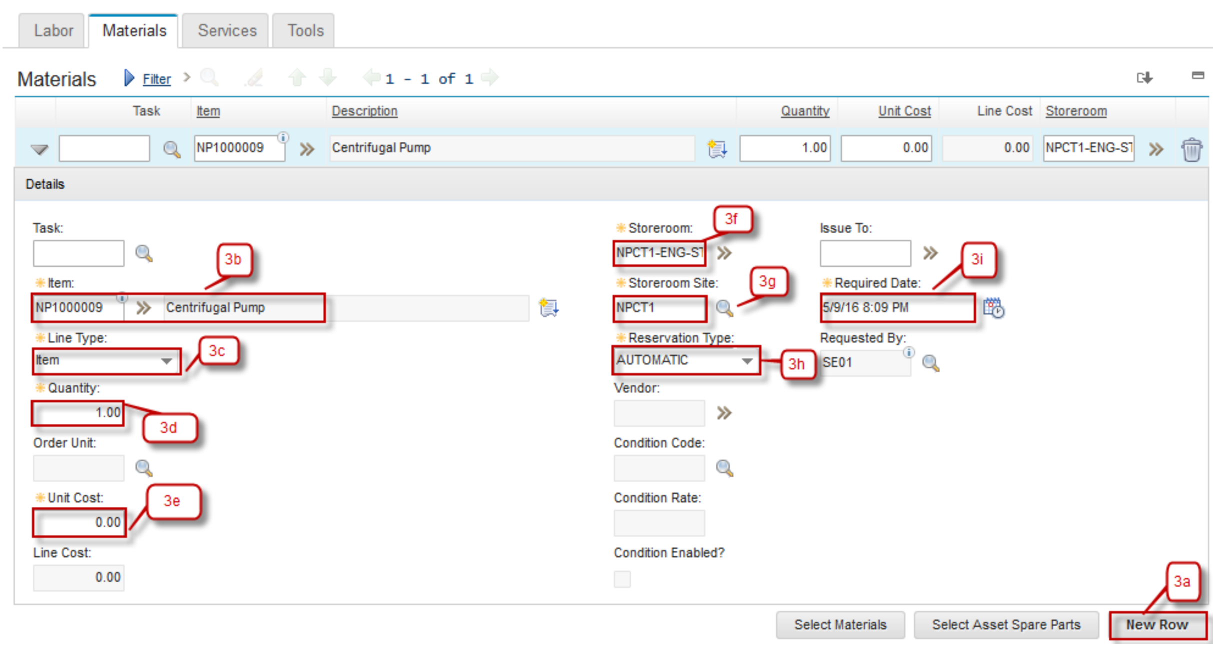
3e. Unit Cost Item, default value of unit cost is 0.00

3f. Select Storeroom where the Item is stored.

3g. Storeroom Site automatically fill when storeroom filled.

3h. Reservation Type automatically fill with default 'AUTOMATIC'.

3i. Required Date automatically with current date.



3.2.2 Labor Planning

Below are step by step to create Labor Planning:

Step	Action

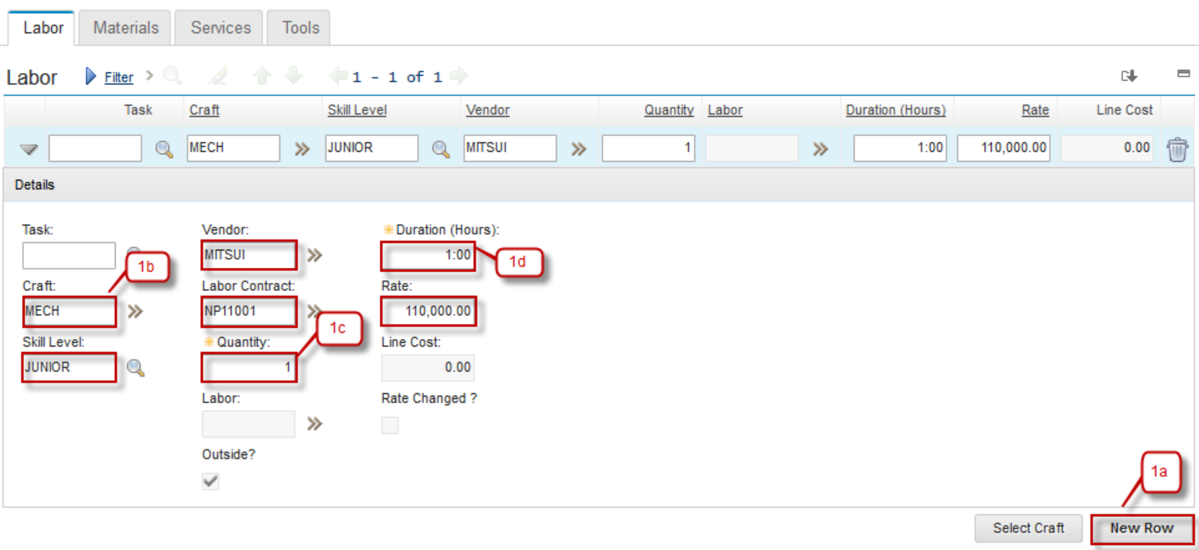
1. Move to Tab **Plan**. Go to Tab **Labor**.

1a. Click **New Row**

1b. If Labor from outside when choose the Craft, field Skill Level, Vendor, Labor Contract and Rate will be automatically fill in accordance with the data on the labor contract.

1c. Input Quantity of Labor.

1d. Input Duration of Labor.



3.2.3 Services Planning

Below are step by step to create Services Planning:

Step	Action

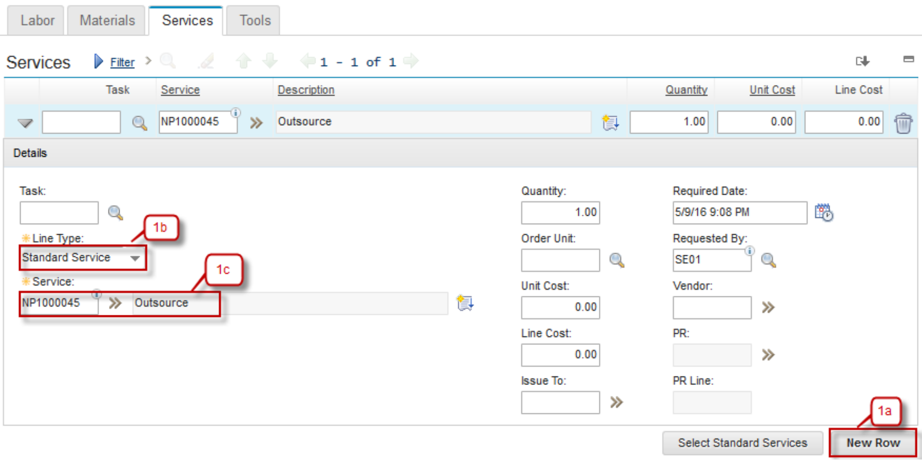
1. Move to Tab **Plan**. Go to Tab **Services**.

1a. Click **New Row**

1b. Select Line Type Service or Standard Service.

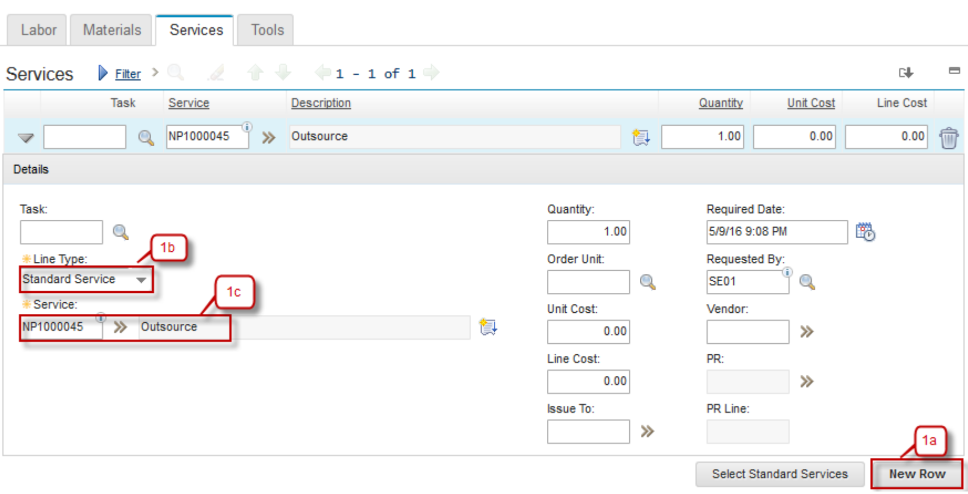
- If Line Type is Service, field Description is required (mandatory).
- If Line Type is Standard Service, choose Service using button »

1c. Select Service will be used.



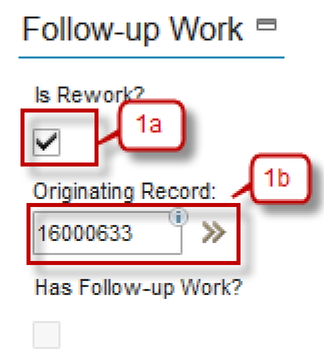
3.2.4 Tools Planning

Below are step by step to create Tools Planning:

Step	Action
1. Move to Tab **Plan**. Go to Tab **Tools**. 1a. Click **New Row** 1b. Select Tool using button » 1c. Input Quantity of Tool. 1d. Input Tool Hours. 1e. Input Rate of Tool. 	

3.2.5 Rework Work Order

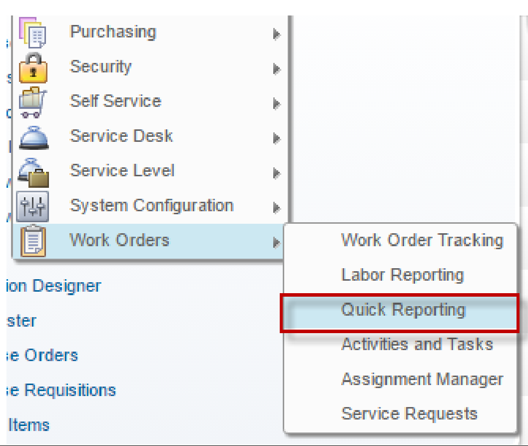
Sometimes a rework need to be performed to a work that already finished, maybe because there is an unfinished work or the result is not as expected. In EMS, the rework should be marked, to measure the maintenance performance. Below are steps to marks a rework work order:

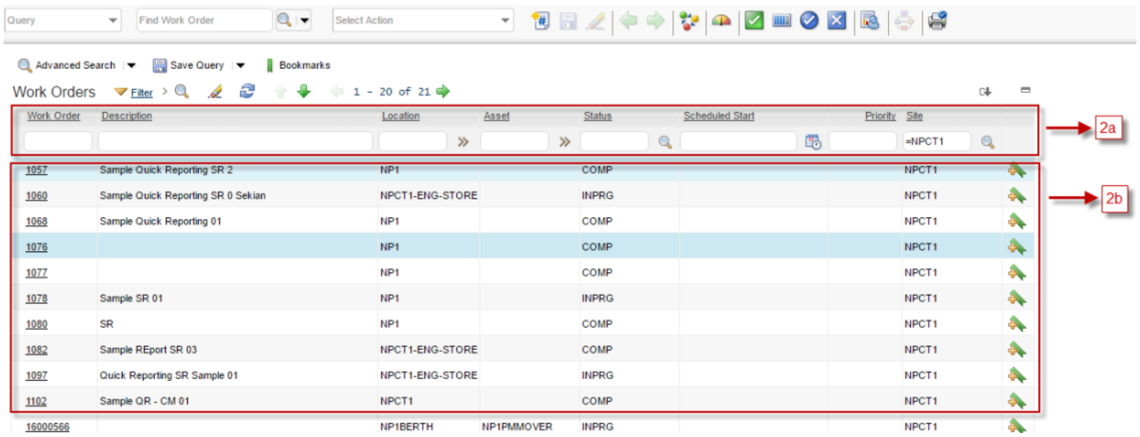
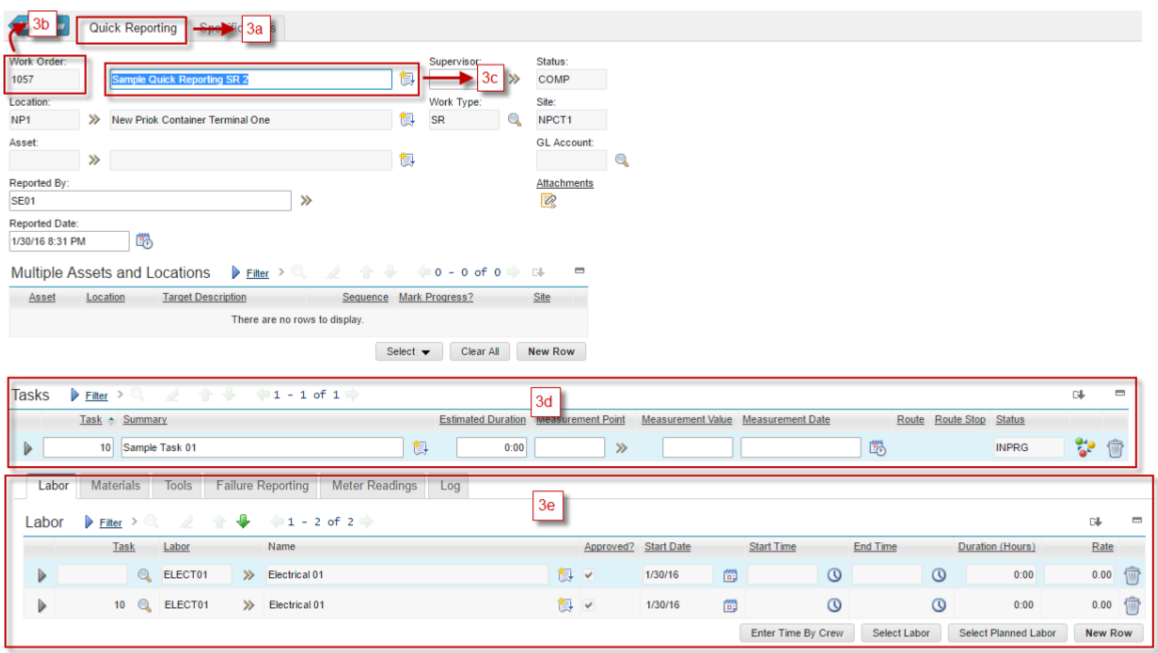
Step	Action
1. Select one of Work Order to be set into a rework work order. 1a. Check box **Is Rework** checked 1b. Select the **Originating Record** that will be used as Work Order originator.	

3.3 Quick Reporting Application

Quick Reporting application is used to report a work that has been performed without any prior work plan. It is usually used for emergency work that should be performed immediately. So, when the work is finished, user should report the work result via **Quick Reporting** application. A work order will be created for the work result that has been reported. The different between creating a work order in **Work Order Tracking** application against **Quick Reporting** application is that work order created **Work Order Tracking** application contains planning information, while work order from **Quick Reporting** application is not.

Below is **Quick Reporting** application accessibility.

1	From **Work Orders** – **Quick Reporting** 

2	Show listed work order  2.a Filter work order by **Work Order** number, **Description**, **Location**, **Asset**, **Status**, **Scheduled Start**, **Priority** and **Site** 2.b **Listed** all work orders
3	 3.a **Quick Reporting** Tab, view update and detail of the work order. 3.b Information about work Order number 3.c Information about description of the work order. 3.d All Tasks listed in the work order. 3.e All Listed Labor, Material, Tools, Failure Reporting and Log in the quick reporting

4 Job Plan

4.1 Job Plan Overview

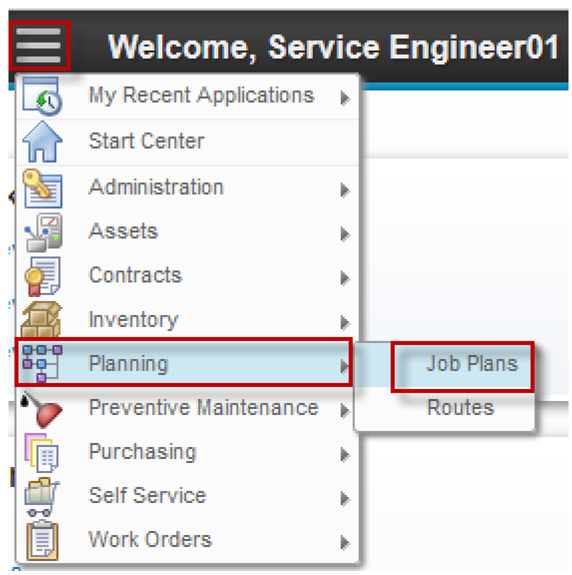
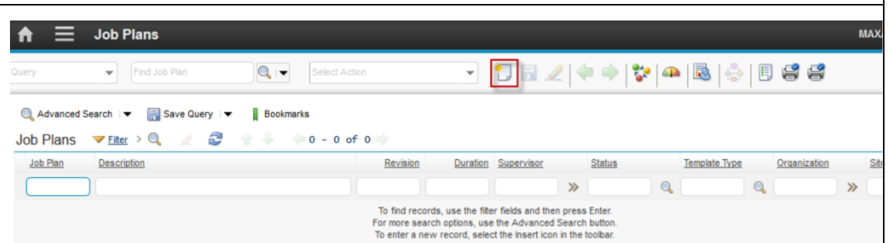
Job Plan is a work template that contains all planning information about a work. Job plans generally contain task (procedures), along with lists of estimated labor, labor hours, materials, services, and tools required for the work. Job plan can be applied to a work order, to automate planning resource and task of the work order.

Note: You also can create a job plan from a work plan that you already defined in a work order, using **Create Job Plan from Work Plan** action in **Work Order Tracking** application.

4.2 Job Plans Application

You use the **Job Plans** application to create, modify, or delete job plan records. A job plan is a detailed description of work to be performed for a work order. You can apply job plans to work orders to automate work planning definition. After a job plan becomes a work plan on a work order, you can modify the work plan without affecting the job plan.

4.2.1 Create Job Plans

Step	Action
1. Go to Labor application by click (GoTo) – **Planning – Job Plans.**	
2. Click **New Job Plan.**	

3A. Input Job Plan Number

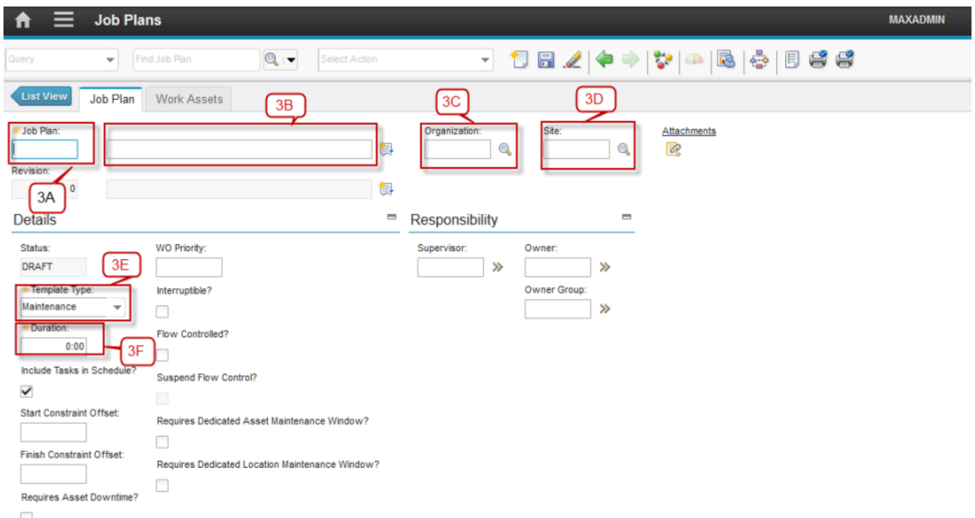
3B. Input Description for Job Plan

3C. Click button and select the Organization – COMP

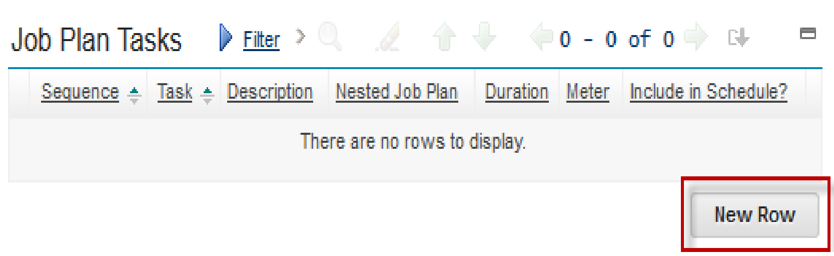
3D. Click button and choose the Sites – COMP1

3E. Select the Job Plan type, Activity / Maintenance / Process

3F. Input the duration of the Job Plan in hourly format



4. Click **New Row** on the Job Plan Tasks for creating new task



5A. Select the Organization with click button , then select COMP

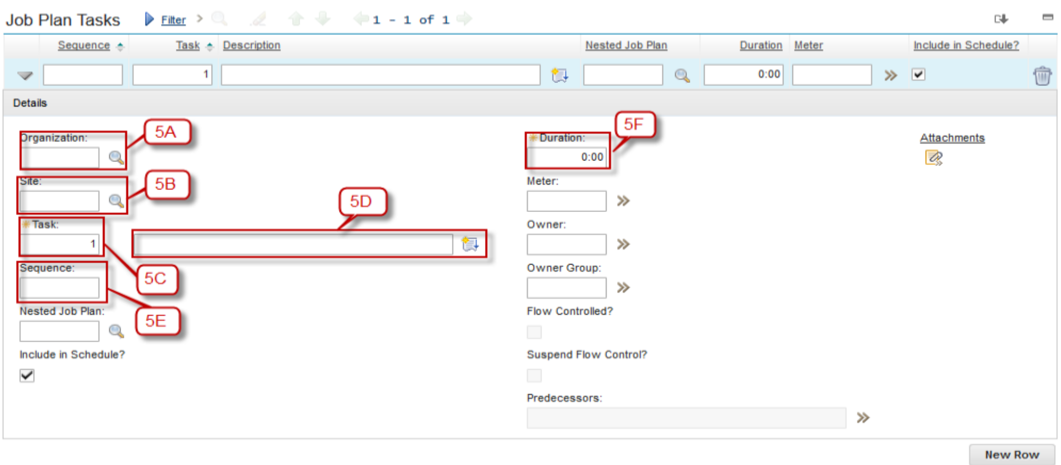
5B. Choose the Site with click button , then select COMP1

5C. Input the number of Task

5D. Input the Task Description

5E. Input Sequence number

5F. Input the duration of this Task



6A. Click on the Labor Tab

6B. Click New Row to add new data of Labor

6C. Click button , then select the Organization

6D. Click button , then select the Site

6E. Click button , then select Task

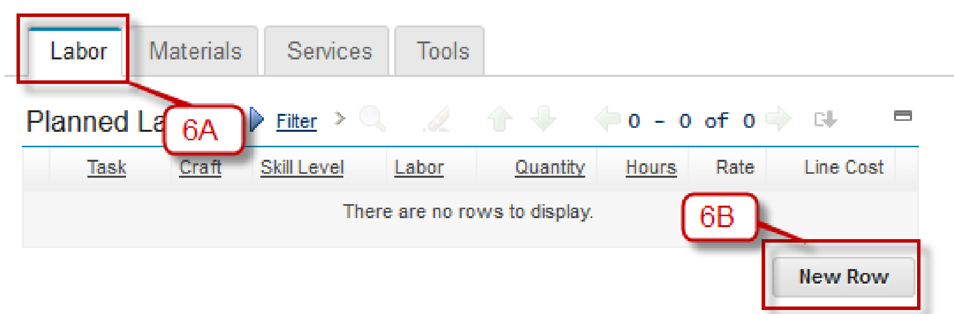
6F. Click button **>>** then click **"Select Value"** then select the Labor

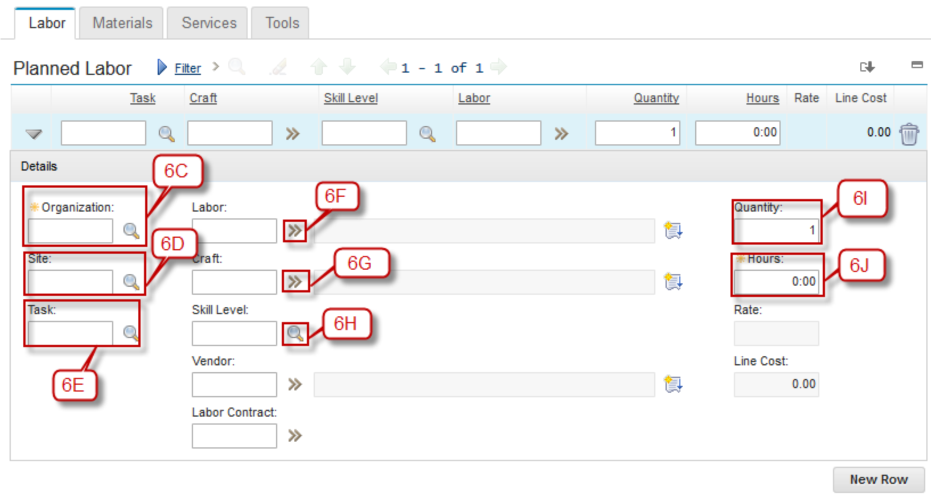
6G. Click button **>>** then click **"Select Value"** then select the Craft

6H. Click button then select the Skill Level

6I. Input Quantity of Labor

6J. Input work hours for Labor





7A. Click on the Tab of Materials to select the Materials

7B. Click **New Row**

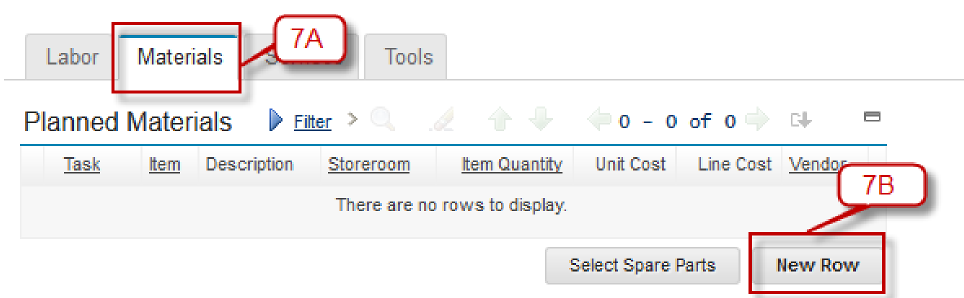
7C. Click on button and select the Task

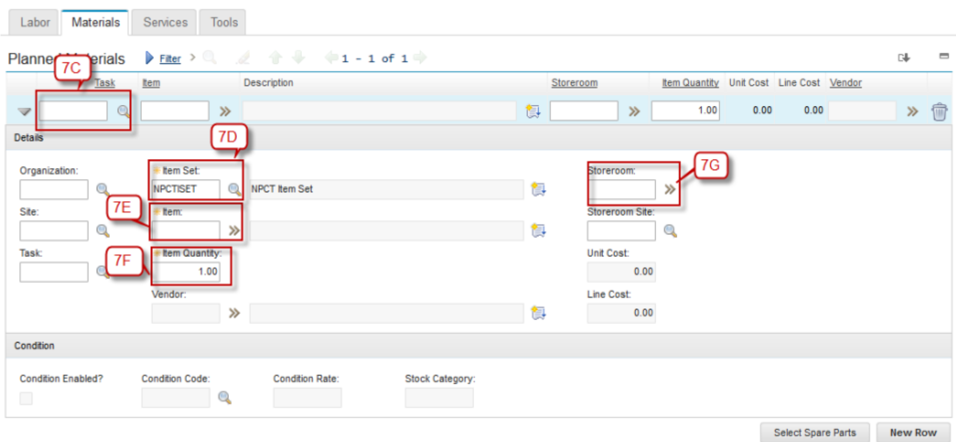
7D. Click button and select the Item Set

7E. Choose Item with click button **>>** then click magnifier icon (select icon)

7F. Then input the Quantity of Item

7G. Select the Storeroom with click **>>** icon





8A. Click on the tab **Services** to create new services or see the services

8B. Click **New Row** to create new record of the services

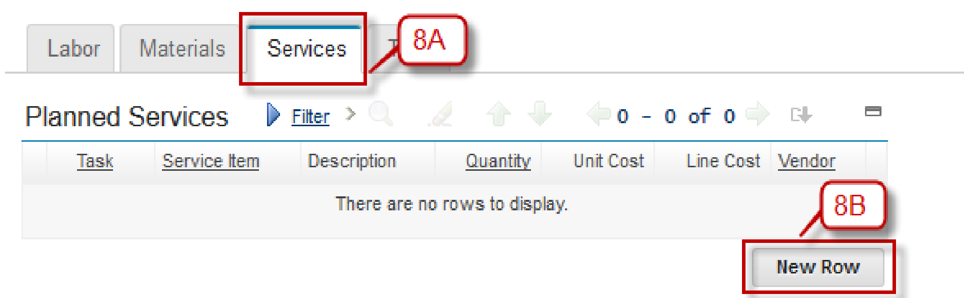
8C. Select the Task with click button

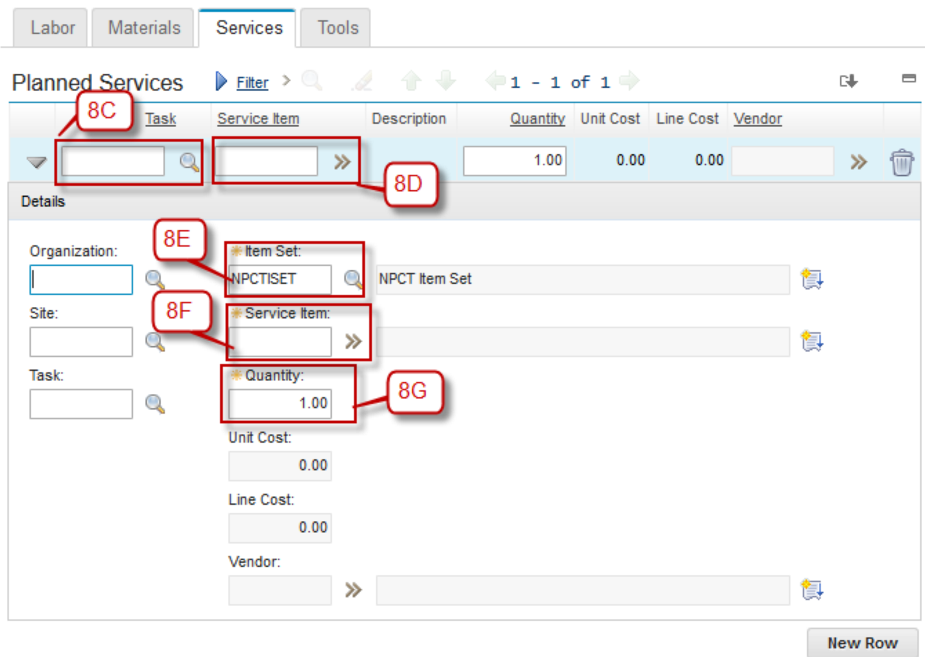
8D. In Service Item field, you can select the Service Item already you have before. But if you want to create new Service Item, please ignore this field.

8E. Click magnifier icon to choose the Item Set, but on the default Item Set was fillin with default COMP1SET.

8F. Click **>>** for choose the Service Item

8G. Input the Quantity of Service item





9A. Click on tab **Tools** to input what tools you used to finished this Job Plan.

9B. Click **New Row** to create a new record.

9C. Select Task for this tools

9D. In Tool field, you can select the Tool already you have before. But if you want to create new Tool, please ignore this field

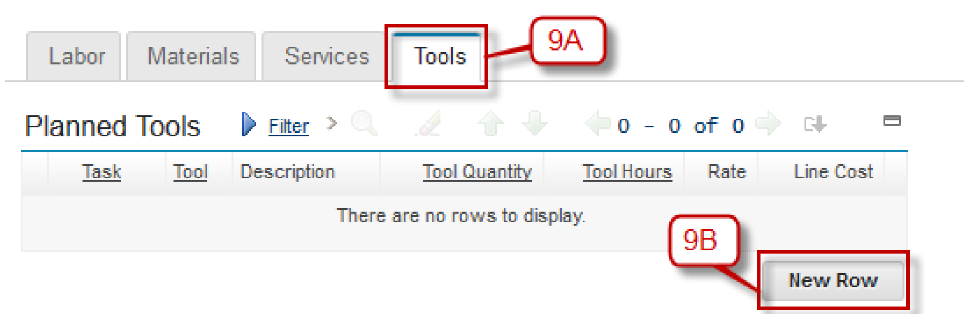
9E. Click button to select the Item Set, but on the default Item Set was fillin with default COMP1SET

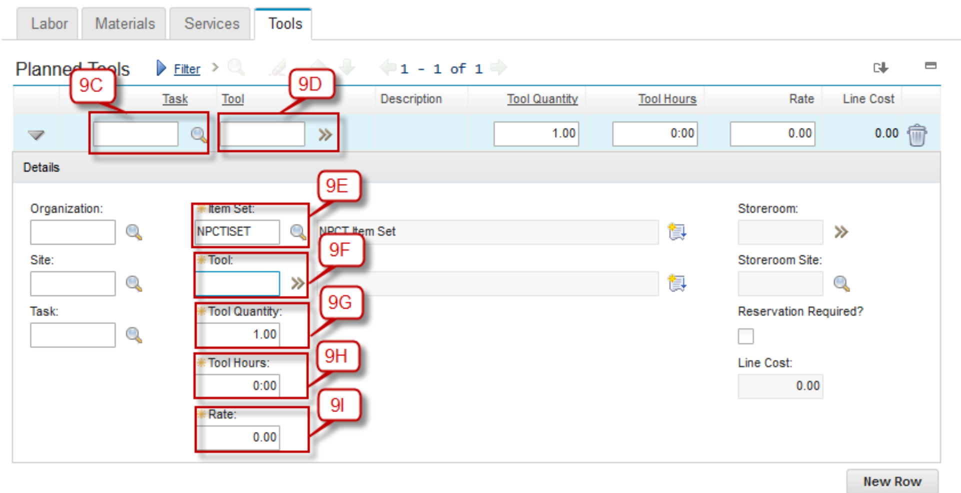
9F. Click **>>** icon to select and see list of tools and then choose

9G. Input Tool Quantity

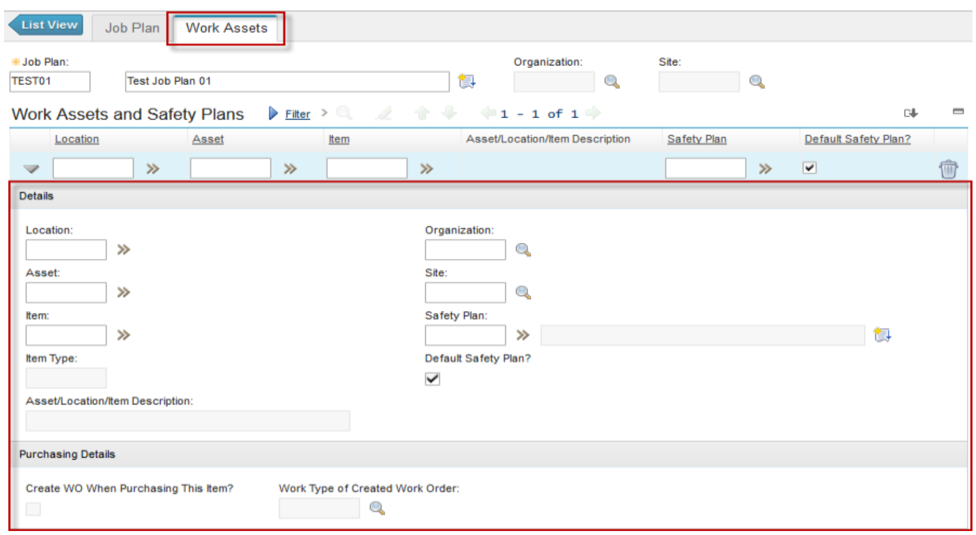
9H. Input Total Hours for tool

9I. Input Rate





10. Beside tab **Job Plan** is **Work Assets**, you can input detailed explanation for a specific asset and job plan. For the example, Job Plan TEST01 is only for ASS01.



4.2.2 Job Plan Approval

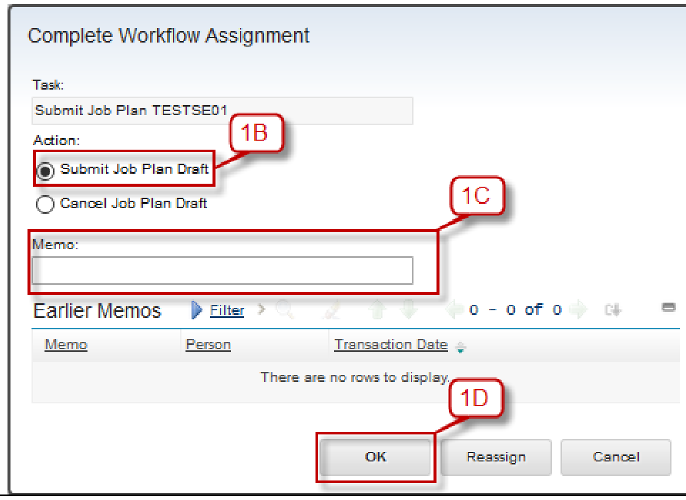
After *Service Engineer* create the Job Plan, Job Plan need to be approved by *CHE Engineer*. Below are step by step to Approval Job Plan:

1A. Click button Route Workflow  to start workflow and then after click the button changed to  its means the workflow is started.

1B. Select the option **“Submit Job Plant Draft”**

1C. Input Memo (if needed).

1D. Then click **OK** .



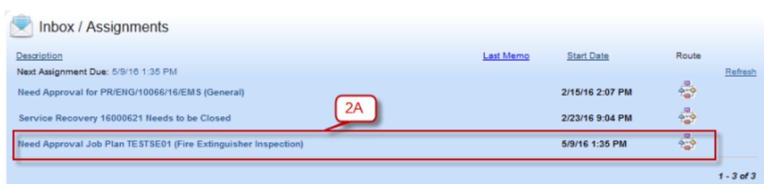
2. Login as CHE Engineer. User name : che01 and password : password.

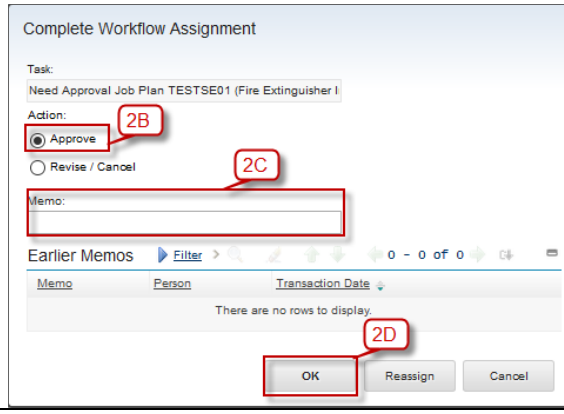
2A. In CHE01 start center in portlet “Inbox / Assignment” CHE01 can see the Job Plan Need Approval, click on the list Need Approval. Then CHE01 will see the detail of Job Plan. After that click  to approval process

2B. Click on approve button

2C. CHE01 can give the memo if any notes for this Job Plan

2D. Click **OK.** Job Plan has been approved.





3. CHE Engineer can **Revise** / **Cancel** a Job Plan was submitted by Service Engineer.

3A. To cancel Job Plan, CHE Engineer must click button **Revise / Cancel.**

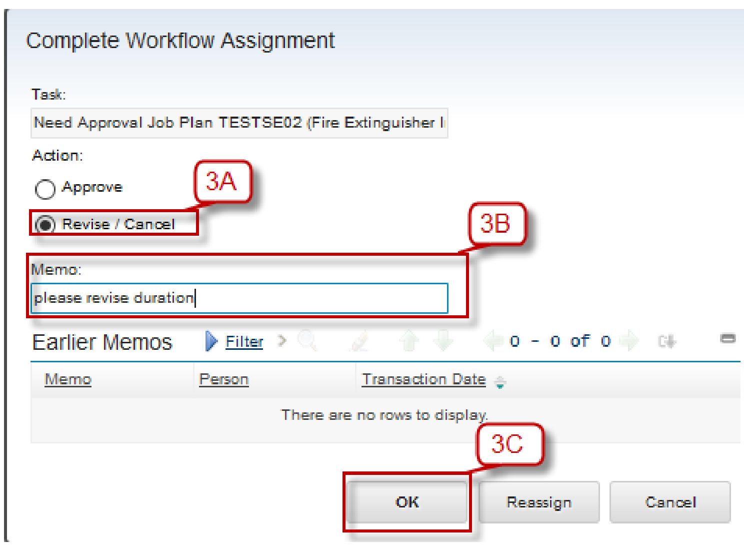
3B. CHE Engineer can give a memo

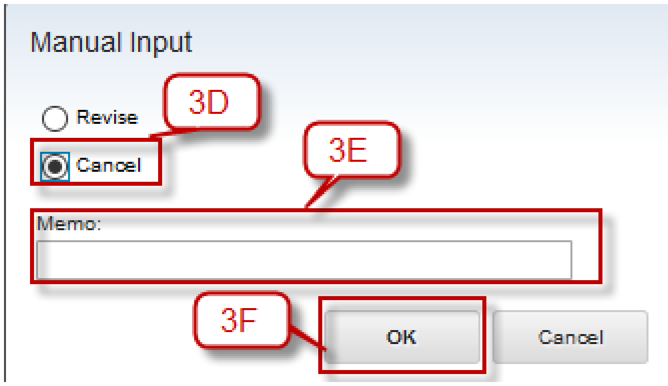
3C. Then click **OK** and dialog box manual input will be appeared.

3D. Click on **Cancel** button

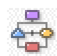
3E. CHE Engineer can give a memo to cancel Job Plan

3F. Click **OK**



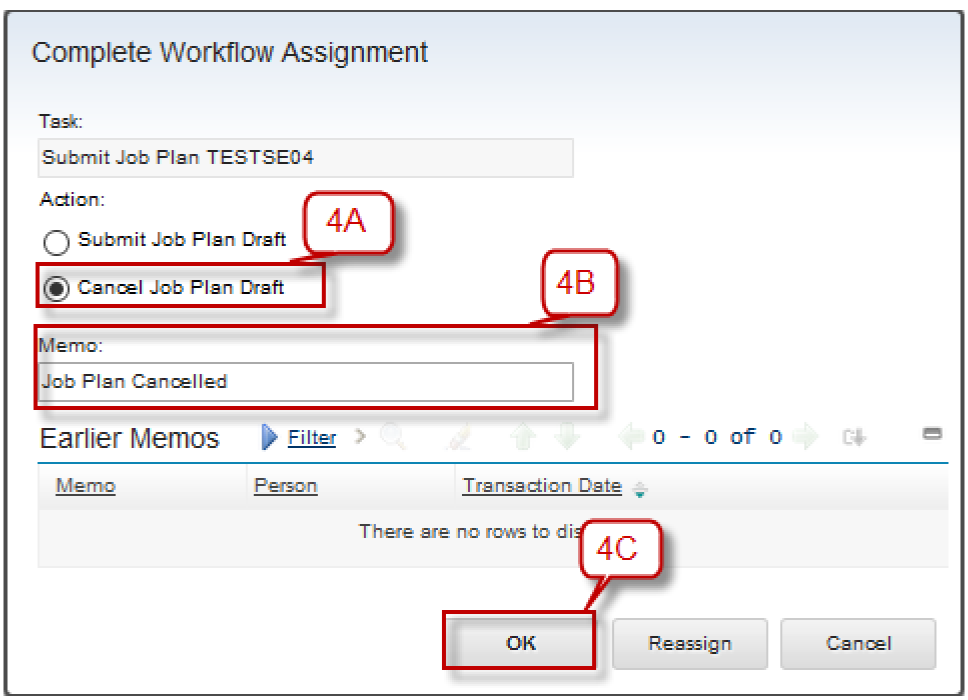


4. Service Engineer can cancel Job Plan with this following step:

4A. Click button Route Workflow  to start workflow and then after click the button changed to  its means the workflow is started.

4B. Seleect the option **"Cancel Job Plant Draft"**

4C. Then click **OK** .



4.2.3 Revise a Job Plan

Below are step by step to Revise a Job Plan:

1. Login as CHE Engineer. User name : che01 and password : password.

1A. In CHE Engineer start center in portlet "Inbox / Assignment" , CHE Engineer can see the Job Plan Need Approval, click on the list Need Approval. Then CHE Engineer will see the detail of Job Plan. After that click  to approval process.

1B. Select **Revise / Cancel**

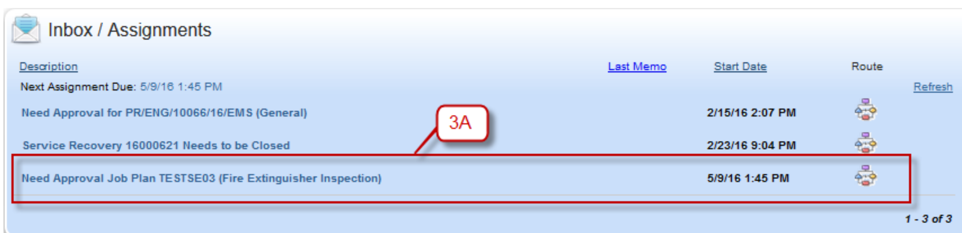
1C. CHE Engineer can give a memo

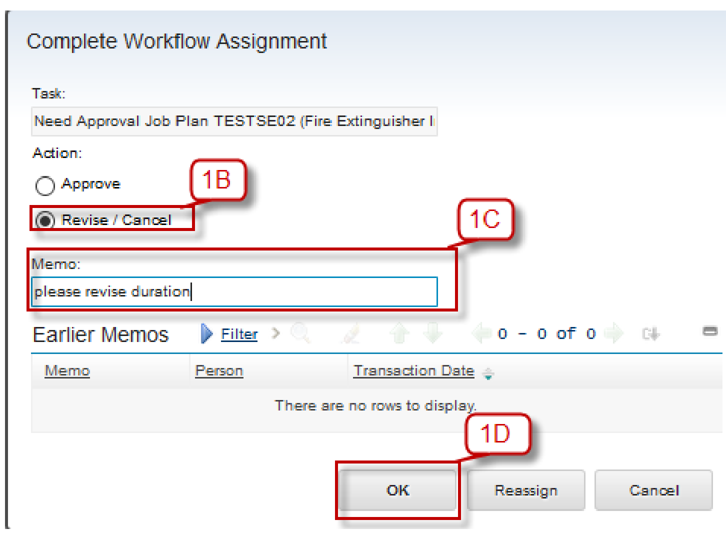
1D. Click **OK** and then dialog box manual input will be appeared.

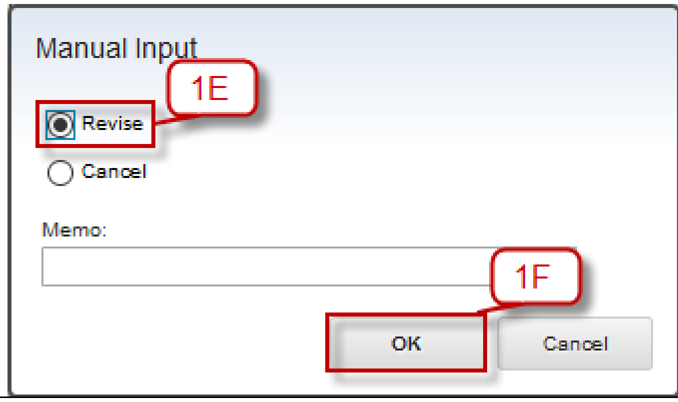
1E. Click **Revise**

1F. Click **OK**

1G. Then the Job Plan was send back to Service Engineer to revised.







2. Login as Service Engineer. User name : se01, password: password.

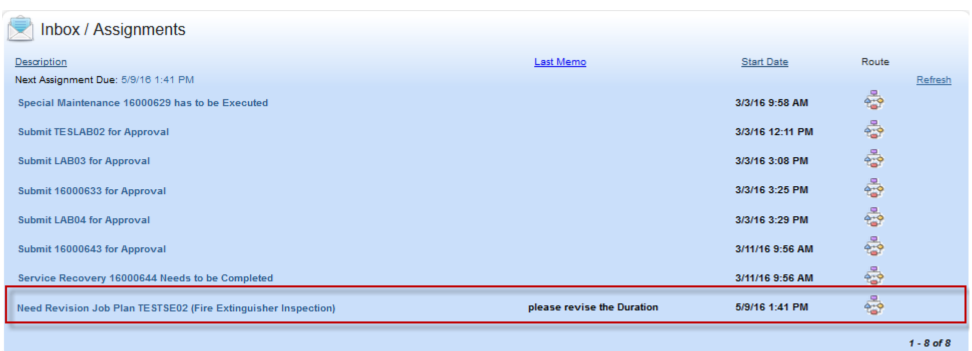
2A. In Start Center Service Engineer, in portlet Inbox / Assignment, Service Engineer can see the Job Plan need to Revision

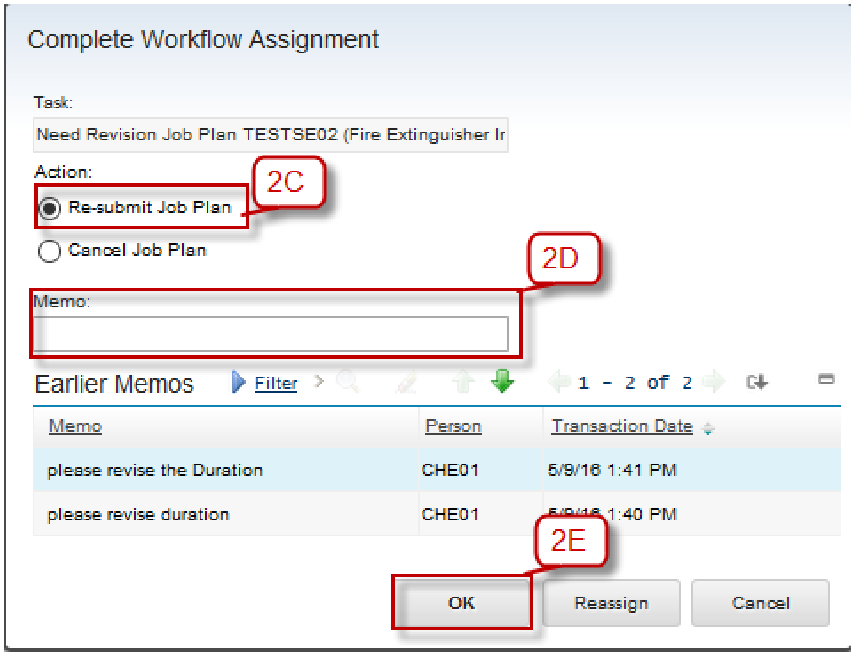
2B. Click Need Revision Job Plan

2C. Click **Re-Submit Job Plan**

2D. Service Engineer can give a memo to this revision

2E. Click **OK.** Job Plan was send back to need approval to CHE Engineer.





3A. In Start Center of CHE Engineer, in portlet Inbox / Assignment, CHE Engineer can see the list to need approval CHE Engineer, click on the Need Approval Job Plan

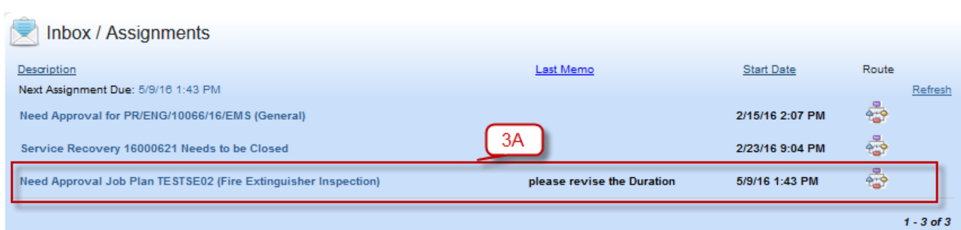
3B. CHE Engineer will check the revision from Service Engineer

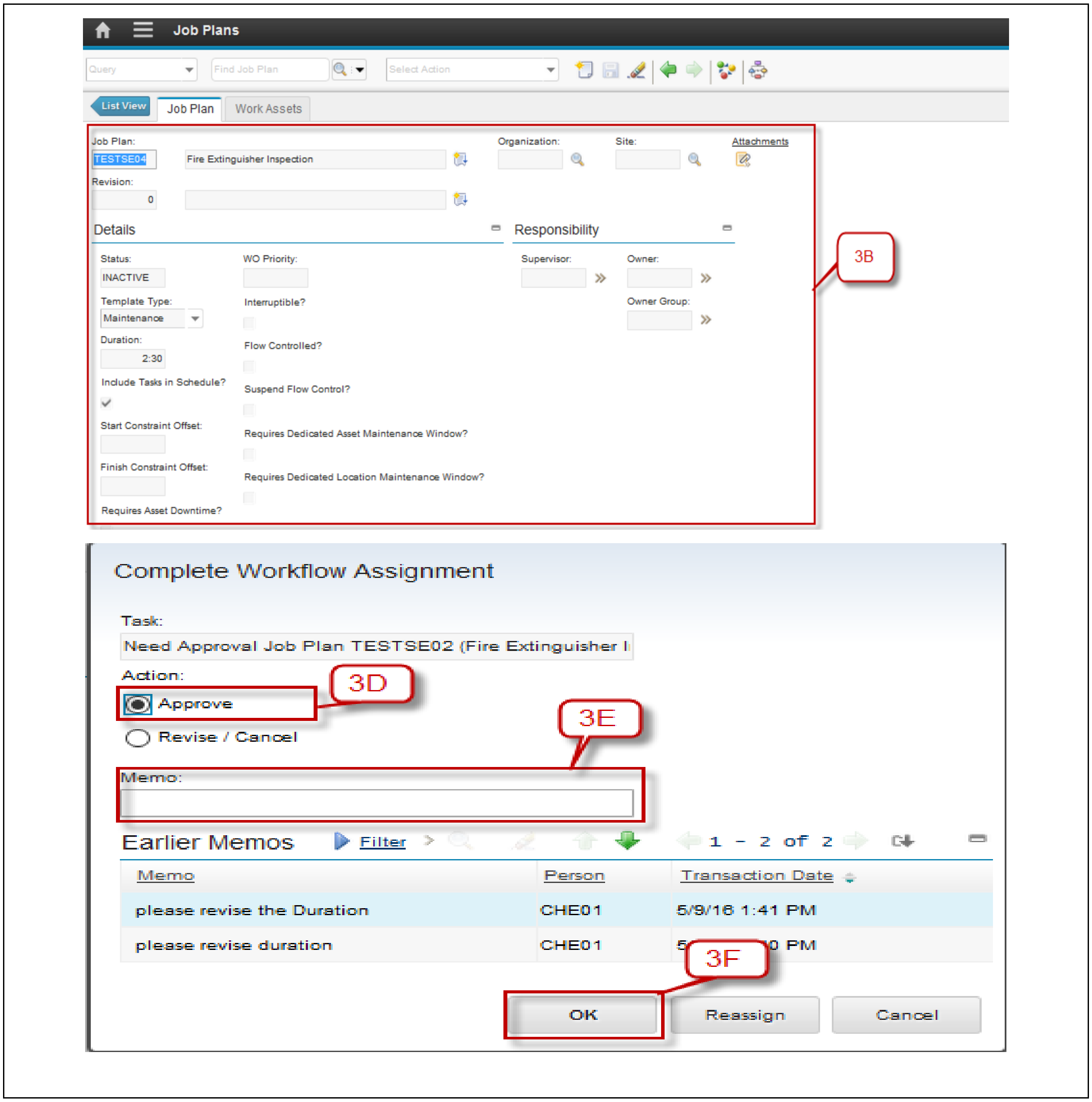
3C. After the revision is ok, then click  to approve the revision

3D. Click **Approve** button

3E. CHE Engineer can give a memo to this result of revision

3F. Then click **OK**





5 Preventive Maintenance

5.1 Preventive Maintenance Overview

Preventive maintenance is a maintenance schedule that is defined to plan a routine maintenance toward assets. Preventive maintenance schedule can be set based on time, or continuous meter that's the asset have.

5.2 Preventive Maintenance Application

You use the **Preventive Maintenance** application to create, modify, and view preventive maintenance plans for work assets and locations. PM records are templates for work orders. On a PM record, you schedule job plans to be performed for preventive maintenance work. Maximo then generates work orders from the PM.

5.2.1 Time Based Preventive Maintenance

Below are step by step to create Time Based Preventive Maintenance:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td colspan="2">

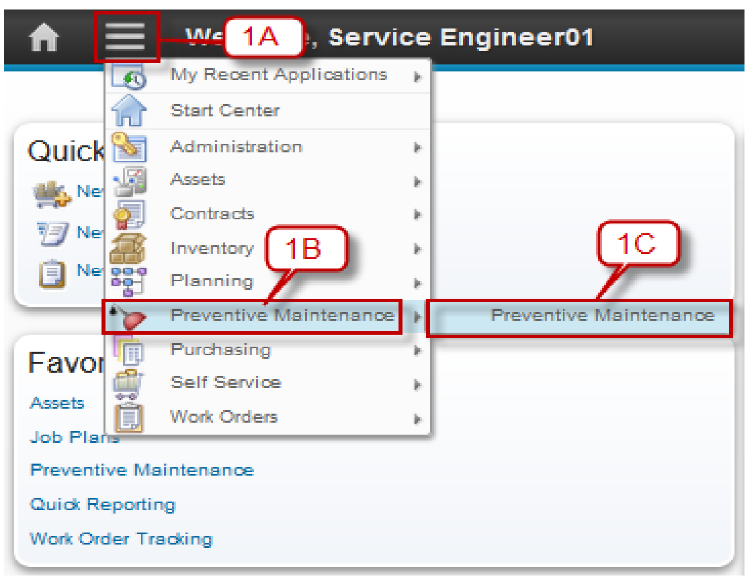
1. Login as Service Engineer. User name : se01, password: password.

1A. Click in icon menu ( )

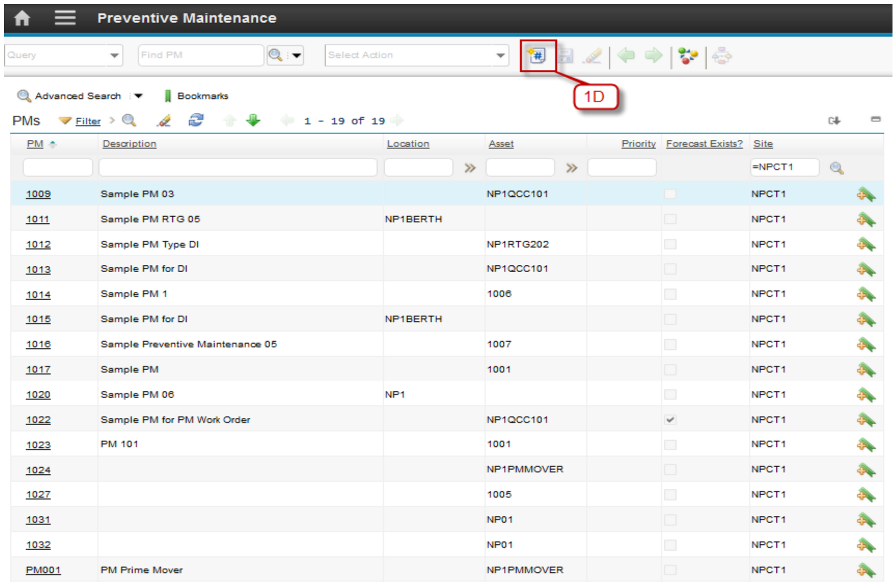
1B. Click **Preventive Maintenance**

1C. Click **Preventive Maintenance**

1D. Click icon New ( )



</td></tr>
</table>



2A. PM number is autonumber. Define description for Preventive Maintenance.

2B. Define Location or Asset or Route

2C. Click **>>** in **Job Plan** field, and select one Job Plan used for this Preventive Maintenance

2D. Click in **Work Type** field and select **DI (Daily Inspection)**

2E. Click in **Work Order Status** field and select **WEXEC (Waiting Execute) .**

- If **Work Type** is **PM (Preventive Maintenance)** status in **Work Order Status** must be **APPR (Approve)**

2F. Click **Save**.

2G. Input detail of Priority, Requires Asset Downtime, Storeroom, Storeroom Site

2H. Click on tab Frequency

2I. Click on Tab **Time Based Frequency**

2J. Define **Frequency** Value with 1

2K. Click in **Frequency units** and select "**DAYS / MONTHS / WEEKS / YEARS**"

2L. Click in **Estimated Next Due Date** and select current date

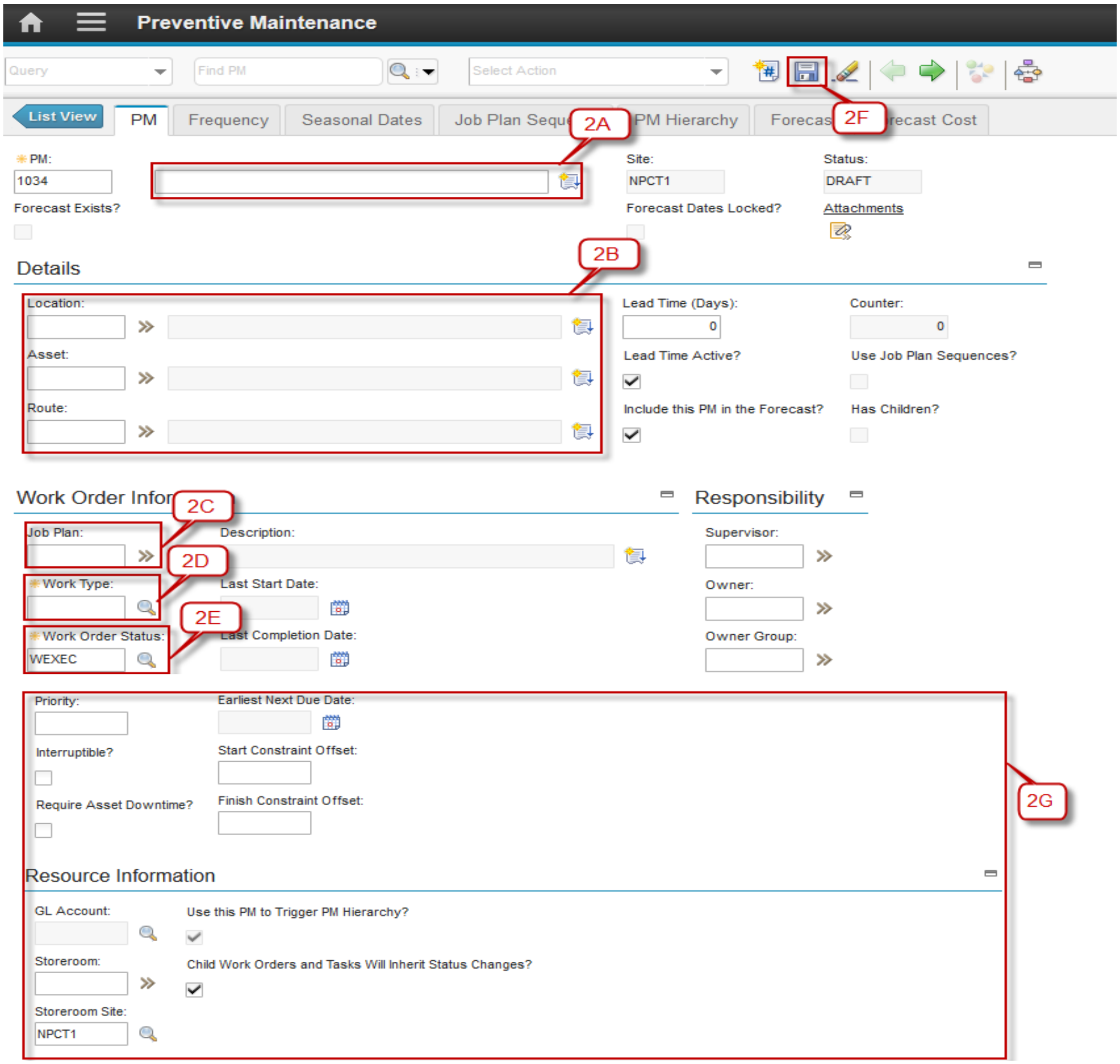
2M. Click to Save PM

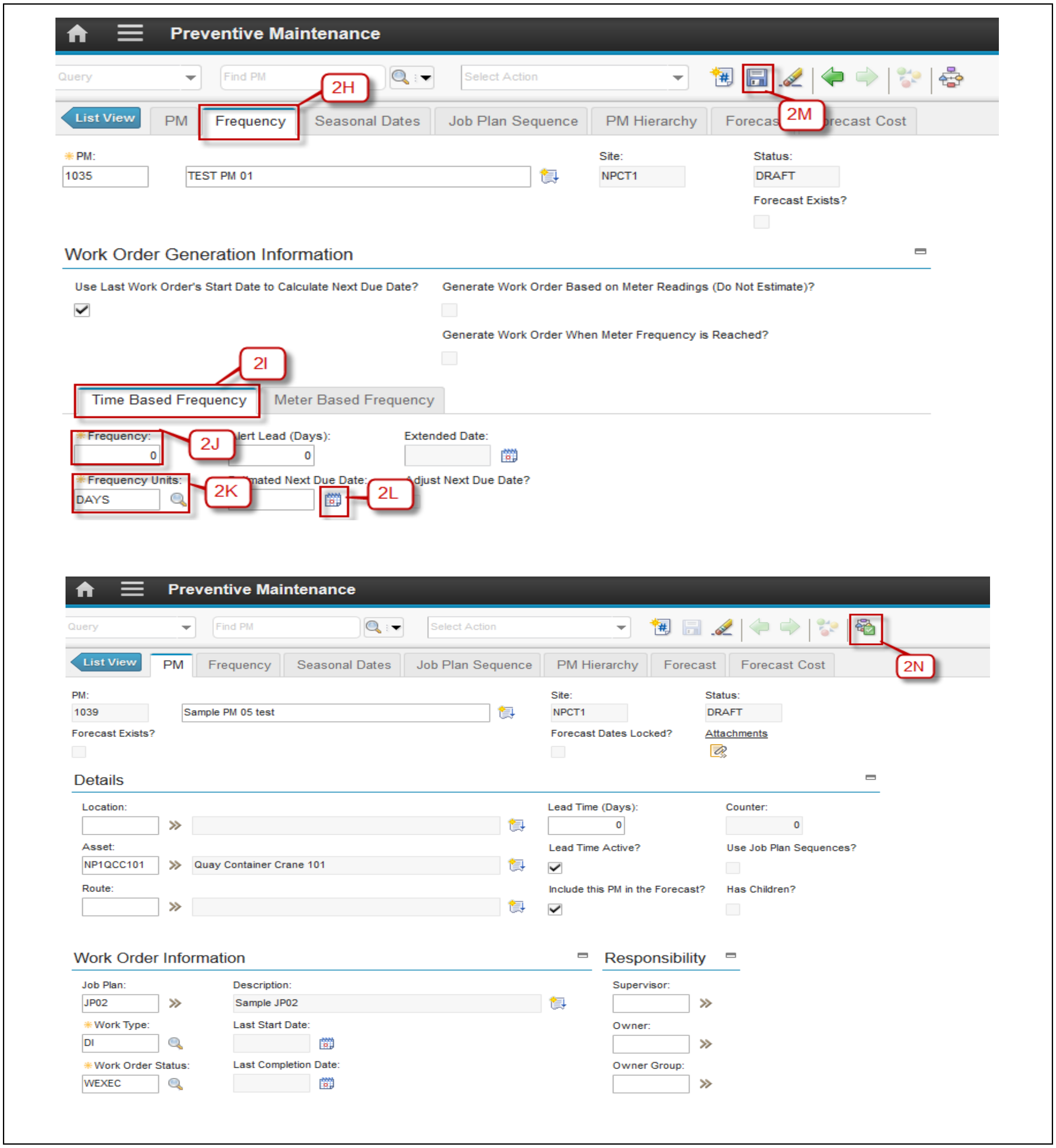
2N. Click to Route Workflow.

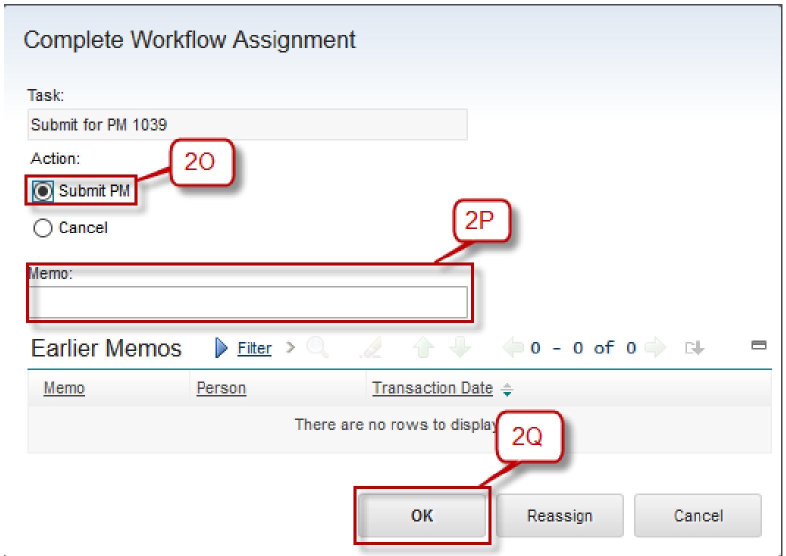
2O. Click on **Submit PM**

2P. Input memo if needed

2Q. Click **OK**




Preventive Maintenance
2H
2M
List View
PM
Frequency
Seasonal Dates
Job Plan Sequence
PM Hierarchy
Forecast Cost
PM:
1035
TEST PM 01
Site:
NPCT1
Status:
DRAFT
Forecast Exists?
Work Order Generation Information
Use Last Work Order's Start Date to Calculate Next Due Date?
Generate Work Order Based on Meter Readings (Do Not Estimate)?
Generate Work Order When Meter Frequency is Reached?
2I
Time Based Frequency
Meter Based Frequency
Frequency:
2J
Extended Date:
Frequency Units:
DAYS
2K
2L
Preventive Maintenance
List View
PM
Frequency
Seasonal Dates
Job Plan Sequence
PM Hierarchy
Forecast
Forecast Cost
2N
PM:
1039
Sample PM 05 test
Site:
NPCT1
Status:
DRAFT
Forecast Exists?
Forecast Dates Locked?
Attachments
Details
Location:
Lead Time (Days):
Counter:
Asset:
NP1QCC101
Quay Container Crane 101
Lead Time Active?
Use Job Plan Sequences?
Route:
Include this PM in the Forecast?
Has Children?
Work Order Information
Responsibility
Job Plan:
JP02
Description:
Sample JP02
Supervisor:
Work Type:
DI
Last Start Date:
Owner:
Work Order Status:
WEXEC
Last Completion Date:
Owner Group:



3. Login as CHE Engineer. User name : che01, password : password.

3A. In CHE Engineer Dashboard in **“Inbox / Assignment”** click **Need Approval for PM**

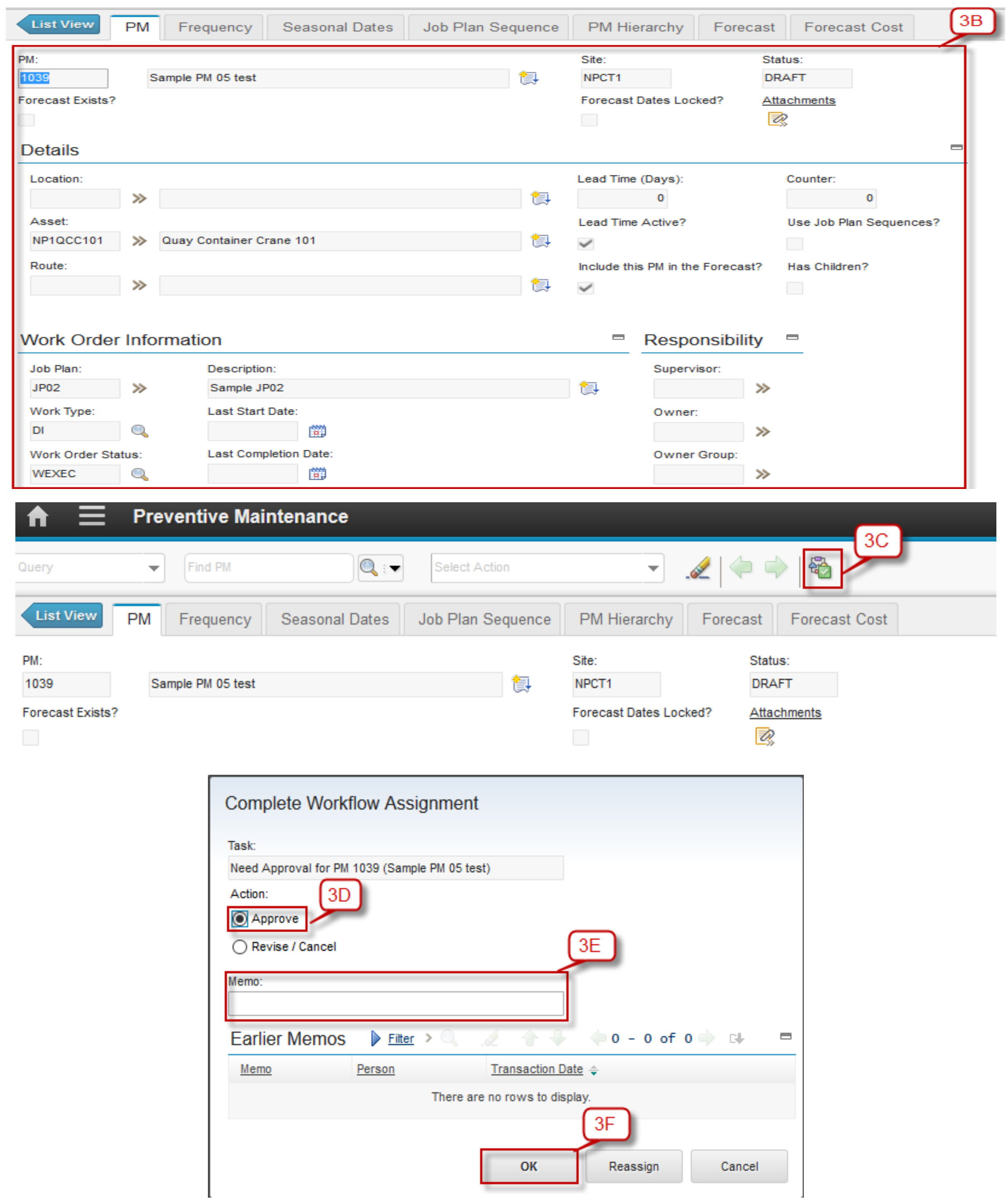
3B. CHE Engineer will review detail of PM

3C. Click  (route workflow)

3D. Click **Approve**

3E. Input Memo if needed

3F. Click **OK**



4. For Revised PM and Cancel PM please see the Revised and Cancel Job Plan, the step is same with Job Plan

5.2.2 Meter Based Preventive Maintenance

Below are step by step to create Meter Based Preventive Maintenance:

Step	Action

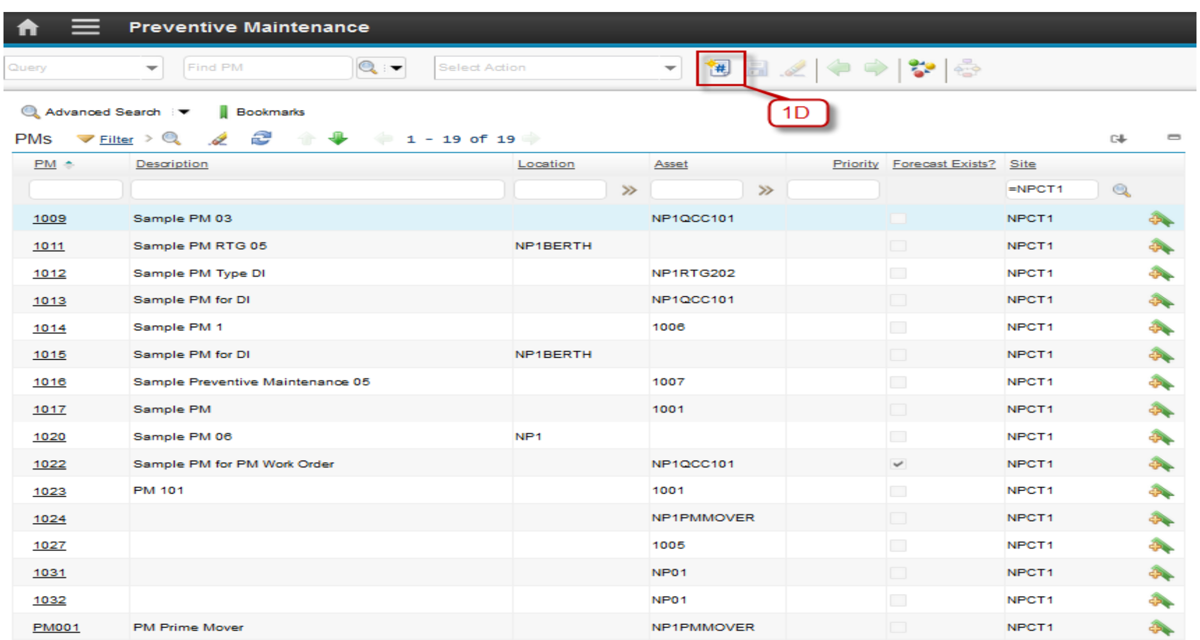
1. Login as Service Engineer. User name : se01, password: password.

1A. Click in icon menu ( )

1B. Click **Preventive Maintenance**

1C. Click **Preventive Maintenance**

1D. Click icon **New** ( )



2A. PM Number is autonumber. Define description for Preventive Maintenance

2B. Define Location or Asset or Route

2C. Click **>>** in **Job Plan** field, and select one Job Plan used for this Preventive Maintenance

2D. Click in **Work Type** field and select **DI (Daily Inspection)**

2E. Click in **Work Order Status** field and select **WEXEC (Waiting Execute) .**

- If **Work Type** is **PM (Preventive Maintenance)** status in **Work Order Status** must be **APPR (Approve)**

2F. Click to save Preventive Maintenance

2G. Input detail of Priority, Requires Asset Downtime, Storeroom, Storeroom Site

2H. Click on tab Frequency

2I. Click on Tab **Meter Based Frequency**

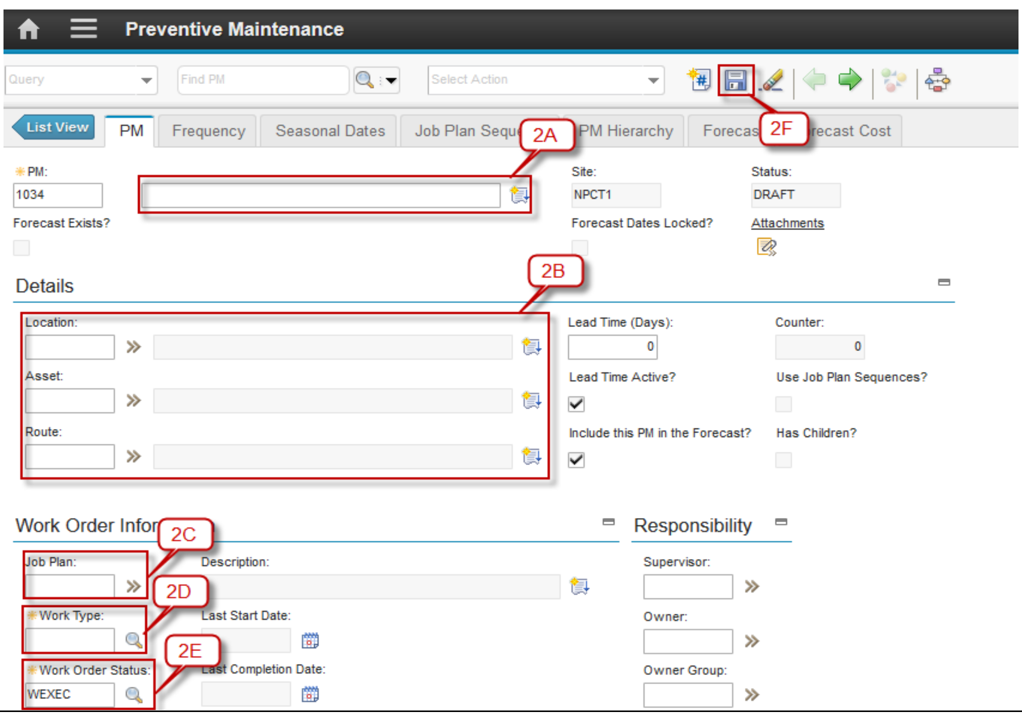
2J. Click **New Row**

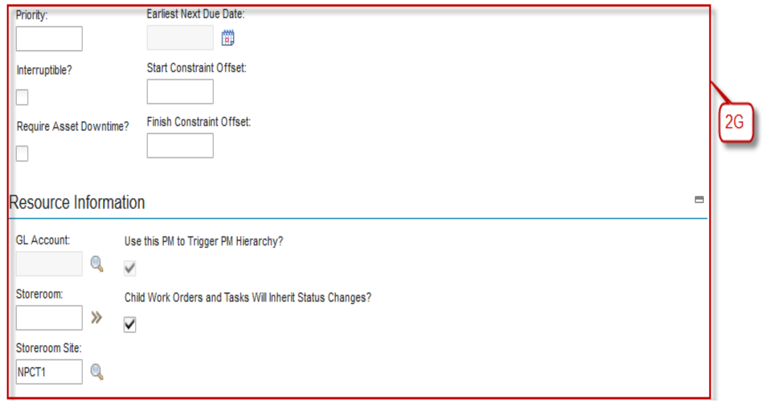
2K. Click in **Meter** fields and choose one from available list

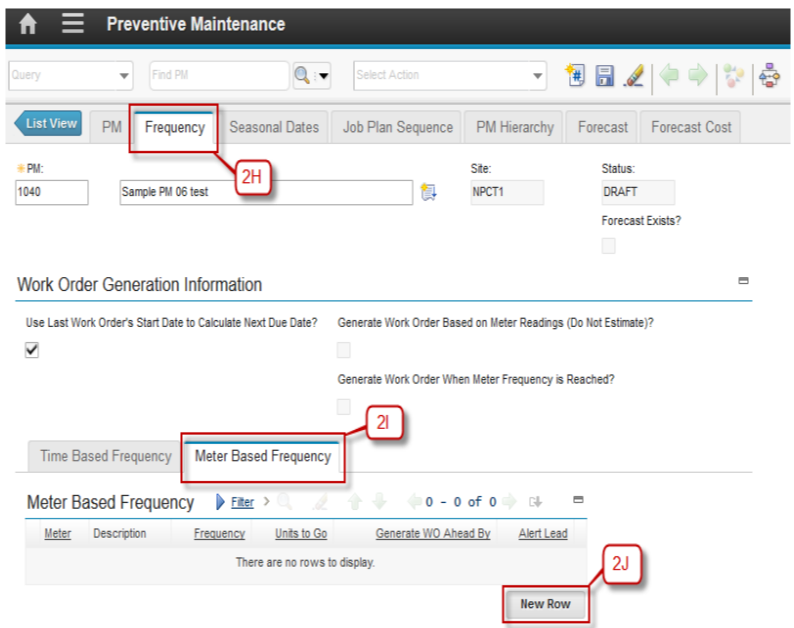
2L. Define **Frequency**

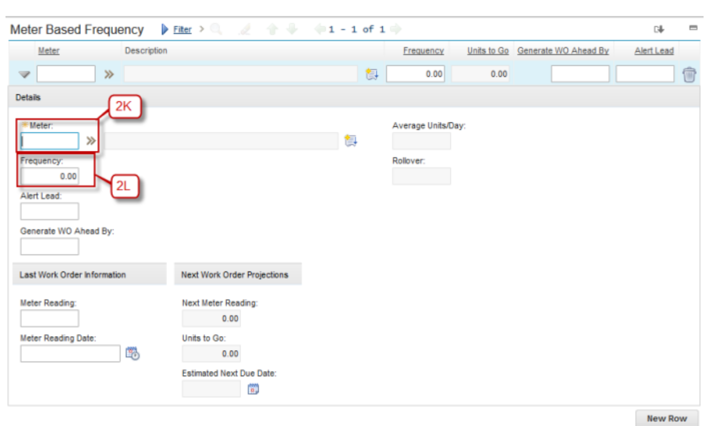
2M. If you want to add another meters repeat 3 step above from **"New Row"**

2N. Click **Save**




Priority:
Earliest Next Due Date:
Interruptible?
Start Constraint Offset:
Require Asset Downtime?
Finish Constraint Offset:
2G
Resource Information
GL Account:
Use this PM to Trigger PM Hierarchy?
Storeroom:
Child Work Orders and Tasks Will Inherit Status Changes?
Storeroom Site:
NPCT1


Preventive Maintenance
Query
Find PM
Select Action
List View
PM
Frequency
Seasonal Dates
Job Plan Sequence
PM Hierarchy
Forecast
Forecast Cost
2H
PM:
1040
Sample PM 06 test
Site:
NPCT1
Status:
DRAFT
Forecast Exists?
Work Order Generation Information
Use Last Work Order's Start Date to Calculate Next Due Date?
Generate Work Order Based on Meter Readings (Do Not Estimate)?
Generate Work Order When Meter Frequency is Reached?
2I
Time Based Frequency
Meter Based Frequency
Meter Based Frequency
Filter
0 - 0 of 0
Meter
Description
Frequency
Units to Go
Generate WO Ahead By
Alert Lead
There are no rows to display.
2J
New Row


Meter Based Frequency
Filter
1 - 1 of 1
Meter
Description
Frequency
Units to Go
Generate WO Ahead By
Alert Lead
0.00
0.00
Details
2K
Meter:
Average Units/Day:
Frequency:
0.00
Rollover:
2L
Alert Lead:
Generate WO Ahead By:
Last Work Order Information
Next Work Order Projections
Meter Reading:
Next Meter Reading:
0.00
Meter Reading Date:
Units to Go:
0.00
Estimated Next Due Date:
New Row

5.2.3 Multiple Job Plans for a Preventive Maintenance

Preventive Maintenance also have multiple Job Plans. Below are step by step to Multiple Job Plans for a Preventive Maintenance.

Step	Action

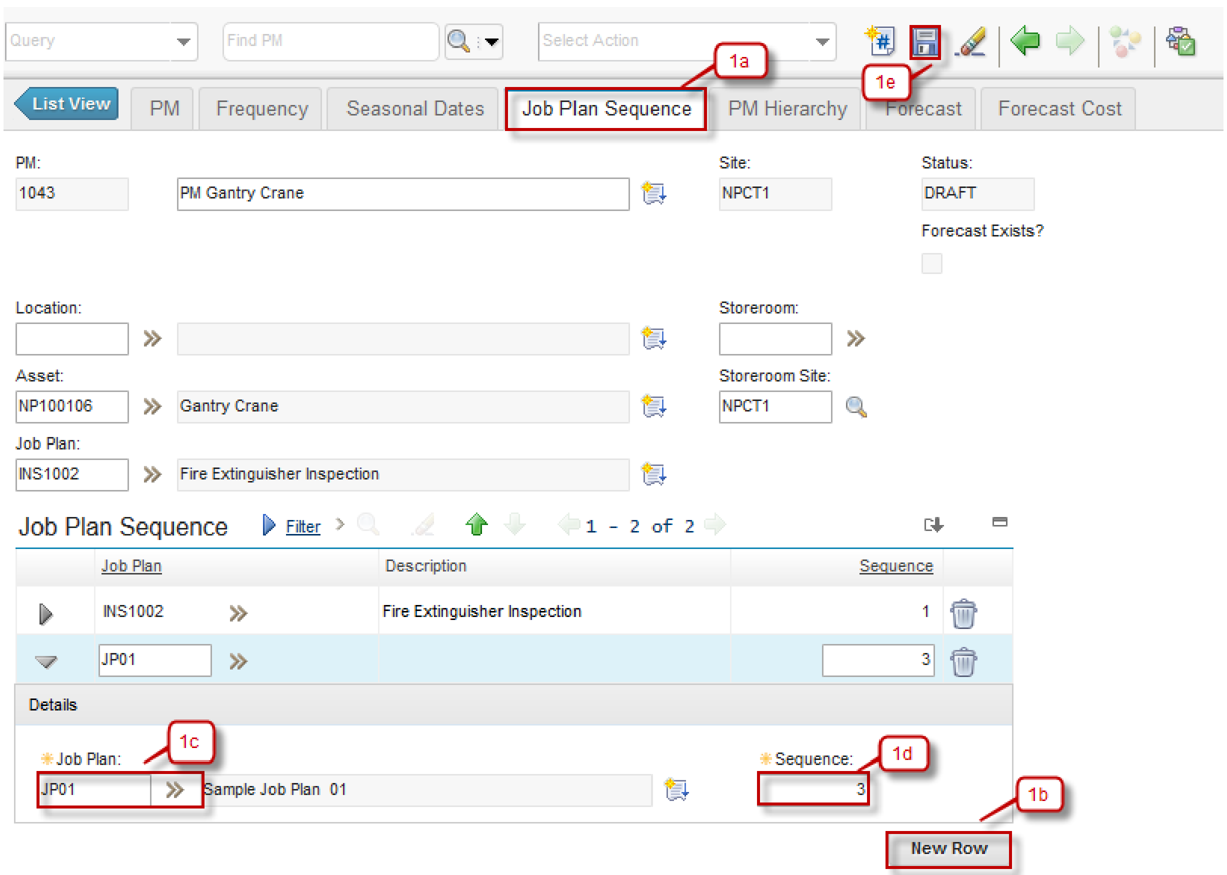
1a. Move to Tab **Job Plan Sequence**. Go to Job Plan Sequence Section on the page.

1b. Click **New Row**

1c. Select the required Job Plan using button »

1d. Input Sequence of Job Plan. Sequence is interval the Job Plan is used.

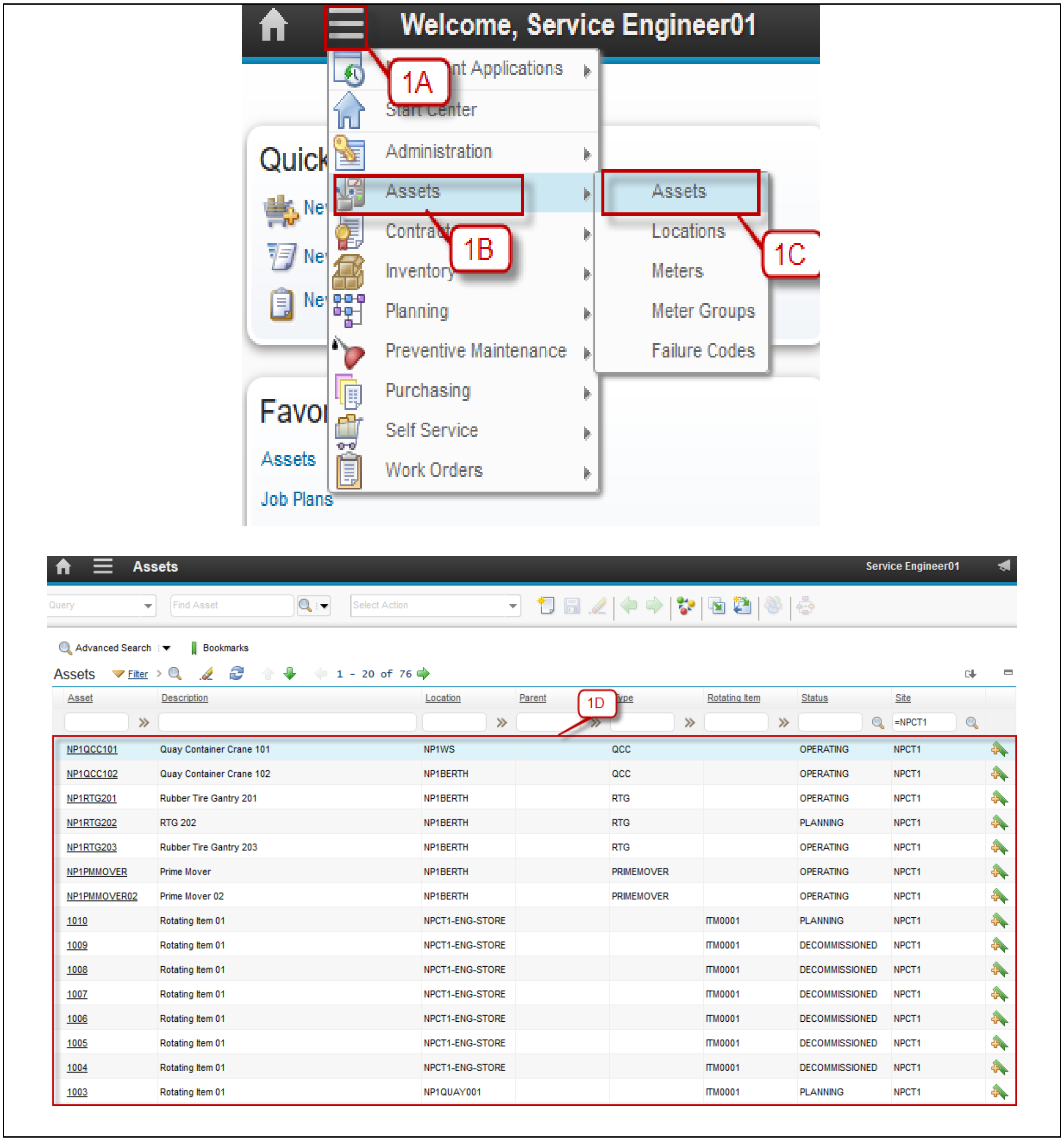
1e. Click **Save**.

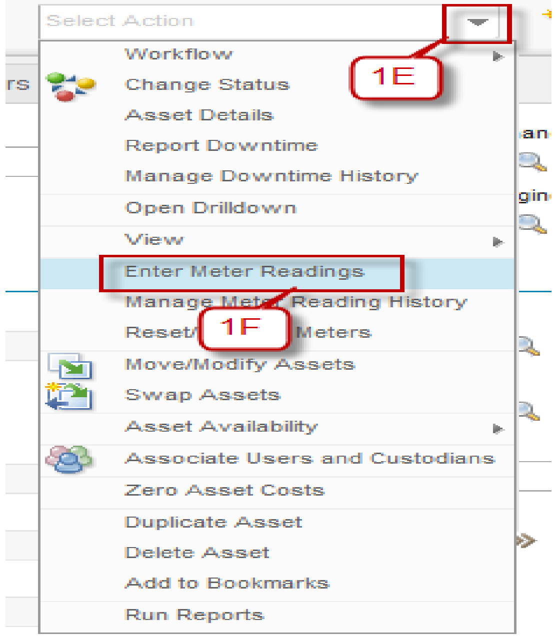


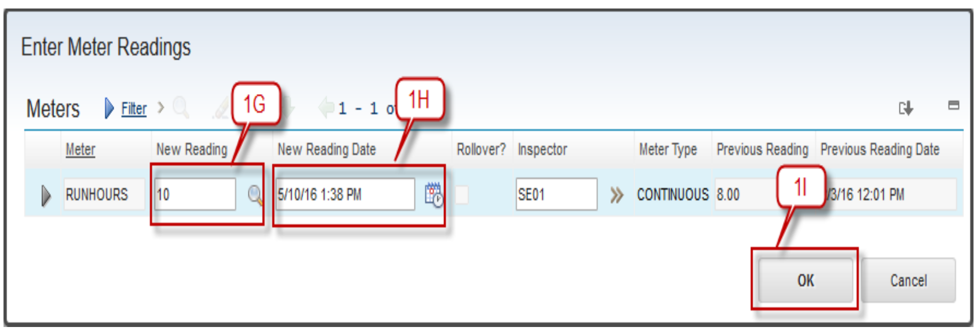
5.2.4 Forecasting – Meter Based

Below are step by step to create Forecasting Meter Based:

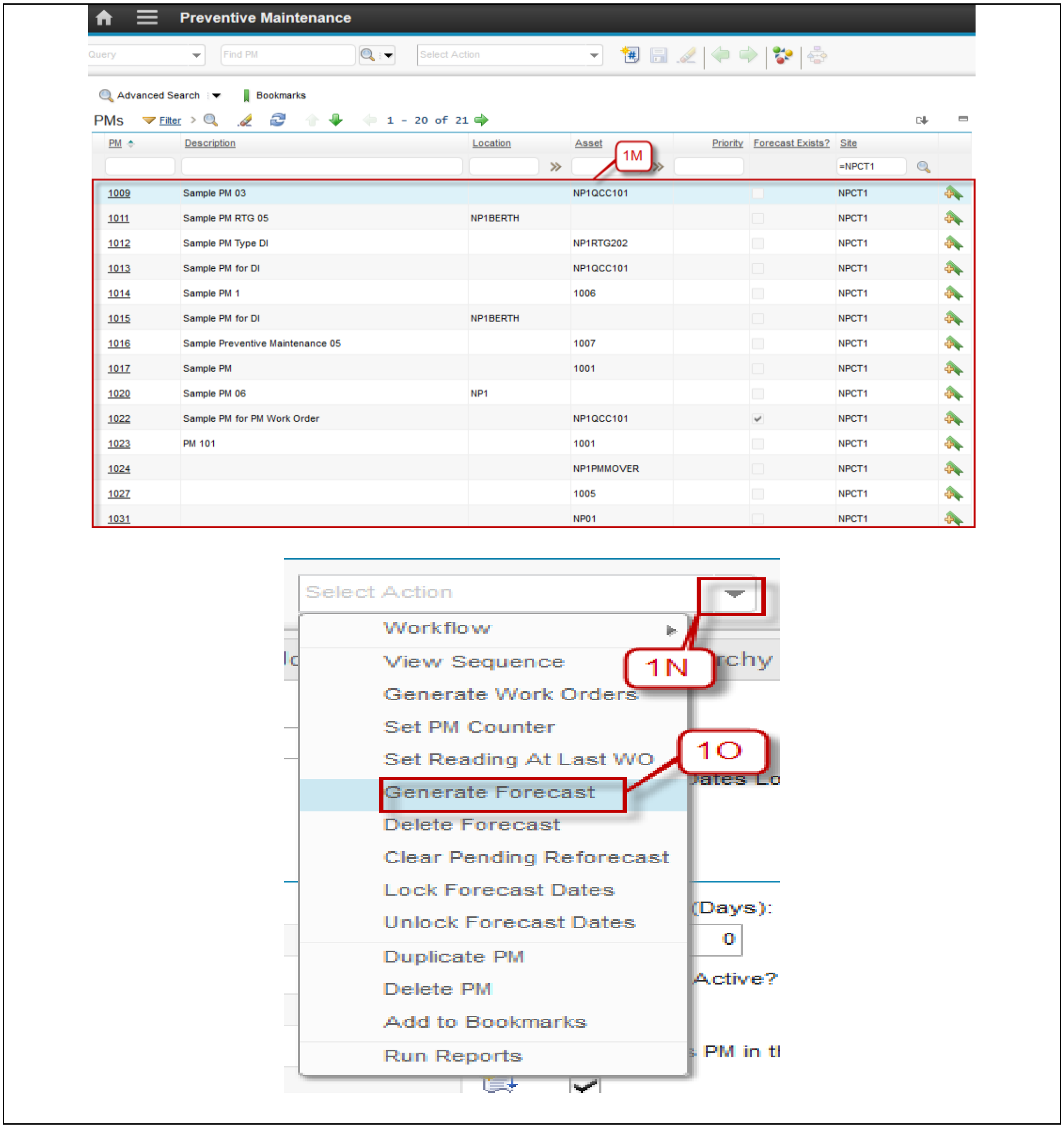
Step	Action
1A. Navigate to **Go To** Application	
1B. Navigate on **Assets**	
1C. Click on **Assets** menu	
1D. Search assets whose meter reading wants to be inputted. Then click Asset Number	
1E. Click **Select Action** with click on drilldown menu	
1F. Click on **Enter Meter Reading** menu	
1G. Input New Meter Reading Value	
1H. Input New Reading Date	
1I. Click **OK**	
1J. Navigate on **Go To**	
1K. Click on **Preventive Maintenance**	
1L. Click on **Preventive maintenance**	
1M. Search PM that wants to be processed. Click PM number	
1N. Click **Select Action** on drilldown icon	
1O. Choose **Generate Forecast**	
1P. Define forecast until or forecast for (days) field. (Defining 1 field will fill the other field automatically)	
1Q. Click **OK**	
1R. Click on **Forecast Tab**	
1S. Click on **Calculation Cost**	
1F. Click on **Select Action** on drilldown icon	
1U. Click **Generate Work Orders**	
1V. Define **Generate Wos Due Today Plus This Number of Days**	
1W. Click **OK**	

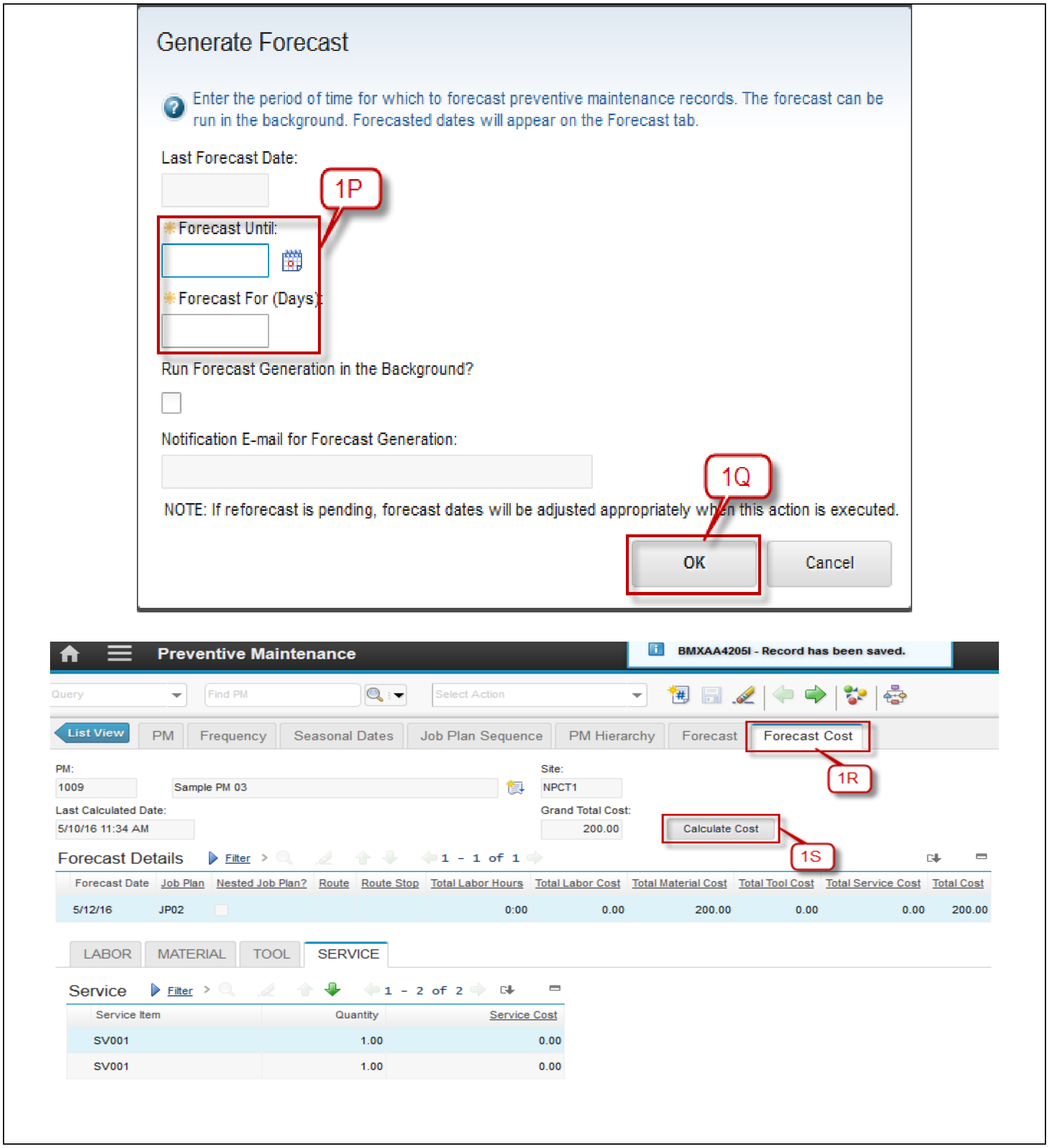

Welcome, Service Engineer01
1A
Start Center
Administration
Assets
1B
Contracts
Inventory
Planning
Preventive Maintenance
Purchasing
Self Service
Work Orders
Assets
Locations
1C
Meters
Meter Groups
Failure Codes
Quick
Favor
Assets
Job Plans
Assets
Service Engineer01
Advanced Search
Bookmarks
Assets
Filter
1 - 20 of 76
Asset
Description
Location
Parent
1D
Type
Rotating Item
Status
Site
=NPCT1
NP1QCC101 Quay Container Crane 101 NP1WS QCC OPERATING NPCT1
NP1QCC102 Quay Container Crane 102 NP1BERTH QCC OPERATING NPCT1
NP1RTG201 Rubber Tire Gantry 201 NP1BERTH RTG OPERATING NPCT1
NP1RTG202 RTG 202 NP1BERTH RTG PLANNING NPCT1
NP1RTG203 Rubber Tire Gantry 203 NP1BERTH RTG OPERATING NPCT1
NP1PMMOVER Prime Mover NP1BERTH PRIMEMOVER OPERATING NPCT1
NP1PMMOVER02 Prime Mover 02 NP1BERTH PRIMEMOVER OPERATING NPCT1
1010 Rotating Item 01 NPCT1-ENG-STORE ITM0001 PLANNING NPCT1
1009 Rotating Item 01 NPCT1-ENG-STORE ITM0001 DECOMMISSIONED NPCT1
1008 Rotating Item 01 NPCT1-ENG-STORE ITM0001 DECOMMISSIONED NPCT1
1007 Rotating Item 01 NPCT1-ENG-STORE ITM0001 DECOMMISSIONED NPCT1
1006 Rotating Item 01 NPCT1-ENG-STORE ITM0001 DECOMMISSIONED NPCT1
1005 Rotating Item 01 NPCT1-ENG-STORE ITM0001 DECOMMISSIONED NPCT1
1004 Rotating Item 01 NPCT1-ENG-STORE ITM0001 DECOMMISSIONED NPCT1
1003 Rotating Item 01 NP1QUAY001 ITM0001 PLANNING NPCT1

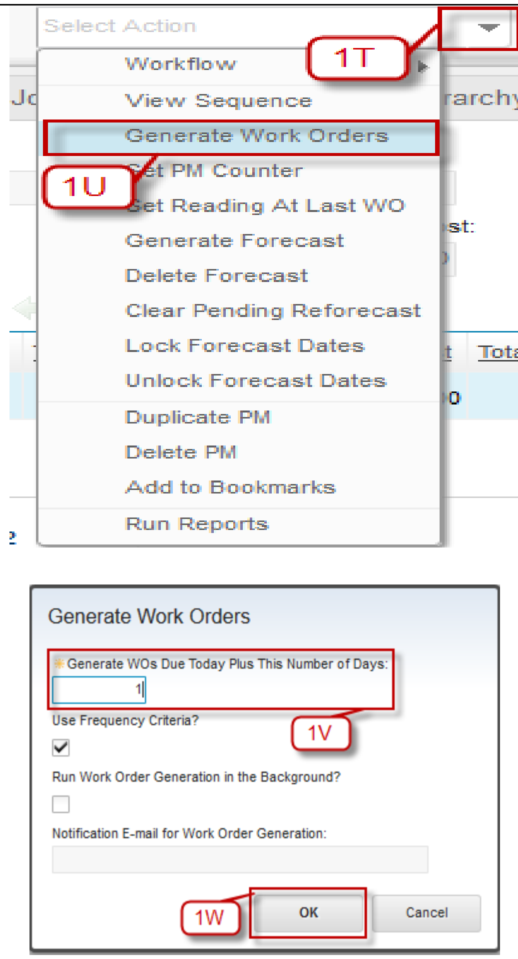

Select Action
1E
Workflow
Change Status
Asset Details
Report Downtime
Manage Downtime History
Open Drilldown
View
Enter Meter Readings
Manage Meter Reading History
1F
Reset/ Meters
Move/Modify Assets
Swap Assets
Asset Availability
Associate Users and Custodians
Zero Asset Costs
Duplicate Asset
Delete Asset
Add to Bookmarks
Run Reports


Enter Meter Readings
Meters
Filter
1G
1H
Meter
New Reading
New Reading Date
Rollover?
Inspector
Meter Type
Previous Reading
Previous Reading Date
RUNHOURS
10
5/10/16 1:38 PM
SE01
CONTINUOUS
8.00
1I
3/16 12:01 PM
OK
Cancel


Welcome, Service Engineer01
1J
My Applications
Start Center
Administration
Assets
Contracts
Inventory
1K
Planning
Preventive Maintenance
1L
Preventive Maintenance
Purchasing
Self Service
Work Orders
Quick
Favor
Assets
Job Plans
Preventive Maintenance
Quick Reporting
Work Order Tracking


Preventive Maintenance
Query
Find PM
Select Action
Advanced Search
Bookmarks
PMs
Filter
1 - 20 of 21
PM
Description
Location
Asset
Priority
Forecast Exists?
Site
=NPCT1
1M
1009 Sample PM 03 NP1QCC101 NPCT1
1011 Sample PM RTG 05 NP1BERTH NPCT1
1012 Sample PM Type DI NP1RTG202 NPCT1
1013 Sample PM for DI NP1QCC101 NPCT1
1014 Sample PM 1 1006 NPCT1
1015 Sample PM for DI NP1BERTH NPCT1
1016 Sample Preventive Maintenance 05 1007 NPCT1
1017 Sample PM 1001 NPCT1
1020 Sample PM 06 NP1 NPCT1
1022 Sample PM for PM Work Order NP1QCC101 NPCT1
1023 PM 101 1001 NPCT1
1024 NP1PMMOVER NPCT1
1027 1005 NPCT1
1031 NP01 NPCT1
Select Action
Workflow
View Sequence
1N
Generate Work Orders
Set PM Counter
Set Reading At Last WO
1O
Generate Forecast
Delete Forecast
Clear Pending Reforecast
Lock Forecast Dates
Unlock Forecast Dates
Duplicate PM
Delete PM
Add to Bookmarks
Run Reports
(Days):
0
Active?


Generate Forecast
Enter the period of time for which to forecast preventive maintenance records. The forecast can be run in the background. Forecasted dates will appear on the Forecast tab.
Last Forecast Date:
Forecast Until:
Forecast For (Days):
Run Forecast Generation in the Background?
Notification E-mail for Forecast Generation:
NOTE: If reforecast is pending, forecast dates will be adjusted appropriately when this action is executed.
OK
Cancel
1P
1Q
Preventive Maintenance
BMXAA4205I - Record has been saved.
List View
PM
Frequency
Seasonal Dates
Job Plan Sequence
PM Hierarchy
Forecast
Forecast Cost
1R
PM:
1009
Sample PM 03
Site:
NPCT1
Last Calculated Date:
5/10/16 11:34 AM
Grand Total Cost:
200.00
Calculate Cost
1S
Forecast Details
Forecast Date
Job Plan
Nested Job Plan?
Route
Route Stop
Total Labor Hours
Total Labor Cost
Total Material Cost
Total Tool Cost
Total Service Cost
Total Cost
5/12/16
JP02
0:00
0.00
200.00
0.00
0.00
200.00
LABOR
MATERIAL
TOOL
SERVICE
Service
Service Item
Quantity
Service Cost
SV001
1.00
0.00
SV001
1.00
0.00



5.2.5 Forecasting – Time Based

Below are step by step to create Forecasting Time Based:

Step	Action
1A. Click **Go To**	
1B. Navigate to **Preventive Maintenance**	
1C. Click on **Preventive Maintenance**	
1D. Search PM that wants to be processed. Click PM number	
1E. Click **Select Action** in drilldown icon	
1F. Choose **Generate Forecast**	
1G. Define **Forecast until** or **Forecast For (Days)** field. (Defining 1 field will fill the other field	

automatically)

1H. Click **OK**

1I. Click **Forecast** Tab

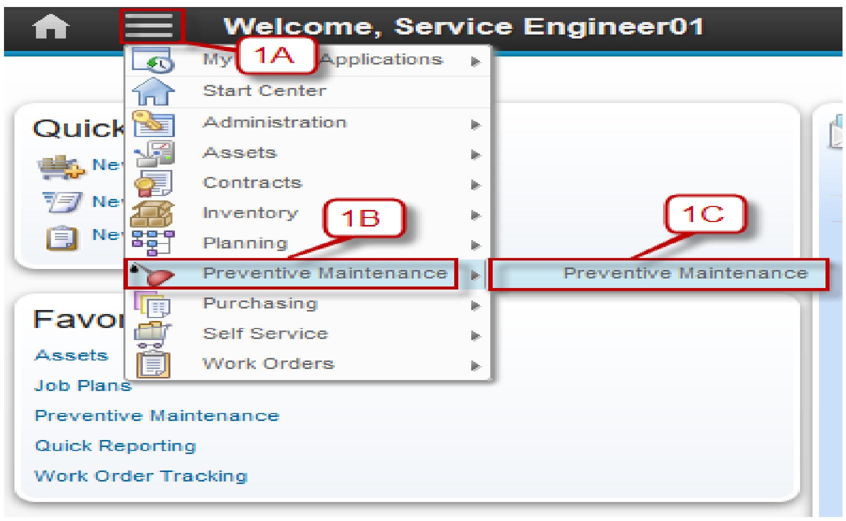
1J. Click **Calculate Cost** Button

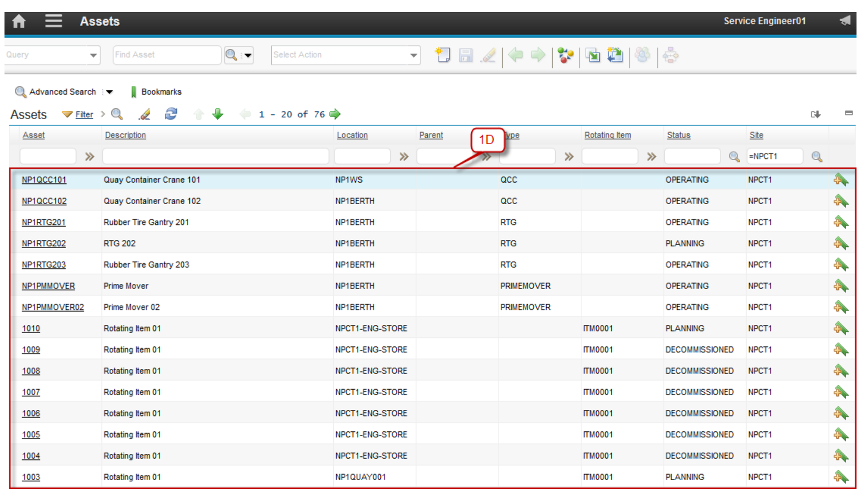
1K. Click **Select Action** in drilldown menu

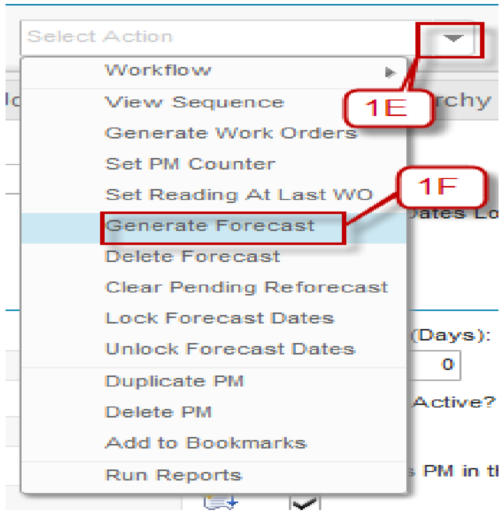
1L. Choose **Generate Work Orders**

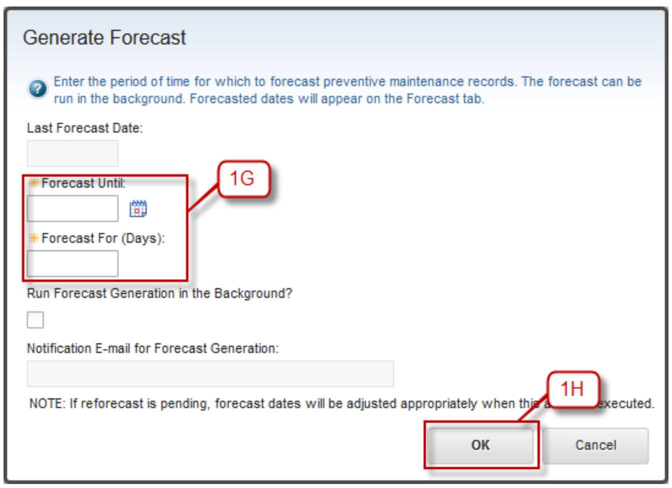
1M. Define **"Generate Wos Due Today Plus This Number of Days"**

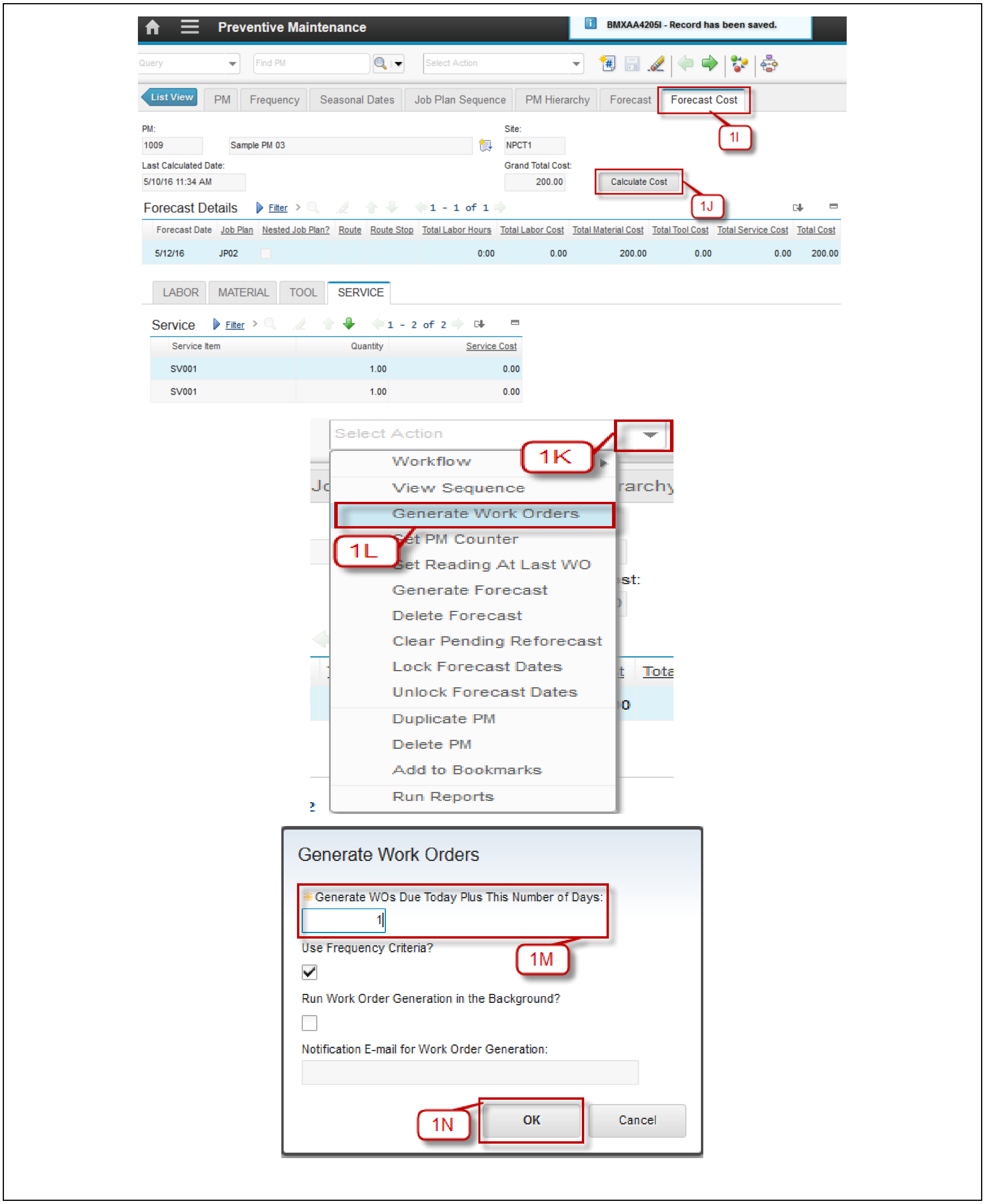
1N. Click **OK**






Select Action
Workflow
View Sequence
Generate Work Orders
Set PM Counter
Set Reading At Last WO
Generate Forecast
Delete Forecast
Clear Pending Reforecast
Lock Forecast Dates
Unlock Forecast Dates
Duplicate PM
Delete PM
Add to Bookmarks
Run Reports
1E
1F


Generate Forecast
Enter the period of time for which to forecast preventive maintenance records. The forecast can be run in the background. Forecasted dates will appear on the Forecast tab.
Last Forecast Date:
Forecast Until:
Forecast For (Days):
1G
Run Forecast Generation in the Background?
Notification E-mail for Forecast Generation:
1H
OK
Cancel


Preventive Maintenance
BMXAA4205I - Record has been saved.
List View
PM
Frequency
Seasonal Dates
Job Plan Sequence
PM Hierarchy
Forecast
Forecast Cost
1I
PM:
1009
Sample PM 03
Site:
NPCT1
Last Calculated Date:
5/10/16 11:34 AM
Grand Total Cost:
200.00
Calculate Cost
1J
Forecast Details
Forecast Date
Job Plan
Nested Job Plan?
Route
Route Stop
Total Labor Hours
Total Labor Cost
Total Material Cost
Total Tool Cost
Total Service Cost
Total Cost
5/12/16
JP02
0:00
0.00
200.00
0.00
0.00
200.00
LABOR
MATERIAL
TOOL
SERVICE
Service
Service Item
Quantity
Service Cost
SV001
1.00
0.00
SV001
1.00
0.00
Select Action
1K
Workflow
View Sequence
Generate Work Orders
1L
Set PM Counter
Set Reading At Last WO
Generate Forecast
Delete Forecast
Clear Pending Reforecast
Lock Forecast Dates
Unlock Forecast Dates
Duplicate PM
Delete PM
Add to Bookmarks
Run Reports
Generate Work Orders
Generate WOs Due Today Plus This Number of Days:
1
Use Frequency Criteria?
1M
Run Work Order Generation in the Background?
Notification E-mail for Work Order Generation:
1N
OK
Cancel

5.3 Routes Application

A route is a list of related work assets, which are considered "stops" along the route. These route stops represent assets or locations. Route simplifies building hierarchies of work orders for inspections. You can use a route in the following ways:

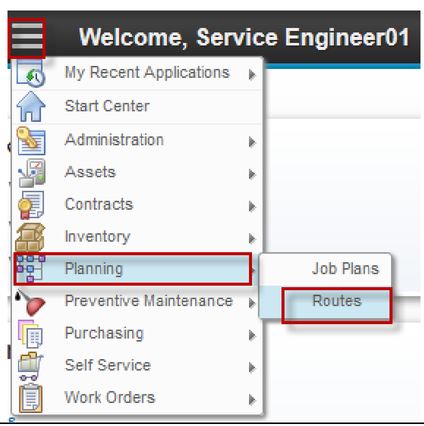
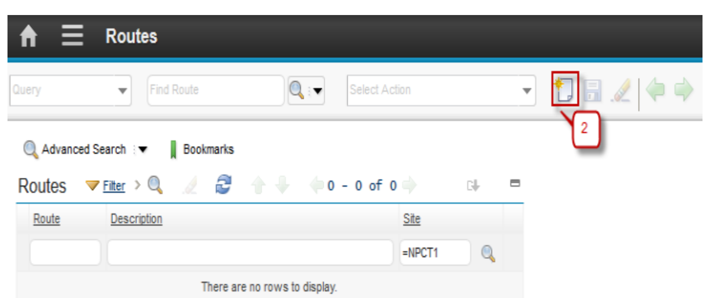
- Apply the route to a preventive maintenance record to generate work orders for all work assets listed as stops on the route.
- Apply the route to a work order and generate child work orders for each work asset listed as a stop on the route.

A route is a list of related asset or location work assets. You can create a route in the **Routes** application of the **Plans** module. You can create a route that lists all assets of the same type, or all assets in a certain location, or both. For example, you might create an inspection route that lists all fire extinguishers in a plant, or all safety equipment in a building.

When you create the route, you can specify a job plan for some or all route stops. Maximo copies these job plans to work orders generated for the route stop. If a safety plan is requires when you use this job plan on the selected work asset, Maximo also copies that information to the work order.

5.3.1 Create Routes

Below are step by step to create Routes:

Step	Action
1. Go to Routes Application. Click button (GoTo) – **Planning** - **Routes**	
2. Click **New Row**	

3a. Input the Route Number.

3b. Input the Route Description.

3c. If you want the status of all child work orders to update simultaneously, select the **Route Stop Inherit Change Status?** checkbox. Otherwise, clear the checkbox.

3d. At the bottom of the **Route Stops** table window, click **New Row**.

3e. In the **Location** or **Asset** field, enter a value or click » (details menu) and select an option.

3f. In the **Job Plan** field, enter a value or click » (details menu) and select an option.

3g. (Optional) In the **Sequence** field, enter a sequence number to correspond to each identifier, or route stop.

3h. Click **Save.**